MIE312 Midterm 2

Control volumes, differential relations and dimensional analysis

Written by a model for Adit. Not Adit's words.

First iteration, 2026-11-10, for the midterm on 2026-11-12

The summary of this document is `content/teach/mie312.md` in the vault; marks go there, section by section, and the next teach pass rewrites both files. The examinable material is White, *Fluid Mechanics*, 8th edition, Sections 3.1–3.7, 4.1–4.3, 4.6, 4.8, 4.10 and 5.1–5.5. Textbook problems are named by number only, as White P3.62; nothing from the book is reproduced. Every worked example is invented for this document.

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1 What the midterm is

Thursday 2026-11-12, 18:10 to 19:40, at the Exam Centre, 255 McCaul Street. Ninety minutes, three questions, fifteen marks each. The instructions ask for three things in every answer: all the work shown, the assumptions stated, and units on every number. Constants you need are printed on the paper, and a crib sheet is handed out with it. Your own crib sheet and every other aid are not allowed.

What is examinable

The course names sections of White, *Fluid Mechanics*, 8th edition, as taught in lecture.

White	Content	This document
3.1–3.2	Basic laws, the Reynolds transport theorem	Section 5
3.3	Conservation of mass	Section 6
3.4	Linear momentum for a control volume	Section 7
3.5	Frictionless flow, Bernoulli	Section 8
3.6	Angular momentum	Section 9
3.7	Energy	Section 10
4.1	The acceleration field	Section 11
4.2	Differential continuity	Section 12
4.3	Differential momentum, Navier–Stokes	Section 13
4.6	Boundary conditions	Section 14
4.8	Vorticity and irrotationality	Section 15
4.10	Exact incompressible viscous flows	Section 16
5.1–5.2	Dimensional homogeneity	Section 17
5.3	The Pi theorem	Section 18
5.4	Nondimensionalizing the equations	Section 19
5.5	Modelling and similarity	Section 20

Not tested: 4.4 (differential angular momentum), 4.5 (differential energy), 4.7 (the stream function) and 4.9 (frictionless irrotational flow). They are in the course's Chapter 4 supplement and worth reading another time. The course says the best preparation is the lecture worked examples, the chapter summaries and the assignments. This document does not have those, so it sits beside them: it derives the theory, gives invented worked examples, and orders textbook problems into short ladders.

What the crib sheet gives, and what it does not

The sheet handed out with the paper is two pages of formulas and a Moody chart, covering the whole course. For this midterm the useful parts are these.

- Section 3 has the Reynolds transport theorem, relative velocity, mass, linear momentum, the force of a jet on a vane, angular momentum in two forms, and energy in head form with the kinetic energy factor α.

- Section 4 has the Pitot relations, which are Bernoulli in disguise.
- Section 12 has the Pi theorem's count, the requirement that repeating variables be independent, and the standard dimensionless groups.
- Section 13 has 2-D incompressible continuity, the vorticity and the element rotation, the strain rates, and the no-slip condition.
- Section 14 has Euler's turbomachine head and torque, which is angular momentum for a pump or turbine.
- Section 16 has the units and dimensions of the common quantities, and section 17 the properties of water by temperature.

Three gaps decide how to study.

1. **No Navier–Stokes.** The sheet does not print the material derivative, continuity in three dimensions or in cylindrical coordinates, the Navier–Stokes equations in any form, or any exact viscous solution except the laminar pipe flow rate. The course announcement says the Navier–Stokes equations need not be memorized. Either the paper prints them, or the question gives what it needs. Plan for neither: each Chapter 4 section here shows how to write its equation from scratch in under two minutes, from Newton's second law and a box. That costs an hour of practice and removes the risk. Asking the course before Thursday which equations the paper prints is cheap.
2. **No assumptions.** Every equation on the sheet is printed without the conditions under which it holds. The marks for "state your assumptions" are exactly the part the sheet cannot give you. Each section here has a box, *On the crib sheet*, that says what the sheet prints for that topic and what you must supply.
3. **No procedure.** The sheet has the Pi theorem's count, not the repeating-variable method, and the momentum equation, not the choice of control volume. Those two choices are the modelling calls of this midterm, and the next section is about them.

How the marks are likely spread

Three questions of fifteen marks in ninety minutes is thirty minutes a question, including the write-up. Questions of this length are usually multi-part, and the first parts carry the set-up marks: the control volume drawn, the assumptions listed, the right equation written down and reduced. A question whose final number is wrong but whose set-up is right usually keeps most of its marks. A question with the right number and no stated assumptions usually does not keep all of them. So the write-up order matters more here than in a problem set: sketch, assumptions, equation, reduction, numbers, units, check.

The three questions will probably follow the three chapters, one each: a control-volume question, a differential question, and a dimensional-analysis question. That is a guess from the announcement's chapter list, not something it says.

2 Reading a question, and the call

Almost every question in this midterm is decided before any algebra, by four choices. The rest is execution. Those four choices are the call.

1. **Which law.** Mass, linear momentum, angular momentum, energy, or Bernoulli; or, in Chapter 4, continuity and Navier–Stokes; or, in Chapter 5, a Pi analysis

and a scaling. A question can need two, as in Bernoulli for a velocity and then momentum for a force.
2. **Which control volume, or which coordinates.** For an integral question: where the control surface cuts, which forces cross it, whether it moves. For a differential question: Cartesian or cylindrical, which velocity components are zero.
3. **Which assumptions.** Steady or not; incompressible or not; frictionless or not; uniform velocity at a cut or a profile; fully developed or not; gravity kept or dropped. Each one is a sentence in the answer and a term struck in the equation.
4. **Which variables.** For dimensional analysis: the list of variables the result depends on, and which three to repeat. A missing variable gives a wrong answer that looks right.

The call line

During practice, before solving a problem, write one line in `calls.md`, domain `fluids`:

```
fluids: White P3.62, x-momentum on a CV cutting the flange and the jet,
steady, uniform exit, gauge p, 20 min. 60
```

The problem, the law, the control volume, the assumptions that matter, the time box, and how sure you are, from 0 to 100. Then solve. Then check the answer in the back of White and settle the line in the same sitting: *held* if the law and control volume you named carried it; *missed* if another choice from the four above was the right one; *blind* if the solution used something you did not consider. A right number reached by a different control volume than the one you named is a miss. That is the point of writing the line first.

The call is the part this midterm marks most and the crib sheet helps with least, so it is the part worth practising in isolation. On the evening before, a useful drill is ten problems from the ladders read for two minutes each, the call written, and nothing solved. Then read each answer in the back of White and ask whether its form agrees with the law you named: a force usually means momentum, a power means energy, a velocity profile means Navier–Stokes, a list of groups means the Pi theorem. Settle those calls that the answer decides, and leave the rest open for a full solve.

Choosing the control volume

Three rules decide most control volumes in Chapter 3.

- Cut where you know things, and cut where you want to know things. A cut where the pressure and velocity are known (a free jet at atmospheric pressure, a reservoir surface) is cheap. A cut through a support, a bolt or a flange exposes the force you are asked for.
- Cut perpendicular to the flow. Then $\mathbf{V} \cdot \mathbf{n}$ is just the speed, the mass flux is $\rho V A$, and the pressure force on the cut is the pressure times the area along the normal.
- Keep the control volume fixed unless something moves at constant speed and you want the force on it. For a moving vane, a control volume moving with the vane makes the flow steady; the price is that every velocity at the surface becomes relative.

For a differential question the corresponding call is the coordinate system and the list of velocity components that are zero. Pipe, annulus, rotating shaft: cylindrical.

Channel, plate, film: Cartesian. Then continuity tells you which derivatives vanish before you touch Navier–Stokes.

Time boxes

Thirty minutes is the exam's own box per question. In practice use twenty for a ladder rung marked easy and thirty for anything else. When the box runs out, stop, write what you would do next in one line, and read the answer. The settle line records whether the call was right even when the arithmetic was not finished.

3 The plan to Thursday

Two days, three sittings and a mock. Each sitting has six core problems from White, in order, each a little harder than the last. Every one has its answer in the back of the book, so every call can be settled the same evening. Write the call line before each problem, as in Section 2, and settle it as soon as you check the answer. Read the matching sections of this document before a sitting, not during it. Optional rungs are for a sitting that runs fast; the per-section ladders list more.

Tuesday 2026-11-10 evening: Chapter 3

Read first: Section 7 and Section 10; skim the rest of Chapter 3.

	Problem	What it is	What it trains
1	White P3.14	A tank with one inlet and two outlets; the level rate, then the outlet speed for a constant level.	Unsteady mass, signs of inflow and outflow.
2	White P3.40	A round jet hits a fixed plate square on; the holding force.	The simplest momentum control volume.
3	White P3.62	A two-branch nozzle with angled jets; the bolt force at the inlet flange.	Pressure force at a cut, vector momentum flux.
4	White P3.88	A pump-jet boat with drag proportional to speed squared; the cruising speed.	Moving control volume, relative velocity.
5	White P3.180	A pump lifts water between reservoirs with losses and an efficiency; the power.	Energy equation in head form.
6	White P3.150	The moment about the inlet flange of an angled elbow.	Angular momentum, $\mathbf{r} \times \mathbf{V}$.
opt	White P3.46	A sheet splits on a frictionless inclined plate; the split fraction.	Zero tangential force as an assumption.
opt	White P3.122	A tank filled steadily while draining through a hole; the limiting depth.	Bernoulli with unsteady mass.

Wednesday 2026-11-11 morning: Chapter 4

Read first: Section 13 and Section 16. Before problem 1, write continuity and the Cartesian Navier–Stokes equations from memory, then check against the box in that section. Do it again before problem 4, in cylindrical coordinates.

	Problem	What it is	What it trains
1	White P4.4	A 2-D converging nozzle field; the accelerations and where the largest is.	The material derivative.
2	White P4.22	Axisymmetric flow with axial velocity linear in z; the general radial velocity.	Cylindrical continuity; the "constant" is a function.
3	White P4.24	A channel with porous walls; a velocity from continuity, the constant from the wall suction.	Continuity plus a boundary condition.
4	White P4.80	An oil film draining down a vertical wall; the profile.	Reducing Navier–Stokes; free-surface condition.
5	White P4.88	A rod sliding along the axis inside a fixed tube; the annular profile.	Cylindrical Navier–Stokes; a $\ln r$ solution.
6	White P4.60	A drain vortex with swirl falling as $1/r$; is it rotational, and the surface depth.	Vorticity, then Bernoulli where it is allowed.
opt	White P4.92	A tank emptying through a laminar pipe; the level history.	Poiseuille inside an unsteady mass balance.
opt	White P4.30	The nozzle field of problem 1: does it satisfy Navier–Stokes, and the pressure field.	Pressure from a given velocity field.

Wednesday 2026-11-11 afternoon: Chapter 5

Read first: Section 18 and Section 20.

	Problem	What it is	What it trains
1	White P5.12	Five variables give one group; show it is a ratio of familiar groups.	Dimensions from definitions.
2	White P5.18	Laminar flow in a triangular pore; the Pi form and the effect of doubling its size.	One group means a scaling law.
3	White P5.38	Spray droplet size against nozzle size, speed and liquid properties.	Repeating variables; Re and We appear.
4	White P5.24	Lift on a missile; the dimensional matrix, its rank, the groups.	Rank, and Mach as a group.
5	White P5.80	A ship and its towing-tank model under Froude scaling; speed, drag and power ratios.	Froude scaling laws.
6	White P5.66	A torpedo's cavitation speed at a new depth.	A similarity group with a pressure difference in it.
opt	White P5.4	A sphere tested in water scaled to a balloon in air.	Reynolds matching across fluids.
opt	White P5.62	A small wind-turbine test scaled to a large machine.	Scaling speed and power together.

Wednesday 2026-11-11 evening: the mock

Ninety minutes, closed book, with only the crib sheet, at the exam's time if the day allows (18:10). Three call lines in the first five minutes, then the protocol of Section 4.

1. White P3.112. An alcohol jet from a lossless nozzle hits a plate, and the holding force is known; the mass flow and the inlet pressure. Momentum then Bernoulli.
2. White P4.84. A liquid film draining down the outside of a vertical rod; the equation, its boundary conditions, the profile and the flow rate. Cylindrical Navier–Stokes with a free surface. If it is too hard at the twenty-minute mark, switch to White P4.36.
3. White P5.52. A Pi analysis of a transition time in an accelerating pipe flow, then a scaling to a new case.

Mark it against the book's answers that evening, and settle the three calls. Then read the Misses section of `readout.md`.

Thursday 2026-11-12

No new problems. In the morning, the call drill from Section 2: ten unsolved problems from the ladders, two minutes each, a call line for each. In the afternoon, the *First moves* and *Trap* boxes of every section, once. Then Section 4.

A warning about the answer key

A few answers in the back of this copy of White do not belong to the problems with the same numbers, apparently from a different printing: P3.6, P4.6, P5.14 and P5.22

among those checked. None of them is on a ladder. If an answer looks unrelated to its problem, it probably is; settle that call *moot*.

4 Exam evening, 2026-11-12

Before

Eat before 17:30. Bring pens, pencils, an eraser, a ruler for sketches, your student card, and the calculator the course allows. Be at 255 McCaul Street by 17:55. Do not study a new topic on Thursday. The useful thing to do that afternoon is to read the *First moves* and *Trap* boxes of every section once, which takes about twenty minutes.

The first five minutes

Read all three questions before writing anything. Then, on the first page of rough work, one line per question: the law, the control volume or coordinates, the two assumptions that matter, and a rank from 1 to 3 by how sure you are that you can finish. Start with rank 1. This is the call from Section 2, done on paper.

Look over the crib sheet once in those five minutes as well, so you know which equations it prints and which you will have to write yourself. If the paper prints the Navier–Stokes equations, the differential question is a reduction exercise; if it does not, write them from the box in Section 13 first.

The eighty-five minutes

Roughly twenty-eight minutes a question. A question not moving after twenty minutes: write down the set-up you have (the control volume, the assumptions, the equation, its reduced form) and move to the next. The set-up carries most of a question's marks, and a finished set-up on all three beats a finished answer on one. Return with whatever time is left.

Leave the last five minutes for units. Every final number needs one, and so does every intermediate number that you box.

The write-up, every question

1. A sketch. The control volume as a dashed line, the inlets and outlets labelled 1 and 2, the coordinate axes with their positive directions, and the forces on the control surface as arrows. For a differential question: the geometry, the coordinates, and the walls with their velocities.
2. The assumptions, as a short list. Steady; incompressible; uniform velocity at each cut; gauge pressure; neglect gravity in x; fully developed; and so on.
3. The governing equation as printed or as written from memory, before any term is removed.
4. The reduced equation, with a reason beside each term removed: "$\partial/\partial t = 0$, steady" and "$v = 0$, from continuity".
5. The algebra, then the numbers, with units carried through.

6. One line of check: the sign of a force against intuition, a velocity at a wall against the no-slip condition, the dimensions of a Pi group.

After

That evening, write the three call lines from memory, as on the exam’s first page, on paper or in a note that does not sync. Keep them out of `calls.md`, which is public on GitHub, until the course has posted solutions or returned the papers, because some students write the midterm later. Then type them in with `2026-11-12` at the start of each and the domain `mie312`, and settle them against the posted solutions or the marked paper. The `mie312` row under Domains in `readout.md` is the record of this test.

5 The Reynolds transport theorem

Newton's second law and the first law of thermodynamics are statements about a fixed lump of matter. A fluid does not stay in lumps: the water in a pump at one instant is downstream a second later. Every equation in this chapter is one of those old laws rewritten for a region of space that the fluid flows through. The rewriting is done once, by the Reynolds transport theorem, and then reused for mass, momentum, angular momentum and energy. This section derives it from a one-dimensional picture, states the general form, and settles the one sign that causes most lost marks.

Recognise it. Every control-volume question in chapter 3 uses it, usually silently. It appears by name when a question asks you to "start from the Reynolds transport theorem", when the control volume moves or changes shape (a cart, a rocket, a syringe, a balloon), when the fluid crosses a boundary with a velocity profile instead of a single speed, or when the property being tracked is unusual (kinetic energy, a concentration, angular momentum).

On the crib sheet. The sheet prints

$$\frac{dB_{\text{sys}}}{dt} = \frac{\partial}{\partial t}\int_{\text{CV}} \rho\beta \, d\forall + \int_{\text{CS}} \rho\beta \, (\mathbf{V}_r \cdot \mathbf{n}) \, dA, \qquad \mathbf{V}_r = \mathbf{V}_f - \mathbf{V}_{\text{cv}}, \qquad \dot{m}_{\text{rel}} = \rho A V_r.$$

Here β is the amount of the property per unit mass; this document writes it b, to keep β free for the momentum correction factor in Section 7. The sheet does not say that $\mathbf{n}$ is the *outward* normal, so that outflow counts positive; that the $\partial/\partial t$ outside the integral is only correct for a control volume that keeps its shape (a deforming one needs d/dt); or what $\mathbf{V}_{\text{cv}}$ means when the surface moves in different places at different speeds. Those are supplied below.

The model

Two ways of looking. A *system* is a fixed collection of matter: the same particles, always, wherever they go. Newton's law $\sum \mathbf{F} = d(m\mathbf{V})/dt$ and the first law $dE = \delta Q - \delta W$ are written for systems. A *control volume* (CV) is a region of space chosen by the analyst, such as the inside of a pipe bend or a box drawn around a jet engine. Fluid flows in and out of it. Its boundary is the *control surface* (CS). The reason to switch is practical: an engineer can measure pressures and speeds at the inlet and outlet of a fixed region, but cannot follow every particle.

Extensive and intensive. Let B be any quantity that adds up over the matter: mass, momentum, energy. Such a quantity is called *extensive*. Its amount per unit mass is $b = dB/dm$, called *intensive*. The four choices used in this chapter are

$$B = m,\ b = 1; \qquad B = m\mathbf{V},\ b = \mathbf{V}; \qquad B = \mathbf{r} \times m\mathbf{V},\ b = \mathbf{r} \times \mathbf{V}; \qquad B = E,\ b = e.$$

The amount of B in a region is $\int b\,\rho\, d\forall$, because $\rho\, d\forall$ is the mass of a small piece of volume $d\forall$.

The one-dimensional derivation. Take a duct with one inlet, section 1, and one outlet, section 2. The CV is the inside of the duct between them and does not move. At time t, choose the system to be exactly the fluid inside the CV. In a short time

dt the system moves downstream. At the inlet, fluid that is *not* part of the system enters the CV: a slab of length $V_1\,dt$, volume $A_1V_1\,dt$, mass $\rho_1A_1V_1\,dt$. At the outlet, a slab of the system leaves the CV, with mass $\rho_2A_2V_2\,dt$. So at $t+dt$,

$$B_{\text{sys}}(t+dt) = B_{\text{CV}}(t+dt) - b_1\rho_1A_1V_1\,dt + b_2\rho_2A_2V_2\,dt.$$

The first term counts everything in the CV; the second removes the inlet slab, which is in the CV but not in the system; the third adds the outlet slab, which is in the system but has left the CV. At t the two coincide: $B_{\text{sys}}(t) = B_{\text{CV}}(t)$. Subtract and divide by dt:

$$\frac{dB_{\text{sys}}}{dt} = \frac{dB_{\text{CV}}}{dt} + b_2\,\rho_2A_2V_2 - b_1\,\rho_1A_1V_1.$$

In words: the rate of change for the system equals the rate of change of what is stored in the CV, plus the rate at which the property is carried out, minus the rate at which it is carried in. The products $\rho AV = \dot{m}$ are mass flow rates, so the last two terms read $\dot{m}_2b_2 - \dot{m}_1b_1$.

The general form. On a general surface the fluid crosses at an angle, and speed varies from point to point. Let $\mathbf{n}$ be the unit vector perpendicular to the surface at each point, pointing *out* of the CV. Only the component of velocity along $\mathbf{n}$ carries fluid across; the component along the surface slides past. The volume crossing a small patch dA per unit time is $(\mathbf{V}\cdot\mathbf{n})\,dA$, so the property carried out per unit time is $b\rho(\mathbf{V}\cdot\mathbf{n})\,dA$. Summing over the surface,

Theorem 5.1 (Reynolds transport theorem, fixed CV)**.** *For a control volume fixed in space,*

$$\frac{dB_{sys}}{dt} = \frac{\partial}{\partial t}\int_{CV} b\,\rho\,d\forall + \int_{CS} b\,\rho\,(\mathbf{V}\cdot\mathbf{n})\,dA.$$

The first term is the *storage* term. The second is the *flux* term: a *flux* is an amount crossing a surface per unit time, so the momentum flux, for example, is the momentum carried across per second. The 1-D result is the case where $\mathbf{V}\cdot\mathbf{n} = +V_2$ at the outlet (fluid leaves along $\mathbf{n}$) and $-V_1$ at the inlet (fluid arrives against $\mathbf{n}$), with b and ρ uniform across each section.

The sign of $\mathbf{V}\cdot\mathbf{n}$. Because $\mathbf{n}$ points out, $\mathbf{V}\cdot\mathbf{n} > 0$ where fluid leaves and < 0 where it enters, and it is zero on solid walls, where no fluid crosses. Nobody has to remember a sign for inlets: the dot product supplies it. For sections where the properties are uniform, the flux integral becomes

$$\int_{\text{CS}} b\,\rho\,(\mathbf{V}\cdot\mathbf{n})\,dA = \sum_{\text{out}} \dot{m}\,b - \sum_{\text{in}} \dot{m}\,b, \qquad \dot{m} = \rho A|V_n|,$$

which is exactly how the sheet writes the mass and momentum equations. Here V_n is the velocity component normal to the section.

Moving and deforming CVs. If the CV travels with a constant velocity $\mathbf{V}_{\text{cv}}$, such as a box riding on a cart, what crosses its surface is decided by the fluid velocity *relative to the surface*, $\mathbf{V}_r = \mathbf{V} - \mathbf{V}_{\text{cv}}$. A swimmer moving at the speed of the current has no water passing her. So the flux term uses $\mathbf{V}_r\cdot\mathbf{n}$. If the CV also changes shape (a balloon inflating, a cylinder with a moving piston), each part of the surface has

its own velocity $\mathbf{V}_s$, and $\mathbf{V}_r = \mathbf{V} - \mathbf{V}_s$ point by point. The storage term then keeps the total derivative outside:

$$\frac{dB_{\text{sys}}}{dt} = \frac{d}{dt}\int_{\text{CV}} b\,\rho\,d\forall + \int_{\text{CS}} b\,\rho\,(\mathbf{V}_r \cdot \mathbf{n})\,dA.$$

The d/dt is needed because the region over which you integrate is itself changing with time; for a fixed region it equals $\partial/\partial t$ moved inside. One caution for momentum: $b = \mathbf{V}$ in the storage term is still the velocity measured in an inertial frame. For a CV moving at constant velocity the frame of the CV is also inertial, so everything may be measured in it. For an accelerating CV it is not; Section 7 handles that case.

Steady flow. Steady means nothing at a fixed point changes with time. For a fixed CV in steady flow the storage term is zero, and the theorem says the rate of change of B for the system equals the net outflow of B. Steady is a statement about a frame: the flow over a moving vane is unsteady to the ground and steady to an observer riding the vane, which is why the vane problems in Section 7 are done in the vane's frame.

First moves. Draw the CV and say whether it is fixed, moving at constant velocity, or deforming. Mark every place fluid crosses the CS and draw $\mathbf{n}$ outward at each. Choose B and write b. Decide whether the storage term vanishes (steady, in the chosen frame). At each crossing, find the velocity relative to the surface and take its normal component; inflow is negative. If a profile is given, keep the integral; if not, use $\dot{m}b$ at each section. Only then substitute the physical law for dB_{sys}/dt: zero for mass, $\sum \mathbf{F}$ for momentum, $\sum \mathbf{M}$ for angular momentum, $\dot{Q} - \dot{W}$ for energy.

Worked examples

Example 5.2. A syringe has a barrel of inside diameter $20\,\text{mm}$ and a needle of inside diameter $0.8\,\text{mm}$. It holds water ($\rho = 998\,\text{kg/m}^3$, $20\,^\circ\text{C}$), and the plunger is pushed in at a steady $5\,\text{mm/s}$. (a) Choose a control volume whose surface follows the plunger, and use the transport theorem for mass to find the jet speed at the needle exit. (b) Repeat with a fixed control volume and show that it agrees. (c) Find the mass flow rate and the time to expel a $40\,\text{mm}$ column of water.

Solution. Assumptions: water incompressible; velocity uniform across the barrel and across the needle exit; the plunger seals perfectly; the needle is short enough that its entry fills it at once.

(a) The CV is all the water in the syringe: its left face is glued to the plunger and moves with it at $V_p = 5\,\text{mm/s}$; the rest of the surface is the barrel and needle wall, fixed, plus the needle exit plane, fixed. This CV deforms, so the storage term is d/dt of the mass inside. With $B = m$, $b = 1$, and $dm_{\text{sys}}/dt = 0$:

$$0 = \frac{d}{dt}(\rho\forall_{\text{CV}}) + \int_{\text{CS}} \rho(\mathbf{V}_r \cdot \mathbf{n})\,dA.$$

At the plunger face the water moves with the face, so $\mathbf{V}_r = 0$: nothing crosses. At the walls nothing crosses. At the exit the face is fixed and $\mathbf{V}_r \cdot \mathbf{n} = +V_j$. The volume shrinks at the rate $A_b V_p$, so $d(\rho\forall)/dt = -\rho A_b V_p$. Hence $0 = -\rho A_b V_p + \rho a V_j$, and

$$V_j = V_p \frac{A_b}{a} = V_p\Big(\frac{D}{d}\Big)^2 = 0.005 \times \Big(\frac{20}{0.8}\Big)^2 = 0.005 \times 625 = 3.125\,\text{m/s}.$$

(b) Now fix the CV in space: a cut across the barrel a little downstream of the plunger, the barrel and needle walls, and the exit plane. The water inside never changes in amount, so the storage term is zero. At the cut, water arrives at V_p against the outward normal, $\mathbf{V}\cdot\mathbf{n} = -V_p$; at the exit $+V_j$. So $0 = -\rho A_b V_p + \rho a V_j$, the same equation. The two CVs move the same physics between the storage term and the flux term.

(c) $a = \frac{\pi}{4}(0.8\times10^{-3})^2 = 5.027\times10^{-7}\,\mathrm{m}^2$, so $Q = aV_j = 1.571\times10^{-6}\,\mathrm{m}^3/\mathrm{s} = 1.57\,\mathrm{mL/s}$ and $\dot m = \rho Q = 1.568\times10^{-3}\,\mathrm{kg/s}$. The column empties when the plunger has travelled $40\,\mathrm{mm}$: $t = 0.040/0.005 = 8.0\,\mathrm{s}$.

Check: the column holds $\frac{\pi}{4}(0.020)^2(0.040) = 1.257\times10^{-5}\,\mathrm{m}^3 = 12.6\,\mathrm{mL}$, and $12.6/1.571 = 8.0\,\mathrm{s}$. A jet of about $3\,\mathrm{m/s}$ from a hand-pushed syringe is the familiar squirt. □

Example 5.3. A boat travels upstream at $6\,\mathrm{m/s}$ relative to the riverbank in a river whose current is $1.5\,\mathrm{m/s}$. A forward-facing intake of area $0.02\,\mathrm{m}^2$ below the waterline scoops water into an onboard pump, which discharges it rearward through a nozzle of exit area $0.012\,\mathrm{m}^2$. Take river water as $1000\,\mathrm{kg/m}^3$. Find (a) the mass flow through the intake, (b) the jet speed relative to the boat and relative to the bank, and (c) the error made by using the boat's ground speed in (a).

Solution. Assumptions: steady in the boat's frame; uniform velocity across intake and nozzle; the water ahead of the intake is undisturbed river water; incompressible.

The CV is the water passage inside the boat, from the intake plane to the nozzle exit plane, moving with the boat. Take $+x$ upstream, the boat's direction of travel. In the bank's frame the boat moves at $\mathbf{V}_{\mathrm{cv}} = +6\,\mathbf{i}\,\mathrm{m/s}$ and the river water at $\mathbf{V}_f = -1.5\,\mathbf{i}\,\mathrm{m/s}$.

(a) Relative to the boat, the water arrives at $\mathbf{V}_r = \mathbf{V}_f - \mathbf{V}_{\mathrm{cv}} = -1.5 - 6 = -7.5\,\mathbf{i}\,\mathrm{m/s}$, that is, at $7.5\,\mathrm{m/s}$ straight into the intake. So $\dot m = \rho A V_r = 1000\times0.02\times7.5 = 150\,\mathrm{kg/s}$.

(b) In the boat's frame the flow is steady, so mass in equals mass out: $150 = 1000\times0.012\times V_{r,2}$, which gives $V_{r,2} = 12.5\,\mathrm{m/s}$ rearward. Relative to the bank, $\mathbf{V}_2 = \mathbf{V}_{r,2} + \mathbf{V}_{\mathrm{cv}} = -12.5 + 6 = -6.5\,\mathrm{m/s}$: the jet moves downstream at $6.5\,\mathrm{m/s}$ as seen from the bank.

(c) Using $6\,\mathrm{m/s}$ in (a) gives $120\,\mathrm{kg/s}$, which is 20% low. The water crossing a moving surface is set by the velocity relative to that surface, and here both the boat and the current contribute.

Check: a pump that takes in water at $7.5\,\mathrm{m/s}$ and expels it at $12.5\,\mathrm{m/s}$ relative to the boat accelerates it backward, which is what a waterjet propulsor must do to push the boat forward. □

Example 5.4. Oil ($\rho = 870\,\mathrm{kg/m}^3$) flows in fully developed laminar flow through a pipe of radius $R = 25\,\mathrm{mm}$ with the profile $u(r) = U_{\max}(1 - r^2/R^2)$, $U_{\max} = 1.2\,\mathrm{m/s}$. Using the transport theorem flux integral with $b = u$, find the rate at which x-momentum crosses a section, and compare it with $\dot m\, V_{\mathrm{avg}}$.

Solution. At a section the flux of x-momentum is $\int b\,\rho\,(\mathbf{V}\cdot\mathbf{n})\,dA = \int_0^R \rho u^2\, 2\pi r\,dr$, because $b = u$ and $\mathbf{V}\cdot\mathbf{n} = u$ at an outlet. The area of a thin ring is $2\pi r\,dr$. Substitute $s = r^2/R^2$, so $2r\,dr = R^2\,ds$:

$$\int_0^R \rho u^2\, 2\pi r\,dr = \rho U_{\max}^2\,\pi R^2 \int_0^1 (1-s)^2\,ds = \frac{\rho U_{\max}^2 \pi R^2}{3}$$
$$= \frac{870\times1.44\times\pi\times6.25\times10^{-4}}{3} = 0.820\,\mathrm{N}.$$

Momentum per second has units $\mathrm{kg\,m/s^2} = \mathrm{N}$. For the mass flow, the same substitution gives $Q = U_{\max}\pi R^2 \int_0^1 (1-s)\,ds = \frac{1}{2}U_{\max}\pi R^2$, so $V_{\text{avg}} = Q/(\pi R^2) = 0.6\,\mathrm{m/s}$, $Q = 1.178 \times 10^{-3}\,\mathrm{m^3/s}$ and $\dot{m} = 1.025\,\mathrm{kg/s}$. Then $\dot{m}V_{\text{avg}} = 0.615\,\mathrm{N}$, and the ratio is $0.820/0.615 = 4/3$.

Check: the ratio is independent of the numbers, because both integrals scale the same way. It is the momentum correction factor $\beta = 4/3$ for laminar pipe flow, used in Section 7. It exceeds 1 because the average of u^2 is always at least the square of the average of u. □

Trap. Taking **n** inward at an inlet and then subtracting the inflow again, so that inflow is counted twice with the wrong sign: either use $\mathbf{V}\cdot\mathbf{n}$ with **n** outward and add every term, or use $\sum_{\text{out}} - \sum_{\text{in}}$ with positive $\dot{m}$, not both. Using the absolute velocity to compute the flow across a moving surface. Writing $\partial/\partial t$ inside the integral for a deforming CV. Forgetting that "steady" depends on the frame. Treating b for momentum as a scalar: the momentum flux at an inlet whose velocity points in $-x$ carries negative x-momentum into the CV, and the sign of the velocity component is separate from the sign of $\mathbf{V}\cdot\mathbf{n}$.

Ladder.

1. White P3.8. Three feed pipes into one outlet; the speeds from a flow clue. The transport theorem as a mass balance over several ports.
2. White P3.12. Time to fill the rest of a surge tank fed and drained by pipes. The unsteady term written as a rate of change of volume.
3. White P3.38. A disk squeezes fluid out radially; the outflow velocity. A deforming control volume.

Answers are in the back of White; settle each call the same sitting.

Where it hides. Every integral relation in this chapter is this theorem with a particular b: mass in Section 6, momentum in Section 7, angular momentum in Section 9, energy in Section 10. Shrink the CV to an infinitesimal box and the same bookkeeping produces the differential continuity and Navier–Stokes equations of chapter 4; the material derivative D/Dt is the transport theorem for a vanishingly small system. The flux integral with a nonuniform profile is where the correction factors α and β come from.

6 Conservation of mass

Mass is neither made nor destroyed in a flow, so whatever enters a region either leaves it or accumulates in it. That one sentence is the whole physics of this section. The work in an exam question is bookkeeping: deciding where the boundaries are, which crossings carry fluid in and which out, whether the amount stored is changing, and how to turn a velocity that varies across a pipe into a single flow rate. It is also usually the first line of a longer momentum or energy question, where a slip costs every mark after it.

Recognise it. A tank filling or draining, a level rising or falling, a pressure in a rigid vessel changing as gas leaks out, a pipe junction (a tee, a manifold, a leak), a change of pipe diameter, a velocity profile given as a formula, "find the average velocity" or "find the flow rate", and any question with two unknown speeds where one equation is clearly missing. In momentum, Bernoulli and energy problems, mass conservation is the equation that links the inlet speed to the outlet speed.

On the crib sheet. The sheet prints $\sum \dot{m}_{\text{in}} = \sum \dot{m}_{\text{out}} + \partial m_{\text{cv}}/\partial t$, the transport theorem, $\dot{m}_{\text{rel}} = \rho A V_r$, and, under pipe networks, $\sum Q_{\text{node}} = 0$. It leaves out: that $\dot{m} = \rho A V$ holds only with V the velocity component *normal* to the section and uniform across it (otherwise integrate); that $\sum Q_{\text{in}} = \sum Q_{\text{out}}$ needs constant density; that the stored mass of a liquid tank is $\rho \int A(h)\, dh$, so the rate of change is $\rho A(h)\, dh/dt$; and that for a gas in a rigid tank the stored mass changes through its density, which is where the ideal gas law $P = \rho R T$ (printed under properties, with $R_{\text{air}} = 287\,\text{J}/(\text{kg K})$) enters. Water densities by temperature are in its water table.

The model

From the transport theorem. Take $B = m$, so $b = 1$. A system has fixed mass, so $dm_{\text{sys}}/dt = 0$, and for a fixed CV

$$0 = \frac{\partial}{\partial t} \int_{\text{CV}} \rho \, d\forall + \int_{\text{CS}} \rho \, (\mathbf{V} \cdot \mathbf{n}) \, dA.$$

The first term is the rate of increase of the mass stored in the CV, dm_{cv}/dt. The second is the net rate at which mass leaves. Where the fluid crosses at sections with uniform properties, $\int \rho(\mathbf{V} \cdot \mathbf{n})\, dA = \sum_{\text{out}} \dot{m} - \sum_{\text{in}} \dot{m}$, which rearranges to the sheet's form

$$\sum \dot{m}_{\text{in}} = \sum \dot{m}_{\text{out}} + \frac{dm_{\text{cv}}}{dt}.$$

Mass flow and volume flow. The *mass flow rate* $\dot{m}$ is the mass crossing a surface per second, in kg/s. The *volume flow rate* Q is the volume crossing per second, in m^3/s. Through a section of area A,

$$Q = \int_A (\mathbf{V} \cdot \mathbf{n})\, dA, \qquad \dot{m} = \int_A \rho (\mathbf{V} \cdot \mathbf{n})\, dA.$$

The *mean* (or average) velocity is defined as the uniform speed that would carry the same volume flow: $V_{\text{avg}} = Q/A$. With constant density, $\dot{m} = \rho Q = \rho A V_{\text{avg}}$ exactly, whatever the profile. That is why the 1-D formulas work for real, nonuniform pipe flow as long as V means the average. Only when a product of velocities is carried, as in momentum (ρu^2) or kinetic energy (ρu^3), does the profile matter again; that is the job of β and α.

Steady flow. If nothing changes in time, $dm_{\text{cv}}/dt = 0$ and $\sum \dot{m}_{\text{in}} = \sum \dot{m}_{\text{out}}$. For one inlet and one outlet, $\rho_1 A_1 V_1 = \rho_2 A_2 V_2$.

Incompressible flow. If the density is constant (liquids, and gases below a Mach number of about 0.3, where Mach number is speed divided by the speed of sound), ρ cancels and volume is conserved too. For a rigid CV completely full of liquid the stored mass cannot change, so $\sum Q_{\text{in}} = \sum Q_{\text{out}}$ even if the flow is unsteady. For one inlet and one outlet, $A_1 V_1 = A_2 V_2$: halve the diameter and the speed rises fourfold.

Tanks with a free surface. A *free surface* is a liquid surface open to a gas, such as the top of the water in a tank. If the tank has cross-sectional area $A(h)$ at height h above the bottom, the liquid volume is $\forall = \int_0^h A(z)\, dz$, and by the fundamental

theorem of calculus $d\forall/dt = A(h)\,dh/dt$. With constant density the mass equation becomes the tank ODE

$$A(h)\,\frac{dh}{dt} = Q_{\text{in}} - Q_{\text{out}}.$$

The CV here is the whole tank up to and including a lid above the surface; air crosses the lid but the analysis tracks only the liquid. If Q_{in} and Q_{out} are constant and A is constant, h changes linearly. If Q_{out} depends on h, as it does for a hole in the bottom ($Q_{\text{out}} = a\sqrt{2gh}$ from Section 8), separate the variables and integrate.

Gas in a rigid tank. A rigid tank of volume $\forall$ holds $m = \rho\forall$. If ρ is uniform inside, $dm_{\text{cv}}/dt = \forall\, d\rho/dt$. With the ideal gas law at constant temperature, $\rho = P/(RT)$, so

$$\frac{\forall}{RT}\,\frac{dP}{dt} = \sum \dot{m}_{\text{in}} - \sum \dot{m}_{\text{out}}.$$

Constant temperature is an assumption that holds for slow leaks from a tank that exchanges heat with its surroundings; a fast blowdown cools the gas, and then the energy equation is needed as well.

Nonuniform profiles. When the velocity varies across a section, integrate. In a round pipe the natural area element is a thin ring of radius r and width dr, with area $dA = 2\pi r\,dr$. In a rectangular channel of width w with velocity varying with height y, $dA = w\,dy$. Two profiles appear repeatedly: laminar pipe flow, $u = U_{\max}(1 - r^2/R^2)$, with $V_{\text{avg}} = U_{\max}/2$; and the turbulent power law $u = U_{\max}(1 - r/R)^{1/n}$ with $n \approx 7$, worked out in an example below.

First moves. Draw the CV and label every crossing. Decide: steady or not, constant density or not. Write $\sum \dot{m}_{\text{in}} = \sum \dot{m}_{\text{out}} + dm_{\text{cv}}/dt$ before putting in numbers. For a tank, write the stored mass as $\rho \int A\,dh$ (liquid) or $\rho\forall$ with $\rho = P/RT$ (gas). For a profile, write dA first and integrate. Convert L/s and mm to SI at the start. At the end, check that the units of every term are kg/s, or m^3/s if density was cancelled.

Worked examples

Example 6.1. A hopper is an inverted cone with its vertex at the bottom and its wall at 30° to the vertical axis. It is filled from empty with water at a constant $2\,\text{L/s}$; the outlet at the vertex is closed. (a) Write the ODE for the depth h. (b) Find dh/dt when $h = 0.5\,\text{m}$. (c) Find the time to reach $h = 0.8\,\text{m}$. (d) At $0.8\,\text{m}$ the outlet opens and drains $3\,\text{L/s}$ while the inflow continues. Find dh/dt at that instant.

Solution. Assumptions: water incompressible; the surface stays flat; flow rates are as stated.

(a) CV: the hopper interior up to a fixed lid above the water. At depth h the surface radius is $r = h\tan 30^\circ$, so $A(h) = \pi h^2 \tan^2 30^\circ = \pi h^2/3$. Mass conservation with constant density: $A(h)\,dh/dt = Q_{\text{in}} - Q_{\text{out}}$, so

$$\frac{\pi h^2}{3}\,\frac{dh}{dt} = Q_{\text{in}} \qquad (\text{outlet closed}).$$

(b) At $h = 0.5\,\text{m}$, $A = \pi(0.25)/3 = 0.2618\,\text{m}^2$, so $dh/dt = 0.002/0.2618 = 7.64 \times 10^{-3}\,\text{m/s} = 7.64\,\text{mm/s}$.

(c) Separate and integrate from $h = 0$ at $t = 0$: $\pi h^3/9 = Q_{\text{in}} t$. At $h = 0.8\,\text{m}$ the volume is $\pi(0.512)/9 = 0.1787\,\text{m}^3$, so $t = 0.1787/0.002 = 89.4\,\text{s}$, about $1.5\,\text{min}$.

(d) Now $Q_{\text{in}} - Q_{\text{out}} = 2 - 3 = -1\,\text{L/s}$ and $A(0.8) = \pi(0.64)/3 = 0.6702\,\text{m}^2$, so $dh/dt = -0.001/0.6702 = -1.49\,\text{mm/s}$: the level falls.

Check: the level rises quickly when the cone is narrow and slowly when it is wide; $7.6\,\text{mm/s}$ at $0.5\,\text{m}$ and $2\,\text{L/s}/0.67\,\text{m}^2 = 3.0\,\text{mm/s}$ at $0.8\,\text{m}$ fits that. The volume of a cone, $\frac{1}{3}\pi r^2 h$ with $r = 0.8/\sqrt{3}$, is again $0.1787\,\text{m}^3$. □

Example 6.2. A rigid $0.4\,\text{m}^3$ tank holds air at $600\,\text{kPa}$ absolute and $300\,\text{K}$. A valve releases air at $0.012\,\text{kg/s}$ at that instant. The tank stays at $300\,\text{K}$. (a) Find the mass and density of the air. (b) Find the rate of change of tank pressure. (c) The valve is choked, so its mass flow is proportional to the tank pressure. How long until the pressure reaches $400\,\text{kPa}$? Use $R = 287\,\text{J/(kg K)}$.

Solution. Assumptions: air an ideal gas; uniform pressure, density and temperature in the tank; isothermal as stated; the only crossing is the valve.

(a) $\rho = P/(RT) = 600\,000/(287 \times 300) = 6.969\,\text{kg/m}^3$, and $m = \rho \forall = 2.787\,\text{kg}$.

(b) CV: the inside of the tank, fixed and rigid. Mass: $dm_{\text{cv}}/dt = -\dot{m}_{\text{out}}$. With $m = P\forall/(RT)$ and T constant,

$$\frac{dP}{dt} = -\frac{RT}{\forall}\dot{m}_{\text{out}} = -\frac{287 \times 300}{0.4} \times 0.012 = -2583\,\text{Pa/s} = -2.58\,\text{kPa/s}.$$

(c) Write $\dot{m}_{\text{out}} = kP$ with $k = 0.012/600\,000 = 2.0 \times 10^{-8}\,\text{kg/(s Pa)}$. Then $dP/dt = -(kRT/\forall)P = -P/\tau$ with $\tau = \forall/(kRT) = 0.4/(2.0 \times 10^{-8} \times 287 \times 300) = 232.3\,\text{s}$. So $P = P_0 e^{-t/\tau}$ and

$$t = \tau \ln \frac{P_0}{P} = 232.3 \ln \frac{600}{400} = 94.2\,\text{s}.$$

Check: at the initial rate of $2.58\,\text{kPa/s}$ the $200\,\text{kPa}$ drop would take $77.4\,\text{s}$. The real rate falls as the pressure falls, so the answer must be longer, and $94\,\text{s}$ is. □

Example 6.3. Water ($998\,\text{kg/m}^3$) flows in a pipe of diameter $D = 0.10\,\text{m}$ with the turbulent profile $u = U_{\max}(1 - r/R)^{1/7}$; a probe on the centreline reads $U_{\max} = 2.4\,\text{m/s}$. (a) Find $V_{\text{avg}}/U_{\max}$ by integration, then Q and $\dot{m}$. (b) The pipe reduces to $60\,\text{mm}$; find the mean velocity there. (c) What error results from taking the centreline reading as the mean?

Solution. Assumptions: steady, incompressible, axisymmetric profile as given.

(a) $Q = \int_0^R u\, 2\pi r\, dr$. Put $y = 1 - r/R$, so $r = R(1 - y)$ and $dr = -R\,dy$; the limits $r = 0, R$ become $y = 1, 0$:

$$Q = 2\pi R^2 U_{\max} \int_0^1 y^{1/7}(1 - y)\, dy = 2\pi R^2 U_{\max}\Big(\frac{7}{8} - \frac{7}{15}\Big) = 2\pi R^2 U_{\max} \cdot \frac{49}{120}.$$

Dividing by πR^2: $V_{\text{avg}}/U_{\max} = 98/120 = 0.817$. So $V_{\text{avg}} = 1.96\,\text{m/s}$, $Q = \frac{\pi}{4}(0.1)^2(1.96) = 0.01539\,\text{m}^3/\text{s}$ and $\dot{m} = 998 \times 0.01539 = 15.36\,\text{kg/s}$.

(b) Steady and incompressible, so Q is the same: $V = 0.01539/(\frac{\pi}{4}(0.06)^2) = 5.44\,\text{m/s}$. Equivalently $1.96 \times (100/60)^2 = 5.44\,\text{m/s}$.

(c) Using $2.4\,\text{m/s}$ as the mean overestimates Q by $2.4/1.96 - 1 = 22\%$.

Check: for general n the same integral gives $V_{\text{avg}}/U_{\max} = 2n^2/[(n + 1)(2n + 1)]$; $n = 7$ gives 0.817, and $n \to \infty$ (a flat profile) gives 1, as it should. A numerical integration of the profile gives the same Q to four figures. □

Trap. Using the diameter where the area belongs, or forgetting to square the diameter ratio. Using $Q_{\text{in}} = Q_{\text{out}}$ for a gas whose density changes. Dropping dm_{cv}/dt in a tank problem because the word "steady" appeared elsewhere in the question. Writing $A\,dh/dt$ with a constant A for a cone or sphere. Taking $\dot{m} = \rho AV$ with a velocity that is not normal to the section: a jet crossing a plane at an angle carries $\rho AV\cos\theta$ through that plane. Integrating a pipe profile with $dA = dr$ instead of $2\pi r\,dr$. Mixing gauge and absolute pressure in $P = \rho RT$; the ideal gas law needs absolute pressure.

Ladder.

1. White P3.10. A porous pipe wall sucks or injects radially; exit and wall speeds. Mass flow through a wall as well as the ends.
2. White P3.14. A tank with one inlet and two outlets; the level rate. Signs of inflow and outflow in the unsteady balance.
3. White P3.16. A plate with a cubic exit profile; the flow leaving through the top. Integrating a nonuniform profile.
4. White P3.36. A fast jet entrains a slow annular stream; the fully mixed speed. Mass with two inlets of different speeds.
5. White P3.38. A disk squeezes fluid out radially; the outflow velocity. Deforming control volume.
6. White P3.24. A cone filled from its tip with a ramping inflow; the height history. Separating and integrating the level equation.

Answers are in the back of White; settle each call the same sitting.

Where it hides. Mass conservation is the first equation in every momentum, Bernoulli and energy problem, where it sets the outlet speed. Shrunk to a point it becomes the continuity equation $\partial\rho/\partial t + \nabla\cdot(\rho\mathbf{V}) = 0$ of chapter 4, and for incompressible flow $\nabla\cdot\mathbf{V} = 0$, which the sheet prints in 2-D. The tank ODE reappears in the draining problems of Section 8, and junction balances $\sum Q = 0$ are the node equations of pipe networks.

7 Linear momentum for a control volume

This is Newton's second law for a region the fluid flows through, and it is the most common 15-mark question in the chapter: find the force needed to hold a bend, a nozzle, a vane or a cart in place. The equation itself is one line. The marks are in the modelling: which control volume to draw, which forces act on its surface, which pressure to use, and which way each momentum flux points. A clear free-body diagram of the CV is worth more than any algebra after it.

Recognise it. "Find the force on the flange bolts", "the force to hold the bend", "the anchor force", "the thrust", "the force of the jet on the plate or vane", "the tension in the cable holding the cart", "the power delivered to the moving blades". Anything with a jet changing direction or a pipe changing direction or area, and asking for a force. Rockets and carts that accelerate because of a jet.

On the crib sheet. The sheet prints $\sum\mathbf{F} = \frac{\partial}{\partial t}\int\mathbf{V}\rho\,d\forall + \sum(\dot{m}\mathbf{V})_{\text{out}} - \sum(\dot{m}\mathbf{V})_{\text{in}}$, the jet-on-vane forms $F_x = \dot{m}(V_{1,x} - V_{2,x})$, $F_y = \dot{m}(V_{1,y} - V_{2,y})$, and $\mathbf{V}_r = \mathbf{V}_f - \mathbf{V}_{\text{cv}}$, $\dot{m}_{\text{rel}} = \rho AV_r$. It does not say that the first equation gives the force *on the fluid in the CV*, while the vane forms give the force *by the fluid on the vane*, which is the negative; that $\sum\mathbf{F}$ must include the pressure forces on the cut sections and gravity; that gauge pressure may be used everywhere because a uniform atmosphere exerts no net force on a closed surface; that the velocities are vectors whose components carry signs; that

1-D fluxes assume uniform velocity (otherwise multiply by β); or which mass flow a moving vane sees. None of the assumptions are printed.

The model

The equation. Put $B = m\mathbf{V}$, so $b = \mathbf{V}$, into the transport theorem. For a system, Newton's law says $d(m\mathbf{V})_{\text{sys}}/dt = \sum \mathbf{F}$, the total external force on the matter. At the instant the system fills the CV, the forces on the system are the forces on whatever is inside the CV. So, for a CV fixed in an inertial frame,

$$\sum \mathbf{F} = \frac{\partial}{\partial t} \int_{\text{CV}} \mathbf{V} \rho \, d\forall + \int_{\text{CS}} \mathbf{V} \rho (\mathbf{V} \cdot \mathbf{n}) \, dA.$$

This is a vector equation, three scalar equations, one per direction. With uniform velocity at each crossing it becomes

$$\sum \mathbf{F} = \frac{\partial}{\partial t} \int_{\text{CV}} \mathbf{V} \rho \, d\forall + \sum_{\text{out}} \dot{m}\mathbf{V} - \sum_{\text{in}} \dot{m}\mathbf{V}.$$

$\dot{m}\mathbf{V}$ is the *momentum flux* at a section: the momentum carried across it per second. It has units kg/s $\times$ m/s = N, the same as a force.

Two separate signs. Each term $\dot{m}\mathbf{V}$ has two signs and they are not the same thing. The sign in front, $+$ for out and $-$ for in, comes from $\mathbf{V} \cdot \mathbf{n}$ and says whether momentum is leaving or arriving. The sign *inside*, of each component of $\mathbf{V}$, comes from the coordinate axes. Water entering a CV while moving in $-x$ contributes $-\dot{m}(-V) = +\dot{m}V$ to the x-equation. Write $\mathbf{V}$ in components with your axes first, then apply the out-minus-in signs. This is the single most common sign error in the chapter.

Which control volume. The CV is a modelling choice, and a good one makes the unknown appear once and everything else known. Three rules. Cut where you know the pressure and velocity (at flanges with gauges, at free jets where the pressure is atmospheric). Cut *through* the thing whose force you want (the bolts, the support, the cable), so that force appears as an external force on the CV. Cut perpendicular to the flow at inlets and outlets, so that $\mathbf{V} \cdot \mathbf{n} = \pm V$ and the pressure acts along the flow direction. For a bend held by flange bolts, the usual CV contains the bend and the water in it, with the surface cut through the bolts at each flange.

Which forces. The forces on the CV are of two kinds. *Body forces* act on every bit of mass inside: gravity, $\int \rho \mathbf{g} \, d\forall$, which is the weight of the fluid and of any solid inside the CV. *Surface forces* act on the CS: pressure, which pushes inward along $-\mathbf{n}$ on every patch, viscous shear, which is usually negligible on short cuts, and the force from any solid the surface cuts through (a support, a bolt, a hand). At an inlet cut the upstream fluid pushes into the CV with force $P_1 A_1$ along the flow; at an outlet cut the downstream fluid pushes back into the CV with $P_2 A_2$, against the flow. Forgetting these pressure forces is the commonest way to get a bend force wrong by a factor of ten.

Why gauge pressure works. A uniform pressure p_a acting over a closed surface gives zero net force: $\oint p_a(-\mathbf{n})\,dA = -p_a \oint \mathbf{n}\,dA = \mathbf{0}$. The reason is geometric: for any direction, the projected areas of the parts of a closed surface facing that way and facing the opposite way are equal. (A cork in still air feels no net push from the atmosphere.) The outside of a pipe bend is surrounded by air at p_a, while the cut sections carry P_1 and P_2. Subtract p_a everywhere: the outside surface now carries zero, and the cuts carry $P_1 - p_a$ and $P_2 - p_a$, the *gauge* pressures. Since the subtracted field gave zero net force, nothing has changed. The force on the support that comes out is the real mechanical force. Use absolute pressure only if part of the CS is not surrounded by atmosphere, such as a nozzle discharging into a vacuum. A free jet in air is at atmospheric pressure, so its gauge pressure is zero.

Nonuniform velocity. With a velocity profile u across a section, the momentum flux is $\int \rho u^2\,dA$, not $\dot{m}V_{\text{avg}}$. Define the *momentum correction factor*

$$\beta = \frac{1}{A}\int_A \left(\frac{u}{V_{\text{avg}}}\right)^2 dA, \qquad \text{so that} \qquad \int \rho u^2\,dA = \beta\,\dot{m}V_{\text{avg}}.$$

For laminar pipe flow $\beta = 4/3$ (derived in Section 5); for turbulent pipe flow $\beta \approx 1.02$, close enough to 1 that it is usually ignored. $\beta \geq 1$ always.

Fixed vanes. A jet of area A and speed V in air strikes a fixed curved vane and leaves at angle θ from its original direction. Along the vane the pressure is atmospheric on the free side, gravity is usually small over the short path, and friction is neglected, so Bernoulli (Section 8) says the speed is unchanged. With x along the incoming jet and the CV around the vane cut through its support,

$$F_x = \dot{m}V(1 - \cos\theta), \qquad F_y = \dot{m}V\sin\theta$$

for the magnitudes of the force by the fluid on the vane: F_x along the incoming jet, F_y away from the side toward which the jet is turned. A flat plate normal to the jet is $\theta = 90^\circ$ and gives $\dot{m}V$; a full reversal, $\theta = 180^\circ$, gives $2\dot{m}V$, the most a jet can deliver.

Moving vanes. Let the vane move away from the nozzle at constant speed u. A frame riding on the vane is inertial, because it does not accelerate, and in that frame the flow is steady. The jet approaches at $V_r = V - u$, slides round the vane at V_r, and leaves at V_r in the vane's direction. So the fixed-vane formulas hold with V replaced by $V - u$, but with a choice of mass flow:

- *A single vane* running away from the nozzle sees the mass flow $\rho A(V - u)$. The jet between nozzle and vane gets longer every second, and the water filling that lengthening jet never reaches the vane. Force $\rho A(V - u)^2(1 - \cos\theta)$.
- *A wheel of many vanes* (a Pelton wheel) always has a vane in the jet, and the nozzle-to-vane distance does not grow, so every kilogram the nozzle emits is turned: mass flow ρAV. Force $\rho AV(V - u)(1 - \cos\theta)$.

In both cases the power is force times vane speed, $\dot{W} = F_x u$. The single vane's power peaks at $u = V/3$; the wheel's at $u = V/2$, where for $\theta = 180^\circ$ it takes all the jet's kinetic energy. The absolute velocity of the leaving water is its relative velocity plus the vane velocity, $\mathbf{V}_2 = \mathbf{V}_{r,2} + \mathbf{u}$.

Accelerating CV. If the CV accelerates with $\mathbf{a}_{\text{rel}}$ (a rocket, a cart pushed by a jet), Newton's law no longer holds in its frame. Measuring velocities relative to the CV, the law gains a fictitious force:

$$\sum \mathbf{F} - \int_{\text{CV}} \mathbf{a}_{\text{rel}}\, \rho\, d\forall = \frac{\partial}{\partial t} \int_{\text{CV}} \mathbf{V}_r \rho\, d\forall + \int_{\text{CS}} \mathbf{V}_r \rho (\mathbf{V}_r \cdot \mathbf{n})\, dA.$$

The second term on the left is mass times acceleration moved to the force side. For a rocket of mass m rising vertically with exhaust leaving at speed V_e relative to it and exit pressure equal to ambient, this gives $m\, dV/dt = \dot{m} V_e - mg -$ drag: the thrust is the exhaust momentum flux. Exam questions on this are short: write this equation, keep $m(t)$ if mass is lost, integrate.

First moves. Draw the CV and cut through the support you want. Draw axes. List assumptions: steady (in which frame), incompressible, uniform velocity at each cut ($\beta = 1$), weight included or neglected and why, friction neglected. Use mass conservation for the unknown speed. Find unknown pressures by Bernoulli or the energy equation, never by assumption. Switch to gauge pressure and say so; a free jet is at zero gauge. Draw every force on the CV: pressure into the CV at each cut, weight, and the unknown support force with assumed positive components. Write each $\mathbf{V}$ in components with signs, then apply $\sum_{\text{out}} \dot{m}\mathbf{V} - \sum_{\text{in}} \dot{m}\mathbf{V}$ one direction at a time. State whether your answer is the force on the fluid, on the device, or by the support. For a moving vane, work in the vane frame and decide single vane or wheel before computing $\dot{m}$.

Worked examples

Example 7.1. A 90° reducing elbow lies in a horizontal plane. Water ($998\,\text{kg/m}^3$) enters flowing in $+x$ through a $150\,\text{mm}$ diameter flange at $180\,\text{kPa}$ gauge and leaves flowing in $+y$ through a $100\,\text{mm}$ flange into a pipe; $Q = 0.05\,\text{m}^3/\text{s}$. Neglect losses in the elbow and the weight of elbow and water (it acts vertically and is carried separately). Find the horizontal force that the supports and flange bolts together must exert on the elbow.

Solution. Assumptions: steady, incompressible, frictionless through the elbow, uniform velocity at each flange ($\beta = 1$), atmosphere outside the elbow, horizontal so no change of elevation.

Mass: $A_1 = \frac{\pi}{4}(0.15)^2 = 0.01767\,\text{m}^2$, $A_2 = \frac{\pi}{4}(0.10)^2 = 7.854 \times 10^{-3}\,\text{m}^2$, so $V_1 = Q/A_1 = 2.829\,\text{m/s}$, $V_2 = Q/A_2 = 6.366\,\text{m/s}$, and $\dot{m} = \rho Q = 49.9\,\text{kg/s}$.

Outlet pressure: Bernoulli between the flanges (steady, incompressible, frictionless, along a streamline, same height):

$$P_2 = P_1 + \tfrac{1}{2}\rho(V_1^2 - V_2^2) = 180\,000 + \tfrac{1}{2}(998)(8.006 - 40.53) = 163.8\,\text{kPa gauge}.$$

CV: the elbow and the water inside, cut across both flanges and through the supports. Gauge pressure is used, so the outside surface carries no force. Forces on the CV: at the inlet the upstream water pushes in $+x$ with $P_1 A_1 = 3181\,\text{N}$; at the outlet the downstream water pushes back into the CV, in $-y$, with $P_2 A_2 = 1286\,\text{N}$; the supports exert the unknown (R_x, R_y). Inflow velocity $(V_1, 0)$, outflow $(0, V_2)$.

x: $R_x + P_1 A_1 = \dot{m}(0 - V_1)$, so $R_x = -49.9(2.829) - 3181 = -141 - 3181 = -3322\,\text{N}$.

y: $R_y - P_2 A_2 = \dot{m}(V_2 - 0)$, so $R_y = 49.9(6.366) + 1286 = 318 + 1286 = 1604\,\text{N}$.

The supports push the elbow with $3.32\,\text{kN}$ in $-x$ and $1.60\,\text{kN}$ in $+y$: magnitude $3.69\,\text{kN}$, at 25.8° to the $-x$ axis toward $+y$. The water pushes the elbow equally and oppositely, in $+x$ and $-y$.

Check: the pressure terms are ten times the momentum terms, which is typical of pipe bends at moderate speed; dropping them would give $0.35\,\text{kN}$, absurdly small for a pipe at 1.8 atmospheres gauge. □

Example 7.2. A water jet ($998\,\text{kg/m}^3$) of diameter $40\,\text{mm}$ leaves a fixed nozzle at $25\,\text{m/s}$ and strikes a vane mounted on a cart that moves directly away from the nozzle at a steady $10\,\text{m/s}$. Relative to the vane, the water leaves at 150° from its approach direction. Neglect friction on the vane and gravity over the vane. (a) Find the force components on the vane and the power delivered. (b) Find the absolute velocity of the water leaving the vane. (c) Repeat (a) for a wheel of vanes with the same blade speed and blade angle, and give its efficiency relative to the jet's kinetic energy flux. (d) At what blade speeds is the power largest in each case?

Solution. Assumptions: steady in the vane frame, which is inertial because the cart's speed is constant; free jet at atmospheric pressure so gauge pressure is zero everywhere on the CS; frictionless, so the relative speed is unchanged along the vane; uniform jet.

(a) $A = \frac{\pi}{4}(0.04)^2 = 1.257\times10^{-3}\,\text{m}^2$. In the vane frame the jet approaches at $V_r = 25-10 = 15\,\text{m/s}$ and leaves at $15\,\text{m/s}$ in the direction $(\cos150^\circ, \sin150^\circ) = (-0.866, 0.5)$. A single vane receives $\dot m_{\text{rel}} = \rho A V_r = 998(1.257\times10^{-3})(15) = 18.81\,\text{kg/s}$. CV around the vane, cut through its mounting. Force on the water: $\mathbf{F} = \dot m_{\text{rel}}(\mathbf{V}_{r,2} - \mathbf{V}_{r,1})$, so

$$F_x = 18.81\,(15\cos150^\circ - 15) = -526.6\,\text{N}, \qquad F_y = 18.81\,(15\sin150^\circ) = 141.1\,\text{N}.$$

The force by the water on the vane is the negative: $526.6\,\text{N}$ in the direction of motion and $141.1\,\text{N}$ in $-y$. Power $= 526.6\times10 = 5.27\,\text{kW}$.

(b) $\mathbf{V}_2 = \mathbf{V}_{r,2} + \mathbf{u} = (10-12.99,\ 7.50) = (-2.99,\ 7.50)\,\text{m/s}$, speed $8.07\,\text{m/s}$, down from $25\,\text{m/s}$.

(c) A wheel turns all the jet: $\dot m = \rho A V = 31.35\,\text{kg/s}$. The relative velocities are unchanged, so $F_x = 31.35(15)(1-\cos150^\circ) = 877.6\,\text{N}$ and power $= 8.78\,\text{kW}$. The jet carries kinetic energy at $\frac12\dot m V^2 = \frac12(31.35)(25)^2 = 9.80\,\text{kW}$, so the efficiency is $8.78/9.80 = 89.6\%$.

(d) Single vane: $\dot W = \rho A(V-u)^2(1-\cos\theta)u$; setting $d\dot W/du = 0$ gives $(V-u)(V-3u)=0$, so $u = V/3 = 8.33\,\text{m/s}$ and $\dot W = 5.42\,\text{kW}$. Wheel: $\dot W = \rho A V(V-u)(1-\cos\theta)u$, largest at $u = V/2 = 12.5\,\text{m/s}$, giving $9.14\,\text{kW}$ and efficiency $(1-\cos\theta)/2 = 93.3\%$.

Check: power at $10\,\text{m/s}$ ($5.27\,\text{kW}$) is just below the single-vane optimum, consistent with 10 being close to 8.33. The wheel can never exceed the jet's $9.80\,\text{kW}$, and does not. □

Example 7.3. A fire nozzle is bolted to a hose of inside diameter $2.5\,\text{in}$ and discharges a $1.0\,\text{in}$ diameter jet into the air. The flow is $0.6\,\text{ft}^3/\text{s}$ of water ($1.94\,\text{slug/ft}^3$). Neglect losses in the nozzle. Find the gauge pressure at the coupling and the force in the coupling bolts, in lbf and N.

Solution. Assumptions: steady, incompressible, frictionless nozzle, uniform velocity at the coupling and in the jet, nozzle horizontal, weight of nozzle and water neglected (it is vertical anyway), jet at atmospheric pressure.

Mass: $A_1 = \frac{\pi}{4}(2.5/12)^2 = 0.03409\,\text{ft}^2$, $A_2 = \frac{\pi}{4}(1/12)^2 = 5.454\times10^{-3}\,\text{ft}^2$. $V_1 = 0.6/0.03409 = 17.60\,\text{ft/s}$, $V_2 = 0.6/0.005454 = 110.0\,\text{ft/s}$, $\dot m = 1.94\times0.6 = 1.164\,\text{slug/s}$.

Pressure: Bernoulli from coupling to jet, jet at zero gauge:

$$P_1 = \tfrac{1}{2}\rho(V_2^2 - V_1^2) = \tfrac{1}{2}(1.94)(12\,102 - 309.8) = 11\,440\,\text{lbf/ft}^2 = 79.4\,\text{psi gauge}.$$

CV: the nozzle and the water inside, cut across the coupling (through the bolts) and across the jet just outside the exit. Take $+x$ along the flow. Forces in x: the hose water pushes in with $P_1A_1 = 11\,440 \times 0.03409 = 389.9\,\text{lbf}$; the jet face carries zero gauge pressure; the bolts exert F_b on the nozzle. Momentum:

$$F_b + P_1A_1 = \dot{m}(V_2 - V_1) = 1.164(110.0 - 17.6) = 107.6\,\text{lbf},$$

so $F_b = 107.6 - 389.9 = -282.3\,\text{lbf}$. The bolts pull the nozzle upstream with $282\,\text{lbf}$, so they are in tension: the water tries to push the nozzle off the hose. In SI, $282.3 \times 4.45 = 1256\,\text{N}$.

Check: the answer is the pressure force minus the momentum gained. The water presses outward on the converging wall, whose inner face looks partly upstream, so a downstream push on the nozzle is the expected sign. $79\,\text{psi}$ is a normal hydrant-fed nozzle pressure. (A slug is $\text{lbf}\,\text{s}^2/\text{ft}$, so slug/s $\times$ ft/s $=$ lbf.) □

Trap. Leaving out P_1A_1 and P_2A_2 at the cut sections, or giving them the wrong direction: pressure always pushes *into* the CV. Using absolute pressure at the cuts while ignoring the atmosphere on the outside. Treating the sign of the momentum flux and the sign of the velocity component as one sign. Reporting the force on the fluid when the force on the device was asked. Using ρAV for a single moving vane (it is $\rho A(V-u)$), or $\rho A(V-u)$ for a wheel (it is ρAV). Using absolute velocities in the vane frame. Assuming $P_2 = P_1$ in a reducer. Forgetting that the jet speed is unchanged over a frictionless vane only in the vane's frame. Writing the jet-on-vane formula as the force on the fluid with the vane's sign.

Ladder.

1. White P3.40. A round jet strikes a fixed plate square on; the holding force. The simplest momentum balance.
2. White P3.64. A jet on a plate with a central hole; the force. Two outflows with different momentum.
3. White P3.42. A sudden contraction discharging to air; the force on the fitting. Pressure force at the inlet cut, in symbols.
4. White P3.62. A two-branch nozzle venting angled jets; the inlet flange bolt force. Vector momentum flux with pressure at a cut.
5. White P3.46. A sheet splits on a frictionless inclined plate; the split fraction. No tangential force as an assumption that closes the problem.
6. White P3.80. A river over a submerged weir; the force on the obstacle. Hydrostatic pressure forces at the cuts, with mass.
7. White P3.88. A pump-jet boat with drag proportional to speed squared; cruising speed. Moving control volume and a quadratic.
8. White P3.82. A jet pushes a model car from rest; the speed after one metre. Accelerating control volume, integrated.
9. White P3.108. A sled braked by a water scoop; the speed history. Moving vane on an accelerating body.

Answers are in the back of White; settle each call the same sitting.

Where it hides. Applied to an infinitesimal box, the same balance of surface forces, body force and momentum flux is the derivation of the Navier–Stokes equations in

chapter 4; the pressure term there is $-\nabla P$, the gradient form of "pressure pushes into the CV". The momentum integral of a boundary layer, from which the sheet's Kármán relation comes, is this equation for a thin CV next to a wall. Along a single streamline the momentum equation becomes Bernoulli (Section 8). Turbomachines use its moment form (Section 9).

8 Bernoulli and the grade lines

Bernoulli's equation trades pressure for speed for height along the path of a fluid particle. It is the quickest tool in the chapter and the most misused, because it is easy to remember and its assumptions are not. This section derives it from Newton's second law so that each assumption appears at the step where it is used, then applies it to the standard devices: the Pitot tube, the Venturi meter, the draining tank and the siphon. The grade lines at the end turn the equation into a picture of where the energy of a pipe flow is.

Recognise it. A question that gives a pressure at one point and asks for a speed at another, or the reverse. A Pitot or Pitot-static tube, a manometer across a flow meter, a Venturi or a nozzle with gauges, a jet from a hole in a tank, a siphon, a free jet's height. "Neglect losses" or "frictionless" in the statement is an invitation. A request to sketch the HGL and EGL. Also the step inside a momentum question where an unknown pressure at a flange must be found.

On the crib sheet. The sheet prints

$$P_0 = P + \tfrac{1}{2}\rho V^2, \quad V = \sqrt{\frac{2(P_0 - P)}{\rho}}, \quad V = \sqrt{2gh\Big(\frac{\rho_m}{\rho_f} - 1\Big)}, \quad Q = C_d A_t \sqrt{\frac{2\Delta P/\rho}{1 - \beta^4}},$$

with $C_d \approx 0.98$ (Venturi) and 0.61 (orifice), $\mathrm{HGL} = z + P/\gamma$, $\mathrm{EGL} = \mathrm{HGL} + \alpha V^2/2g$, and six drawing rules for the grade lines. Its water table gives the vapour pressure P_{vap} by temperature (needed for siphons and cavitation), its properties section gives $a = \sqrt{kRT}$ for the Mach-number check, and its turbomachinery section prints $\mathrm{NPSH_A}$. It does not print Bernoulli's equation itself, $P + \frac{1}{2}\rho V^2 + \rho g z = \text{const}$; you get it from the energy equation on the sheet by setting $h_p = h_t = h_L = 0$ and $\alpha = 1$. It prints none of the assumptions, and nothing about the difference between "along a streamline" and "everywhere".

The model

Streamlines. A *streamline* is a curve drawn in the flow that is tangent to the velocity at every point along it, so that fluid moves along streamlines and never across them. In steady flow, the pattern does not change, and a streamline is the actual path of a particle. A bundle of neighbouring streamlines forms a *streamtube*: a pipe with walls made of flow, through which no fluid crosses the sides.

Newton's law along a streamline. Let s measure distance along a streamline, and take a small cylinder of fluid lying along it, with cross-section dA and length ds. Its mass is $\rho\, dA\, ds$. Three things could push it along s.

- Pressure: P on the back face and $P + (\partial P/\partial s)ds$ on the front, so a net $-(\partial P/\partial s)\, ds\, dA$. Pressure on the side walls is perpendicular to s and does not contribute.

- Gravity: the weight $\rho g\,dA\,ds$ has the component $-\rho g\,dA\,ds\,(dz/ds)$ along s, where z is height and dz/ds is the sine of the streamline's slope.
- Shear from viscosity on the side walls. *Assume it is zero*: the flow is *frictionless*, also called inviscid.

The particle's acceleration along s is $a_s = \partial V/\partial t + V\,\partial V/\partial s$: the first part is the speed changing in time at a fixed point, the second the particle moving to where the speed is different. *Assume the flow is steady*, so the first part is zero. Newton's law, divided by the volume $dA\,ds$:

$$\rho V\frac{dV}{ds} = -\frac{dP}{ds} - \rho g\frac{dz}{ds}.$$

Since $V\,dV/ds = d(V^2/2)/ds$, *assume* ρ *constant* (incompressible) and move it inside the derivative:

$$\frac{d}{ds}\Big(P + \tfrac{1}{2}\rho V^2 + \rho g z\Big) = 0.$$

Integrating along the streamline gives

Theorem 8.1 (Bernoulli)**.** *In steady, incompressible, frictionless flow, along a streamline,*

$$P + \tfrac{1}{2}\rho V^2 + \rho g z = \textit{const}, \qquad \textit{or} \qquad \frac{P_1}{\rho g} + \frac{V_1^2}{2g} + z_1 = \frac{P_2}{\rho g} + \frac{V_2^2}{2g} + z_2.$$

The fourth assumption, *along a streamline*, is that both points lie on the same streamline, since the integration was along one. A fifth is implicit: between the two points there is no pump or turbine adding or removing energy (no *shaft work*) and no heat transfer. A pump is a device that pushes on the fluid with blades, a force that was not in the list above.

Head. Dividing by $\rho g = \gamma$, the *specific weight*, makes every term a length. Each is energy per unit weight of fluid, called a *head*: P/γ is the *pressure head*, $V^2/2g$ the *velocity head*, z the *elevation head*. A pressure head of $10\,\text{m}$ means the pressure could lift the fluid $10\,\text{m}$. Heads are the natural units for pipes and pumps.

Across streamlines. Bernoulli says nothing about how the constant changes from one streamline to the next. Two facts fill the gap. If the streamlines are straight and parallel, the pressure across them varies hydrostatically, $P + \rho g z = \text{const}$ across, because there is no acceleration across. So the pressure across a straight free jet in air is atmospheric, and across a straight pipe section it is hydrostatic. And if every streamline starts in the same uniform reservoir, the constant is the same on all of them; this is the usual justification for using Bernoulli "anywhere" in a tank-fed flow. (A flow with no rotation of fluid elements, called irrotational, has the same constant everywhere; chapter 4 proves this.)

Static, dynamic and stagnation pressure. The *static pressure* P is the thermodynamic pressure of the fluid, the one a gauge flush with a wall reads, since it does not disturb the flow. The *dynamic pressure* $\frac{1}{2}\rho V^2$ is the kinetic energy per unit volume. A *stagnation point* is a point where the flow is brought to rest, such as the nose of a body or the open end of a tube facing the flow. Bernoulli from far upstream to a stagnation point at the same height gives the *stagnation pressure*

$$P_0 = P + \tfrac{1}{2}\rho V^2.$$

A *Pitot tube* is a tube facing into the flow; its open end is a stagnation point and it reads P_0. A *Pitot-static tube* also has holes in its side, which read P. The difference gives $V = \sqrt{2(P_0 - P)/\rho}$. Connected to a U-tube manometer with liquid of density ρ_m and reading h, the hydrostatics of the two legs give $P_0 - P = (\rho_m - \rho)gh$, which is the sheet's form. The -1 in it matters when both fluids are liquids and not when the flowing fluid is air.

Venturi meter. A *Venturi meter* is a smooth contraction to a throat of area A_t and back. Bernoulli from the pipe (area A, speed V) to the throat (speed V_t), with mass conservation $AV = A_t V_t$ and $\beta = d/D$, gives $P - P_t = \frac{1}{2}\rho V_t^2(1 - \beta^4)$, so

$$Q = A_t V_t = A_t \sqrt{\frac{2(P - P_t)}{\rho(1 - \beta^4)}}.$$

A real meter delivers a little less; the discharge coefficient C_d on the sheet corrects for friction and for the profile. (The β here is a diameter ratio, unrelated to the momentum factor.)

Draining tanks. A tank with a free surface at height h above a small hole in the bottom or side. Take a streamline from the surface (pressure atmospheric, speed V_s) to the jet just outside the hole (pressure atmospheric, speed V). Bernoulli gives $V^2 - V_s^2 = 2gh$; mass gives $AV_s = aV$; so $V = \sqrt{2gh/(1 - a^2/A^2)} \approx \sqrt{2gh}$ when $a \ll A$. This is *Torricelli's result*: the jet leaves as fast as a stone dropped from the surface. The flow is not steady, since h falls, but if the hole is small the level falls slowly and the steady formula holds at each instant; this is called *quasi-steady*. Feeding it into the tank ODE of Section 6 gives the draining time.

Siphons. A *siphon* is a tube that lifts liquid over a high point and discharges below the free surface. Bernoulli from the free surface to the outlet sets the speed: $V = \sqrt{2g\Delta z}$ with Δz the drop to the outlet. Bernoulli from the surface to the summit then sets the summit pressure, which is below atmospheric. If it falls to the vapour pressure P_v the liquid boils, the column breaks, and the siphon stops. This is *cavitation*; P_v is on the sheet's water table.

Grade lines. Plot two heights along a pipe. The *hydraulic grade line* (HGL) is $P/\gamma + z$: the height to which liquid would rise in a thin vertical tube (a piezometer) tapped into the pipe. The *energy grade line* (EGL) is $P/\gamma + V^2/2g + z$, the total head; it lies $V^2/2g$ above the HGL (with α for a real profile, Section 10). In Bernoulli flow the EGL is horizontal. Real pipes lose head to friction, so the EGL slopes down in the direction of flow; it jumps up at a pump and down at a turbine. Where the pipe narrows, V rises and the gap between the lines grows. At a free surface or free jet the pressure is atmospheric, so the HGL is at the surface or the jet centreline. Wherever the HGL falls below the pipe, the pressure is below atmospheric.

When Bernoulli is the wrong model. Across a pump, fan or turbine (shaft work; use the energy equation). Along a long pipe, through a valve, a sudden expansion or any region where the flow separates from a wall (friction is not negligible; add a head loss). Between points on different streamlines in a flow with rotation, such

as across a boundary layer or a wake. In gases above a Mach number of about 0.3 (density changes). In flows that change quickly in time: start-up, valve closure, oscillation (the $\partial V/\partial t$ term). And with heat addition. Bernoulli holds in a contraction (accelerating flow is well-behaved); it fails in an expansion, where the decelerating flow separates. That is why a Venturi recovers its pressure only with a long, gentle diffuser.

First moves. Pick two points where most quantities are known: a free surface ($P = 0$ gauge, $V \approx 0$ for a large tank), a free jet ($P = 0$ gauge), a stagnation point ($V = 0$). Check they are on one streamline, or that both streamlines start in the same reservoir. Write the four assumptions and check each: steady (or quasi-steady), incompressible, frictionless between the points, no pump or turbine between them. Pick a datum for z and say where it is. Add mass conservation for the second unknown speed. Use gauge or absolute consistently; use absolute when comparing with vapour pressure. For a manometer, do the hydrostatics leg by leg.

Worked examples

Example 8.2. A Pitot-static tube on the centreline of a ventilation duct carries air at $25\,^{\circ}\mathrm{C}$ and $101.3\,\mathrm{kPa}$. Its two pressure lines go to a U-tube manometer filled with water ($998\,\mathrm{kg/m^3}$), which shows a level difference of $32\,\mathrm{mm}$. (a) Find the air density. (b) Find the air speed. (c) Justify the incompressible assumption. (d) What error comes from dropping the -1 in the manometer formula?

Solution. Assumptions: steady, frictionless along the stagnation streamline, incompressible (checked below); air an ideal gas with $R = 287\,\mathrm{J/(kg\,K)}$; the static holes read the free-stream pressure; manometer lines filled with air.

(a) $\rho = P/(RT) = 101\,300/(287 \times 298.15) = 1.184\,\mathrm{kg/m^3}$.

(b) Manometer: the higher pressure P_0 pushes the water in its leg down by h relative to the other leg. Equate pressures at the level of that lower meniscus. On the stagnation side it is P_0 plus the weight of the air column above. On the static side it is P plus the same air column, less a height h of air, plus a height h of water. So $P_0 - P = (\rho_m - \rho)gh = (998 - 1.184)(9.81)(0.032) = 312.9\,\mathrm{Pa}$. Then

$$V = \sqrt{\frac{2(P_0 - P)}{\rho}} = \sqrt{\frac{2 \times 312.9}{1.184}} = 23.0\,\mathrm{m/s}.$$

(c) The speed of sound is $a = \sqrt{kRT} = \sqrt{1.4 \times 287 \times 298.15} = 346\,\mathrm{m/s}$, so the Mach number is $23.0/346 = 0.066$, far below 0.3. Density changes are of order $\frac{1}{2}\mathrm{Ma}^2 \approx 0.2\%$.

(d) Without the -1, $V = \sqrt{2gh\rho_m/\rho} = 23.006\,\mathrm{m/s}$, a 0.06% difference: negligible for air over water, but not for water over mercury, where $\rho_m/\rho - 1 = 12.6$ instead of 13.6.

Check: $23\,\mathrm{m/s}$ in a duct is a fast but normal ventilation speed, and $313\,\mathrm{Pa}$ is about $3\,\mathrm{cm}$ of water, which matches the reading. □

Example 8.3. A siphon of inside diameter $25\,\mathrm{mm}$ draws water ($20\,^{\circ}\mathrm{C}$: $998\,\mathrm{kg/m^3}$, $P_v = 2.34\,\mathrm{kPa}$) from a large open tank. The highest point of the tube is $1.8\,\mathrm{m}$ above the tank's free surface, and the tube discharges into the air $3.0\,\mathrm{m}$ below the free surface. Atmospheric pressure is $101.3\,\mathrm{kPa}$. Neglecting friction, find (a) the flow rate, (b) the absolute and gauge pressure at the summit, (c) the greatest summit height, with the outlet unchanged, before the water cavitates.

Solution. Assumptions: steady (the tank is large, so its level is fixed), incompressible, frictionless, one streamline from the free surface through the tube; uniform speed in the tube; surface speed negligible. Datum $z = 0$ at the free surface.

(a) Surface ($P = P_a$, $V = 0$, $z = 0$) to outlet jet ($P = P_a$, $z = -3.0$): $V = \sqrt{2g(3.0)} = 7.672\,\text{m/s}$. $Q = \frac{\pi}{4}(0.025)^2(7.672) = 3.77 \times 10^{-3}\,\text{m}^3/\text{s} = 3.77\,\text{L/s}$.

(b) Surface to summit ($z = 1.8$, the same V by mass conservation in a uniform tube):

$$P_s = P_a - \rho g z_s - \tfrac{1}{2}\rho V^2 = 101\,300 - 998(9.81)(1.8) - \tfrac{1}{2}(998)(58.86) = 54.3\,\text{kPa abs},$$

which is $-47.0\,\text{kPa}$ gauge.

(c) Set $P_s = P_v$ with V fixed by the outlet:

$$z_{s,\max} = \frac{P_a - P_v}{\rho g} - \frac{V^2}{2g} = \frac{101\,300 - 2340}{998 \times 9.81} - 3.0 = 10.11 - 3.0 = 7.11\,\text{m}.$$

Check: the summit pressure is lowered both by height and by speed; the $10.1\,\text{m}$ is the familiar limit of a water barometer, less the $3.0\,\text{m}$ of velocity head. Friction would lower the flow; the frictionless answer is an upper bound. □

Example 8.4. An open cylindrical tank of diameter $1.5\,\text{m}$ drains through a hole of diameter $40\,\text{mm}$ in its flat bottom. (a) Find the time for the depth to fall from $2.0\,\text{m}$ to $0.5\,\text{m}$, treating the hole as ideal. (b) Repeat for a sharp-edged orifice with $C_d = 0.61$. (c) Find the initial flow rate in case (a).

Solution. Assumptions: incompressible; quasi-steady (justified because the area ratio $A/a = (1500/40)^2 = 1406$ makes the surface fall over a thousand times slower than the jet); frictionless from the surface to the jet; jet at atmospheric pressure; surface velocity head negligible ($a^2/A^2 \approx 5 \times 10^{-7}$).

Bernoulli, surface to jet: $V = \sqrt{2gh}$. Mass, CV the tank (Section 6): $A\,dh/dt = -aV = -a\sqrt{2gh}$. Separate:

$$\int_{h_0}^{h_1} \frac{dh}{\sqrt{h}} = -\frac{a}{A}\sqrt{2g}\,t \quad \Rightarrow \quad t = \frac{A}{a}\sqrt{\frac{2}{g}}\left(\sqrt{h_0} - \sqrt{h_1}\right).$$

(a) $t = 1406 \times \sqrt{2/9.81} \times (1.414 - 0.707) = 449\,\text{s} = 7.5\,\text{min}$.

(b) The real discharge is $C_d a\sqrt{2gh}$, so the time divides by C_d: $449/0.61 = 736\,\text{s} = 12.3\,\text{min}$.

(c) $V_0 = \sqrt{2 \times 9.81 \times 2.0} = 6.26\,\text{m/s}$, $Q_0 = \frac{\pi}{4}(0.04)^2(6.26) = 7.87 \times 10^{-3}\,\text{m}^3/\text{s} = 7.87\,\text{L/s}$.

Check: draining from $2.0\,\text{m}$ to empty would take $898\,\text{s}$, so the first $1.5\,\text{m}$ takes exactly half the total time, because $\sqrt{2} - \sqrt{0.5} = \sqrt{0.5}$. The flow slows as the head falls, so the last half metre is slow. The volume drained, $\frac{\pi}{4}(1.5)^2(1.5) = 2.65\,\text{m}^3$, over $449\,\text{s}$ is an average of $5.9\,\text{L/s}$, between the initial 7.87 and the final $3.94\,\text{L/s}$. □

Trap. Applying Bernoulli across a pump, fan or turbine. Applying it between points on different streamlines in a flow with rotation, for example from the free stream to a point inside a boundary layer. Applying it through a sudden expansion or a partly closed valve. Forgetting that the outlet speed in a siphon or nozzle is set by mass conservation, not chosen. Using gauge pressure when comparing with vapour pressure. Dropping the -1 in a liquid-over-liquid manometer. Taking the jet pressure as the tank pressure. Using $V = \sqrt{2gh}$ with h measured from the bottom of the tank instead of from the free surface to the hole. Forgetting the hydrostatic change in pressure across a pipe when the two taps are at different heights.

Ladder.

1. White P3.130. A gasoline nozzle with an elevation rise and no losses; the upstream pressure. Bernoulli with mass, nothing else.
2. White P3.116. A jet from a hole in a tank wall; the landing distance and the best hole height. Torricelli plus projectile motion.
3. White P3.114. A nozzle jet on a plate with a known force; speeds and a manometer reading. Bernoulli and momentum together.
4. White P3.122. A tank filled steadily while draining; the limiting depth. Bernoulli inside an unsteady mass balance.
5. White P3.142. A tank drains through a bottom plug, keeping the surface speed; the drain time. Quasi-steady Bernoulli, then an ODE.
6. White P3.144. A hose jet rises to a ceiling with a growing area; the flow from a measured force. Bernoulli along the jet, then momentum.

Answers are in the back of White; settle each call the same sitting.

Where it hides. Bernoulli is the energy equation of Section 10 with no pump, turbine or loss and $\alpha = 1$; it is also the momentum equation of Section 7 applied to a streamtube. In chapter 4 it comes out of the Navier–Stokes equations with the viscous term dropped (Euler's equation), and for irrotational flow it holds everywhere, which is the basis of potential flow. Inside a momentum problem it supplies the flange pressure; inside a pump problem it supplies the suction pressure and the cavitation check (NPSH on the sheet). The stagnation pressure reappears in the drag and pressure coefficients of external flow.

9 Angular momentum

Linear momentum gives forces; angular momentum gives moments. It is needed whenever the question asks for a torque or a bending moment, and whenever something rotates: sprinklers, pump impellers, turbine runners, fans. For rotating machines it produces the Euler turbomachine equation, the single formula behind every pump and turbine head on the crib sheet. The new ingredients are the cross product, which supplies the lever arm and the sense of rotation, and the velocity of a fluid leaving a moving outlet, which is the relative velocity plus the velocity of the outlet itself.

Recognise it. "Find the moment (or torque) at the base", "the bending moment at the support", "the torque to hold the sprinkler stationary", "the rotation rate of the sprinkler", "the torque and power of the pump impeller", "the head developed by the pump", "blade angle", "velocity triangle". A pipe that discharges at a distance from its support, especially if the jet is not aimed at the support.

On the crib sheet. The sheet prints the stationary-CV form $\sum \mathbf{M}_o = \frac{\partial}{\partial t}\int (\mathbf{r}\times\mathbf{V})\rho\, d\forall + \sum(\mathbf{r}\times\mathbf{V})_{\text{out}}\dot{m}_{\text{out}} - \sum(\mathbf{r}\times\mathbf{V})_{\text{in}}\dot{m}_{\text{in}}$, the steady 1-D form about an axis $M_z = \sum_{\text{out}} \dot{m}\, r V_\theta - \sum_{\text{in}} \dot{m}\, r V_\theta$, $\mathbf{V} = \mathbf{V}_r + \mathbf{U}$ with $\mathbf{U} = \boldsymbol{\omega}\times\mathbf{r}$ at a rotating outlet, and, in its turbomachinery section, Euler's head $H = (U_2 V_{t2} - U_1 V_{t1})/g$ and torque $T = \dot{m}(r_2 V_{t2} - r_1 V_{t1})$, with $P = \omega T_o$ under pipe flow. The same section also prints the system curve $H_{\text{sys}} = \Delta z + \Delta P/\gamma + h_L(Q)$, the affinity laws, the dimensionless specific speed and NPSH_A, which belong to pump selection rather than to this section. The nomenclature table gives the units of torque (N m) and power (W). It leaves out: that $\mathbf{V}$ in all of these is the *absolute* velocity (measured in the fixed frame), that V_θ and V_t are the tangential components with a sign fixed by the sense of rotation, that $\sum \mathbf{M}_o$ includes moments of the pressure forces and weights about the same point, and how to find V_{t2} from the blade angle.

The model

Moment and angular momentum. The *moment* of a force $\mathbf{F}$ applied at position $\mathbf{r}$ about a point O is $\mathbf{M} = \mathbf{r} \times \mathbf{F}$. In a plane with $\mathbf{r} = (x, y)$ and $\mathbf{F} = (F_x, F_y)$ it points along z with $M_z = xF_y - yF_x$; positive is counterclockwise looking down the z-axis. Its size is the force times the perpendicular distance from O to the force's line of action. The *angular momentum* of a mass m moving with $\mathbf{V}$ at $\mathbf{r}$ is $\mathbf{r} \times m\mathbf{V}$, built the same way from momentum instead of force. Newton's law for a system, crossed with $\mathbf{r}$, gives $d(\mathbf{H})_{\text{sys}}/dt = \sum \mathbf{M}_o$, where $\mathbf{H} = \int \mathbf{r} \times \mathbf{V}\, dm$.

The CV equation. With $B = \mathbf{H}$, $b = \mathbf{r} \times \mathbf{V}$ in the transport theorem, for a fixed CV,

$$\sum \mathbf{M}_o = \frac{\partial}{\partial t} \int_{\text{CV}} (\mathbf{r} \times \mathbf{V}) \rho \, d\forall + \int_{\text{CS}} (\mathbf{r} \times \mathbf{V}) \rho (\mathbf{V} \cdot \mathbf{n}) \, dA.$$

The flux $\int (\mathbf{r} \times \mathbf{V}) \rho (\mathbf{V} \cdot \mathbf{n}) \, dA$ is the angular momentum carried out per second. With uniform jets it is $\sum_{\text{out}} \dot{m} (\mathbf{r} \times \mathbf{V}) - \sum_{\text{in}} \dot{m} (\mathbf{r} \times \mathbf{V})$, where $\mathbf{r}$ runs from O to the centre of each section. The left side is the moment, about the same O, of everything that acts on the CV: the support moment you want, the support force (choose O at the support so its moment is zero), pressure forces at the cuts, weights acting at their centres of mass. As in Section 7, gauge pressure may be used and a free jet carries none.

Choosing the point. Put O where the unknown force acts, often the base of a pipe or the axis of a rotor. Then the unknown force has no moment, and an inlet that passes through O contributes neither flux nor pressure moment because its $\mathbf{r}$ is zero or parallel to $\mathbf{V}$.

The steady 1-D form about an axis. In a rotating machine everything is measured about the shaft axis z. Use polar coordinates: at radius r, the velocity has a radial part V_r (away from the axis), a tangential part V_θ (around the axis, positive in the sense of rotation) and an axial part. Only V_θ has a moment about z: $(\mathbf{r} \times \mathbf{V})_z = rV_\theta$. So for steady flow

$$\sum M_z = \sum_{\text{out}} \dot{m}\, rV_\theta - \sum_{\text{in}} \dot{m}\, rV_\theta.$$

This is the sheet's equation. The CV is a fixed cylinder around the rotor: the rotor turns inside it, but in steady operation the angular momentum inside does not change, so the storage term vanishes.

Absolute velocity at a moving outlet. An observer riding on a rotor sees the fluid leave a nozzle or blade with the *relative* velocity $\mathbf{V}_r$, set by the nozzle or blade geometry and by mass conservation in the rotor. The nozzle itself moves with the *blade speed* $U = \omega r$ tangentially. The velocity in the fixed frame, the *absolute* velocity, is the vector sum

$$\mathbf{V} = \mathbf{V}_r + \mathbf{U}.$$

The angular momentum equation is written in the fixed frame, so it needs $\mathbf{V}$, not $\mathbf{V}_r$. For a sprinkler arm of radius R whose jet leaves backward relative to the arm at V_r, the absolute tangential velocity is $V_\theta = -(V_r - \omega R)$, measured in the sense of rotation; the water leaves backward more slowly than it does from a stationary arm.

Velocity triangles. The vector sum $\mathbf{V} = \mathbf{V}_r + \mathbf{U}$ drawn as a triangle at a blade's outlet is the *velocity triangle*. At the outer radius r_2 of a pump impeller of width b_2, mass conservation gives the radial (normal) part, the same in both frames: $V_{n2} = Q/(2\pi r_2 b_2)$. The relative velocity runs along the blade, which leaves at angle β_2 measured from the tangent, so its tangential part is $V_{n2}\cot\beta_2$, pointing backward against the rotation for a backward-curved blade. Then

$$V_{t2} = U_2 - V_{n2}\cot\beta_2.$$

The same construction at the inlet gives V_{t1}; often the inlet has no swirl, so $V_{t1} = 0$.

Euler's turbomachine equation. For a pump, the torque the shaft applies to the fluid is the only moment about the axis (pressure acts radially on a cylinder, and gravity is parallel to the axis or symmetric). So

$$T = \dot{m}(r_2 V_{t2} - r_1 V_{t1}).$$

The shaft turns at ω, so the power delivered to the fluid is $P = T\omega = \dot{m}(U_2 V_{t2} - U_1 V_{t1})$, using $U = \omega r$. The head added, energy per unit weight, is power divided by the weight flow $\dot{m}g$:

$$H = \frac{U_2 V_{t2} - U_1 V_{t1}}{g}.$$

This is the ideal head: it assumes the fluid leaves exactly along the blades and that no energy is lost. Real pumps deliver less, and the ratio is the *hydraulic efficiency*. For a turbine, the fluid gives up angular momentum and the same equations give a negative T on the fluid, which is a positive torque on the shaft.

First moves. Choose O at the support or the axis. Draw axes and fix the positive sense of rotation. List every crossing with its $\mathbf{r}$ and its absolute $\mathbf{V}$; at a moving outlet build $\mathbf{V} = \mathbf{V}_r + \mathbf{U}$ first. Compute $\dot{m}$ from mass conservation, with relative velocities in the rotor. List every moment on the CV: the unknown, weights at their centroids, pressure forces at cuts not through O. Write $\sum M_z = \sum_{\text{out}} \dot{m} r V_\theta - \sum_{\text{in}} \dot{m} r V_\theta$ with signs. For a machine, write $T = \dot{m}(r_2 V_{t2} - r_1 V_{t1})$, then $P = T\omega$ and $H = P/(\dot{m}g)$, and say that H is ideal.

Worked examples

Example 9.1. A pipe rises vertically from a flange O at floor level to a height of $1.0\,\text{m}$, turns horizontal and runs $1.5\,\text{m}$ to a nozzle of exit diameter $25\,\text{mm}$ that points straight down, discharging $8\,\text{L/s}$ of water ($998\,\text{kg/m}^3$) to the air. The horizontal run, with its water, weighs $6\,\text{kg}$ per metre. (a) Find the moment the flange must exert on the pipe. (b) and (c) Repeat with the nozzle pointing horizontally away from the riser, then straight up.

Solution. Assumptions: steady; incompressible; uniform jet at zero gauge pressure; the riser's weight and the inlet's pressure and momentum act along the vertical through O, so they have no moment about O; the nozzle's own length is negligible.

Axes: x along the horizontal run, y up, z out of the plane, counterclockwise positive. $a = \frac{\pi}{4}(0.025)^2 = 4.909\times10^{-4}\,\text{m}^2$, $V = Q/a = 16.30\,\text{m/s}$, $\dot{m} = 998\times0.008 = 7.984\,\text{kg/s}$. The exit is at $\mathbf{r} = (1.5, 1.0)\,\text{m}$.

Moments on the CV (pipe and its water) about O: the flange moment M_O; the weight of the run, $6 \times 9.81 \times 1.5 = 88.3\,\mathrm{N}$ at $x = 0.75\,\mathrm{m}$, moment $-0.75 \times 88.3 = -66.2\,\mathrm{N\,m}$. The inlet has $\mathbf{r} = 0$, so its flux is zero.

(a) $\mathbf{V} = (0, -16.30)$, so $(\mathbf{r} \times \mathbf{V})_z = xV_y - yV_x = 1.5(-16.30) = -24.45\,\mathrm{m^2/s}$, and

$$M_O - 66.2 = 7.984 \times (-24.45) = -195.2 \quad \Rightarrow \quad M_O = -129.0\,\mathrm{N\,m}.$$

The flange holds the pipe down with $129\,\mathrm{N\,m}$ clockwise. The jet acts like a rocket: it pushes the tip up with $\dot{m}V = 130.1\,\mathrm{N}$, more than the weight of the run.

(b) $\mathbf{V} = (16.30, 0)$: $(\mathbf{r} \times \mathbf{V})_z = -1.0 \times 16.30$, flux $-130.1\,\mathrm{N\,m}$, so $M_O = -130.1 + 66.2 = -63.9\,\mathrm{N\,m}$. The lever arm is now the height, $1.0\,\mathrm{m}$.

(c) $\mathbf{V} = (0, 16.30)$: flux $+195.2\,\mathrm{N\,m}$, so $M_O = 195.2 + 66.2 = +261.4\,\mathrm{N\,m}$, counterclockwise: the jet now pushes the tip down, adding to the weight.

Check: in each case the jet's reaction is $130.1\,\mathrm{N}$ opposite to the jet, and its moment is that force times the perpendicular distance from O to the jet line (1.5, 1.0, $1.5\,\mathrm{m}$). Momentum and moment agree. □

Example 9.2. A lawn sprinkler has two arms, each $0.2\,\mathrm{m}$ from the vertical axis to the nozzle. Each nozzle has a $5\,\mathrm{mm}$ exit and points horizontally, perpendicular to its arm, so the jets leave tangentially in opposite senses. Water ($998\,\mathrm{kg/m^3}$) enters along the axis at $0.5\,\mathrm{L/s}$ in total. Find (a) the torque to hold the rotor still, (b) the rotation rate with no friction, (c) the rotation rate when the bearing exerts a friction torque of $0.2\,\mathrm{N\,m}$, and the absolute tangential speed of the jets in that case.

Solution. Assumptions: steady operation at constant ω; incompressible; uniform jets at zero gauge pressure; flow divides equally; water enters along the axis, so it brings in no angular momentum.

CV: a fixed cylinder enclosing the rotor, cut by the inlet pipe on the axis and by the jets. Positive sense: the direction the rotor turns, opposite to the jets. Each nozzle area $a = \frac{\pi}{4}(0.005)^2 = 1.963 \times 10^{-5}\,\mathrm{m^2}$. Relative to the arm, $V_r = Q/(2a) = 0.0005/(3.927 \times 10^{-5}) = 12.73\,\mathrm{m/s}$, which does not depend on ω because the rotor's volume is fixed. Total $\dot{m} = 0.499\,\mathrm{kg/s}$. The absolute tangential velocity at each nozzle is $V_\theta = -(V_r - \omega R)$. The only moment on the CV about the axis is the torque T from the stand or bearing, positive in the rotation sense. Angular momentum:

$$T = \dot{m} R V_\theta = -\dot{m} R (V_r - \omega R).$$

(a) $\omega = 0$: $T = -0.499 \times 0.2 \times 12.73 = -1.271\,\mathrm{N\,m}$. The stand must apply $1.27\,\mathrm{N\,m}$ against the rotation sense.

(b) No friction, $T = 0$: $\omega = V_r/R = 63.7\,\mathrm{rad/s} = 608\,\mathrm{rpm}$. The jets leave with zero absolute velocity, since the nozzle runs forward exactly as fast as the water leaves it backward, and the water simply drops, carrying no angular momentum.

(c) $T = -0.2\,\mathrm{N\,m}$: $V_r - \omega R = 0.2/(0.499 \times 0.2) = 2.004\,\mathrm{m/s}$, so $\omega = (12.73 - 2.00)/0.2 = 53.6\,\mathrm{rad/s} = 512\,\mathrm{rpm}$, and each jet moves backward at $2.00\,\mathrm{m/s}$ in the fixed frame. The friction absorbs $0.2 \times 53.6 = 10.7\,\mathrm{W}$.

Check: a sprinkler at several hundred rpm without friction is why real ones have brakes. The friction torque is 16% of the stall torque and slows the rotor by 16%, as the linear relation between T and ω requires. □

Example 9.3. A centrifugal water pump ($998\,\mathrm{kg/m^3}$) has an impeller with inlet radius $50\,\mathrm{mm}$, outlet radius $150\,\mathrm{mm}$, outlet width $20\,\mathrm{mm}$, and backward-curved blades

at 30° to the tangent at the outlet. It turns at 1750 rpm and delivers $0.03\,\mathrm{m}^3/\mathrm{s}$ with no swirl at the inlet. (a) Draw the outlet velocity triangle in words and find V_{t2}. (b) Find the ideal head, torque and power. (c) A test at this flow measures a head of $52\,\mathrm{m}$ and a shaft power of $22\,\mathrm{kW}$. Find the pump efficiency and the hydraulic efficiency.

Solution. Assumptions: steady; incompressible; flow leaves exactly along the blades; uniform around the circumference; no inlet swirl, $V_{t1} = 0$.

(a) $\omega = 1750 \times 2\pi/60 = 183.3\,\mathrm{rad/s}$, so $U_2 = \omega r_2 = 27.49\,\mathrm{m/s}$. The radial component, by mass conservation through the outlet cylinder: $V_{n2} = Q/(2\pi r_2 b_2) = 0.03/(2\pi \times 0.15 \times 0.02) = 1.592\,\mathrm{m/s}$. The relative velocity leans back along the blade at 30° to the tangent, so its tangential part is $V_{n2}\cot 30^\circ = 2.757\,\mathrm{m/s}$ against the rotation. The triangle: $\mathbf{U}_2$ tangential, forward, $27.49\,\mathrm{m/s}$; $\mathbf{V}_{r2}$ from its tip, backward and outward at 30°; $\mathbf{V}_2$ closing the triangle. So $V_{t2} = 27.49 - 2.76 = 24.73\,\mathrm{m/s}$.

(b) $\dot{m} = 29.94\,\mathrm{kg/s}$. Torque $T = \dot{m} r_2 V_{t2} = 29.94 \times 0.15 \times 24.73 = 111.1\,\mathrm{N\,m}$. Power $= T\omega = 111.1 \times 183.3 = 20.36\,\mathrm{kW}$. Head $H = U_2 V_{t2}/g = 27.49 \times 24.73/9.81 = 69.3\,\mathrm{m}$. Check: $\rho g Q H = 998 \times 9.81 \times 0.03 \times 69.3 = 20.36\,\mathrm{kW}$, the same.

(c) Power to the water at the measured head: $\rho g Q H = 998 \times 9.81 \times 0.03 \times 52 = 15.27\,\mathrm{kW}$. Pump efficiency $= 15.27/22 = 69.4\%$. Hydraulic efficiency $= 52/69.3 = 75.0\%$. The remaining gap, 20.36 ideal versus 22 shaft, is friction in bearings and seals and leakage.

Check: the ideal head is below $U_2^2/g = 77\,\mathrm{m}$, the limit for radial blades, as it must be for backward blades. □

Trap. Using the relative velocity in the angular momentum flux of a rotating outlet. Mixing signs: fix the positive sense once and give V_θ a sign relative to it. Taking moments about a point other than the support, which brings in the unknown support force. Forgetting the moments of weights and of pressure at cuts not through O. Using $\mathbf{r}$ to the wrong point: it runs from O to the centre of the exit section, and only the perpendicular distance to the jet line matters. Using the blade angle from the radial direction when it was given from the tangent, or the reverse. Converting rpm to rad/s with 60 but not 2π. Calling the Euler head the measured head.

Ladder.

1. White P3.150. The moment about the inlet flange of an angled elbow. The moment arm of each momentum flux.
2. White P3.162. A many-bucket waterwheel driven by a jet; power and best speed. Relative velocity and the optimum blade speed.
3. White P3.156. A disk with two tangential exit ducts drives a load; the mass flow. Rotating outlets, absolute velocity.
4. White P3.164. A bend feeding a uniformly porous leg; the holding moment. Integrating the moment of a distributed outflow.

Answers are in the back of White; settle each call the same sitting.

Where it hides. Euler's equation is the source of every pump curve: with $V_{t2} = U_2 - V_{n2}\cot\beta_2$ and $V_{n2} \propto Q$, the ideal head falls linearly with flow, and the affinity laws on the sheet scale it with speed and size. The power $T\omega = \rho g Q H$ joins this section to pump heads in Section 10. In chapter 4, the angular momentum balance on an infinitesimal element is what makes the stress tensor symmetric ($\tau_{xy} = \tau_{yx}$). For a pipe held at one support, the linear momentum equation gives the support force and the angular momentum equation, with the same CV, gives the support moment; a question that asks for both needs both.

10 The energy equation

The first law of thermodynamics for a control volume, written in the units pipe engineers use, is the equation for every pipe system with a pump or turbine in it. It looks like Bernoulli with three extra terms, and that is exactly what it is: Bernoulli plus the head a pump adds, minus the head a turbine takes, minus the head lost to friction. This section derives it from the first law so that each extra term has a physical meaning, explains the kinetic energy correction factor α, and connects heads to power.

Recognise it. A pump or turbine between two points. "Find the pump head", "the power required", "the shaft power", "the efficiency", "the power generated", "the head loss", "the temperature rise". Two reservoirs joined by a pipe. Pressure gauges on the inlet and outlet of a machine. Any Bernoulli-shaped question where the statement mentions friction, losses or a machine. A laminar flow where kinetic energy terms matter ($\alpha = 2$).

On the crib sheet. The sheet prints the steady head form

$$\frac{P_1}{\gamma} + \alpha_1 \frac{V_1^2}{2g} + z_1 + h_p = \frac{P_2}{\gamma} + \alpha_2 \frac{V_2^2}{2g} + z_2 + h_t + h_L,$$

$\alpha = 2.0$ (laminar), $\alpha \approx 1.05$ (turbulent), EGL $=$ HGL $+ \alpha V^2/2g$, $P = \gamma Q h$, $P = \omega T_o$, $\eta = P_{\text{pump}}/P_{\text{input}}$, the head loss formula $h_L = (fL/D + \sum K_L)V^2/2g$ from chapter 6, and, under turbomachinery, the system curve $H_{\text{sys}} = \Delta z + \Delta P/\gamma + h_L(Q)$ and Euler's head. It leaves out the assumptions (steady, one inlet and one outlet, incompressible, uniform properties across each section apart from α), that point 1 must be upstream of point 2 for h_L to be positive, that α multiplies the *mean* velocity, that h_p is the head given to the fluid (not the motor's), and that a turbine's efficiency multiplies rather than divides.

The model

The first law for a system. The *first law* says the energy E of a fixed mass rises by the heat added minus the work done by it: $dE/dt = \dot{Q} - \dot{W}$. Per unit mass the energy is $e = \hat{u} + \frac{1}{2}V^2 + gz$: *internal energy* $\hat{u}$ (the molecular kinetic and potential energy, which shows as temperature), kinetic energy, and potential energy.

For a control volume. Put $B = E$, $b = e$ in the transport theorem:

$$\dot{Q} - \dot{W} = \frac{\partial}{\partial t} \int_{\text{CV}} e\rho \, d\forall + \int_{\text{CS}} e\rho (\mathbf{V} \cdot \mathbf{n}) \, dA.$$

Now split the work done by the fluid in the CV on its surroundings into three kinds. *Shaft work* $\dot{W}_s$ is work through a rotating shaft that crosses the CS: a turbine takes it out ($\dot{W}_s > 0$), a pump puts it in ($\dot{W}_s < 0$). *Pressure work* is done where fluid crosses the CS: the fluid inside pushes the fluid ahead of it out of the outlet against pressure P, which costs $P(\mathbf{V} \cdot \mathbf{n}) \, dA$ per unit time, and the upstream fluid pushes fluid in at the inlet, which is negative work. *Viscous work* is done by shear at the CS; it is zero on walls, where the fluid does not move, and negligible on inlet and outlet cuts perpendicular to the flow. Moving the pressure work to the right side, where it joins the flux, gives

$$\dot{Q} - \dot{W}_s = \frac{\partial}{\partial t} \int_{\text{CV}} e\rho \, d\forall + \int_{\text{CS}} \Big(\hat{u} + \frac{P}{\rho} + \frac{V^2}{2} + gz\Big)\rho (\mathbf{V} \cdot \mathbf{n}) \, dA.$$

The term P/ρ is the *flow work* per unit mass, the cost of pushing fluid across a boundary. Combined with $\hat{u}$ it is the enthalpy $h = \hat{u} + P/\rho$ of thermodynamics.

The steady 1-D form. Assume steady flow, one inlet 1 and one outlet 2 with the same $\dot{m}$, and P, $\hat{u}$ and z uniform across each section (true for z and P in a straight pipe up to the small hydrostatic change). Velocity is the one property that varies strongly across a section. The kinetic energy flux is $\int \frac{1}{2}\rho u^3\, dA$, and the *kinetic energy correction factor* is defined by

$$\int_A \tfrac{1}{2}\rho u^3\, dA = \alpha\, \dot{m}\frac{V_{\text{avg}}^2}{2}, \qquad \alpha = \frac{1}{A}\int_A \left(\frac{u}{V_{\text{avg}}}\right)^3 dA.$$

For laminar pipe flow, $u/V_{\text{avg}} = 2(1 - r^2/R^2)$; with $s = r^2/R^2$, $dA = \pi R^2\, ds$, so $\alpha = 8\int_0^1 (1-s)^3\, ds = 2$. Turbulent profiles are flatter, and α is 1.03 to 1.1, commonly 1.05 or simply 1. Divide the energy balance by $\dot{m}g$:

$$\frac{P_1}{\gamma} + \alpha_1\frac{V_1^2}{2g} + z_1 = \frac{P_2}{\gamma} + \alpha_2\frac{V_2^2}{2g} + z_2 + \frac{\dot{W}_s}{\dot{m}g} + \frac{(\hat{u}_2 - \hat{u}_1) - \dot{Q}/\dot{m}}{g}.$$

Every term is now a head, in metres. Name the last two.

Pump and turbine heads. Split the shaft term as $\dot{W}_s/(\dot{m}g) = h_t - h_p$, where $h_p \geq 0$ is the head a pump adds to the fluid and $h_t \geq 0$ the head a turbine extracts. Moving h_p to the left gives the sheet's form. Since $\dot{m}g = \rho g Q = \gamma Q$, the power is

$$\dot{W}_{\text{to fluid}} = \gamma Q h_p, \qquad \dot{W}_{\text{from fluid}} = \gamma Q h_t.$$

Machines are not perfect. A pump's motor delivers shaft power $\dot{W}_{\text{shaft}} = \gamma Q h_p/\eta_p$: the shaft must supply more than the fluid receives. A turbine's shaft delivers $\dot{W}_{\text{shaft}} = \eta_t \gamma Q h_t$: the shaft receives less than the fluid gives up. In both, $\eta < 1$, and a generator or motor efficiency multiplies on top.

Head loss. The last term, $h_L = [(\hat{u}_2 - \hat{u}_1) - \dot{Q}/\dot{m}]/g$, is the *head loss*. Friction in the fluid converts mechanical energy (pressure, kinetic and potential) into internal energy, which either warms the fluid or leaves as heat. The second law of thermodynamics says this conversion runs one way only, so $h_L \geq 0$ in the direction of flow: friction is an *irreversibility*. The energy is not destroyed, but it is no longer available to do work. Because a head of a few metres corresponds to a temperature rise of thousandths of a kelvin ($\Delta T = g h_L/c$), the loss is computed from friction correlations in chapter 6, not measured as heating.

The relation to Bernoulli. With $h_p = h_t = h_L = 0$ and $\alpha = 1$, the energy equation is Bernoulli. So the energy equation is the general statement: Bernoulli plus pump, minus turbine, minus loss. It holds between two *sections* of a pipe (averaged over each), not along one streamline; it carries the shaft work that Bernoulli forbids; and it carries friction. Conversely, wherever the energy equation's extra terms are zero, the two give the same answer.

Grade lines with machines. The EGL is now $P/\gamma+\alpha V^2/2g+z$. It slopes down along pipes at the rate of head loss per length, falls suddenly at a fitting or a turbine, and rises suddenly by h_p at a pump. Between two free surfaces open to the air, $P = 0$ and $V \approx 0$ at each, and the equation becomes the system equation on the sheet: $h_p = (z_2 - z_1) + h_L$.

First moves. Number the points in the direction of flow. Prefer free surfaces ($P = 0$ gauge, $V \approx 0$) and free jets ($P = 0$ gauge). Write the assumptions: steady, incompressible, one inlet and one outlet, α value (2 laminar, about 1 or 1.05 turbulent: check Reynolds number if the fluid is viscous). Write the full head equation with every term, then strike the ones that vanish and say why. Get speeds from mass conservation. Convert pressures to heads with $\gamma = \rho g$. Solve for the unknown head, then power $\gamma Q h$, then shaft or electrical power with the efficiency on the correct side. Check that h_L came out positive.

Worked examples

Example 10.1. A pump moves $0.04\,\mathrm{m^3/s}$ of water ($998\,\mathrm{kg/m^3}$) from a $150\,\mathrm{mm}$ suction pipe to a $100\,\mathrm{mm}$ discharge pipe. A gauge on the suction pipe reads $-25\,\mathrm{kPa}$ (a vacuum) and one on the discharge pipe reads $310\,\mathrm{kPa}$; the discharge gauge is $0.4\,\mathrm{m}$ higher. (a) Find the head added by the pump. (b) Find the power given to the water and the shaft power at a pump efficiency of 72%. (c) The pump lifts water from an open sump to an open tank whose surface is $26\,\mathrm{m}$ higher. Find the total head loss in the pipes.

Solution. Assumptions: steady, incompressible, turbulent flow with $\alpha = 1$ (using 1.05 changes the kinetic term by $0.05\,\mathrm{m}$), no loss between the gauges other than inside the pump, which is counted in h_p as the head actually given to the water.

(a) Speeds: $V_s = 0.04/(\tfrac{\pi}{4}0.15^2) = 2.264\,\mathrm{m/s}$, $V_d = 0.04/(\tfrac{\pi}{4}0.10^2) = 5.093\,\mathrm{m/s}$. Energy from suction gauge (1) to discharge gauge (2):

$$h_p = \frac{P_2 - P_1}{\gamma} + \frac{V_2^2 - V_1^2}{2g} + (z_2 - z_1) = \frac{335\,000}{998 \times 9.81} + \frac{25.94 - 5.124}{19.62} + 0.4.$$

The terms are $34.22 + 1.06 + 0.40 = 35.68\,\mathrm{m}$.

(b) $\dot{W} = \gamma Q h_p = 998 \times 9.81 \times 0.04 \times 35.68 = 13.97\,\mathrm{kW}$. Shaft power $= 13.97/0.72 = 19.41\,\mathrm{kW}$.

(c) Energy from the sump surface to the tank surface: both at zero gauge pressure and negligible speed, so $h_p = (z_2 - z_1) + h_L$, and $h_L = 35.68 - 26 = 9.68\,\mathrm{m}$.

Check: the pressure rise dominates the head, as it should for a pump with nearly equal pipe sizes. A loss of $9.7\,\mathrm{m}$ out of $35.7\,\mathrm{m}$ is large; it would come from long or rough pipes. Using absolute pressures gives the same h_p, since only the difference enters. □

Example 10.2. A small hydroelectric plant takes water ($1000\,\mathrm{kg/m^3}$) from a reservoir whose surface is $60\,\mathrm{m}$ above the tailwater surface. The flow is $2.0\,\mathrm{m^3/s}$. Friction in the intake and penstock costs $4.0\,\mathrm{m}$ of head, and the water leaves the draft tube into the tailwater at $3.0\,\mathrm{m/s}$, its kinetic energy then being lost. The turbine is 92% efficient and the generator 95%. Find the turbine head, the power extracted from the water, the electrical output, and the overall efficiency relative to the gross head.

Solution. Assumptions: steady, incompressible, $\alpha = 1$, both surfaces at atmospheric pressure and at rest, exit kinetic energy fully lost (an exit loss with $K = 1$).

Energy from reservoir surface (1) to tailwater surface (2): $z_1 = h_t + h_L$, with $h_L = 4.0 + V^2/2g = 4.0 + 9/19.62 = 4.0 + 0.459\,\text{m}$. So $h_t = 60 - 4.459 = 55.54\,\text{m}$.

Power from the water: $\gamma Q h_t = 1000 \times 9.81 \times 2.0 \times 55.54 = 1.090\,\text{MW}$. Electrical: $0.92 \times 0.95 \times 1.090 = 0.952\,\text{MW}$.

Gross available, $\gamma Q z_1 = 1.177\,\text{MW}$, so the overall efficiency is $0.952/1.177 = 80.9\%$.

Check: the penstock costs $4.0/60 = 6.7\%$, the exit 0.8%, the machines $1 - 0.874 = 12.6\%$; the product 0.926×0.874 is 0.809. Efficiencies multiply here because the turbine receives less than the water gives. □

Example 10.3. Oil ($870\,\text{kg/m}^3$, $\mu = 0.1\,\text{Pa s}$, $c = 1900\,\text{J/(kg K)}$) flows at $0.1\,\text{L/s}$ through a horizontal pipe that contracts gradually from $20\,\text{mm}$ to $10\,\text{mm}$ diameter. Pressure taps in the fully developed regions upstream and downstream show $P_1 - P_2 = 2.4\,\text{kPa}$. (a) Show the flow is laminar. (b) Find the head loss between the taps. (c) Find the error made by taking $\alpha = 1$. (d) Find the power dissipated and the temperature rise of the oil.

Solution. Assumptions: steady, incompressible, fully developed laminar profile at both taps, no heat transfer, taps at the same height.

(a) $V_1 = 10^{-4}/(\tfrac{\pi}{4}0.02^2) = 0.318\,\text{m/s}$, $V_2 = 1.273\,\text{m/s}$. Reynolds number $\text{Re} = \rho V D/\mu$: $870(0.318)(0.02)/0.1 = 55$ and $870(1.273)(0.01)/0.1 = 111$, both far below 2300. So $\alpha_1 = \alpha_2 = 2$.

(b) Energy, 1 to 2, no machines, same height:

$$h_L = \frac{P_1 - P_2}{\gamma} + \alpha\frac{V_1^2 - V_2^2}{2g} = \frac{2400}{870 \times 9.81} + 2 \times \frac{0.1013 - 1.6211}{19.62} = 0.2812 - 0.1549 = 0.126\,\text{m}.$$

(c) With $\alpha = 1$: $h_L = 0.2812 - 0.0775 = 0.204\,\text{m}$, which is 61% too high. The pressure drop pays both for friction and for accelerating the oil; with the real profile, accelerating it costs twice as much as the mean speed suggests, so less is left for friction.

(d) Power dissipated: $\gamma Q h_L = 870 \times 9.81 \times 10^{-4} \times 0.126 = 0.108\,\text{W}$. Temperature rise: $\Delta T = g h_L/c = 9.81 \times 0.126/1900 = 6.5 \times 10^{-4}\,\text{K}$.

Check: $h_L > 0$ in the direction of flow, as the second law requires. The temperature rise is far too small to measure, which is why losses are found from pressures, not thermometers. □

Trap. Putting the efficiency on the wrong side: pump shaft power is $\gamma Q h_p/\eta$, turbine shaft power is $\eta\gamma Q h_t$. Numbering the points against the flow, which makes h_L negative. Dropping the velocity heads at gauges on pipes of different sizes. Using $\alpha = 1$ for laminar flow. Treating h_p as the motor's power divided by γQ. Mixing gauge pressure at one point with absolute at the other. Forgetting the exit loss when a pipe discharges into a reservoir: the kinetic energy of the jet is not recovered. Applying Bernoulli across the pump, which gives $h_p = 0$. Using $\gamma = 9.81$ for a fluid that is not water.

Ladder.

1. White P3.178. A horizontal pump between two pipe sizes, no losses; the power. Pump head from the change in pressure and speed.
2. White P3.180. A pump between reservoirs with a loss coefficient and an efficiency. Head form with losses and efficiency.

3. White P3.174. Pumped storage: turbine output by day, pump input by night. Losses add to a pump's head and subtract from a turbine's.
4. White P3.176. A fireboat pump; the motor power. Elevation, nozzle speed, loss and efficiency together.
5. White P3.126. Kerosene from an air-pressurized tank, with and without pipe loss. Pressurized surface; the loss as a known head.
6. White P3.184. A dam turbine with a loss rising as speed squared; the two flows for a target power. A cubic in the flow rate and choosing a root.

Answers are in the back of White; settle each call the same sitting.

Where it hides. Bernoulli (Section 8) is the special case with no machines, no loss and $\alpha = 1$. Euler's turbomachine equation (Section 9) supplies h_p from the impeller geometry, and $\gamma Q h = T\omega$ connects the two. Chapter 6 supplies h_L: Darcy's fL/D for straight pipe and K_L for fittings, which turns this equation into the three pipe-problem types on the sheet. The pump's system curve, $H_{\text{sys}} = \Delta z + \Delta P/\gamma + h_L(Q)$, is this equation between two reservoirs. In chapter 4, the differential energy equation includes the viscous dissipation function, the local version of h_L.

11 The acceleration field

Chapter 3 drew a box around a region and balanced what crossed its faces. Chapter 4 shrinks the box to a point, so that the balances hold everywhere in the flow and become equations for the velocity field itself. The first thing Newton's second law needs is an acceleration, and in a fluid that is less obvious than it looks: the velocity is known at fixed points in space, but Newton's law is about a fixed lump of fluid that moves through those points. This section builds the acceleration of a fluid particle out of a velocity field given at fixed points. Everything later in the chapter uses it.

Recognise it. A velocity field given as a formula, $\mathbf{V} = u(x,y,z,t)\,\mathbf{i} + v(\dots)\,\mathbf{j} + w(\dots)\,\mathbf{k}$, and a question asking for "the acceleration" at a point, the largest acceleration in a region, the acceleration along a streamline or centreline, or whether the flow is steady. A nozzle or diffuser modelled by a one-dimensional $u(x)$. Any later question that needs the left side of the Navier–Stokes equation, because that left side is this acceleration times density.

On the crib sheet. Not printed. Section 13 of the sheet gives only the notation for a 2-D field, $\mathbf{V} = u(x,y)\,\mathbf{i} + v(x,y)\,\mathbf{j}$. The material derivative, the split into local and convective parts and the component formulas below have to come from you. They are a chain rule; the model below rebuilds them in under a minute.

The model

Two ways to describe a moving fluid. A *Lagrangian* description follows individual particles: particle number 7 is at position $\mathbf{r}_7(t)$, and its velocity and acceleration are $\dot{\mathbf{r}}_7$ and $\ddot{\mathbf{r}}_7$, exactly as in dynamics. A *Eulerian* description stands at fixed points and records what passes: $\mathbf{V}(x,y,z,t)$ is the velocity of whichever particle happens to be at (x,y,z) at time t. A weather map, a wind-tunnel probe and a CFD result are Eulerian. Fluid mechanics works almost entirely in the Eulerian picture, because tracking every particle is hopeless while a field over space is something you can measure and write down. Here a *field* just means a quantity given as a function of position and time.

The cost is that $\partial\mathbf{V}/\partial t$, the rate of change at a fixed point, is not the particle's acceleration. A fixed point sees a sequence of different particles go past.

The material derivative, by the chain rule. Follow one particle. At time t it is at (x, y, z); a short time dt later it is at $(x+u\,dt,\ y+v\,dt,\ z+w\,dt)$, because it moves with the local velocity. Its velocity changes because time passed and because it moved to a place where the field is different. The chain rule for a function of four variables gives the change:

$$d\mathbf{V} = \frac{\partial \mathbf{V}}{\partial t}\,dt + \frac{\partial \mathbf{V}}{\partial x}\,dx + \frac{\partial \mathbf{V}}{\partial y}\,dy + \frac{\partial \mathbf{V}}{\partial z}\,dz, \qquad dx = u\,dt,\ dy = v\,dt,\ dz = w\,dt.$$

Divide by dt. The acceleration of the particle is

$$\mathbf{a} = \frac{D\mathbf{V}}{Dt} = \underbrace{\frac{\partial \mathbf{V}}{\partial t}}_{\text{local}} + \underbrace{u\frac{\partial \mathbf{V}}{\partial x} + v\frac{\partial \mathbf{V}}{\partial y} + w\frac{\partial \mathbf{V}}{\partial z}}_{\text{convective}} = \frac{\partial \mathbf{V}}{\partial t} + (\mathbf{V}\cdot\nabla)\mathbf{V}. \tag{1}$$

The operator $D/Dt = \partial/\partial t + u\,\partial/\partial x + v\,\partial/\partial y + w\,\partial/\partial z$ is the *material derivative*: the rate of change seen by an observer riding on a particle of material. It applies to any field, not only velocity. DT/Dt is how fast a particle's temperature changes, and $D\rho/Dt$ how fast its density changes. The symbol $\mathbf{V}\cdot\nabla$ is shorthand for $u\,\partial_x + v\,\partial_y + w\,\partial_z$, the rate of change in the direction of motion scaled by the speed.

The two parts have names worth keeping straight.

- *Local* acceleration, $\partial\mathbf{V}/\partial t$: the field itself changing in time at a fixed point. A valve opening, a pump spinning up. It is zero in a *steady* flow, which is a flow whose field does not depend on t.
- *Convective* acceleration, $(\mathbf{V}\cdot\nabla)\mathbf{V}$: the particle being carried ("convected") to a place where the velocity is different. Water speeding up through a steady nozzle has only this kind.

So a steady flow can accelerate. Steady means the picture does not change; it does not mean each particle keeps its velocity. Every particle going through a fixed garden-hose nozzle speeds up.

The components. Written out in Cartesian coordinates, with each line the material derivative of one velocity component:

$$\begin{aligned}
a_x &= \frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} + w\frac{\partial u}{\partial z},\\
a_y &= \frac{\partial v}{\partial t} + u\frac{\partial v}{\partial x} + v\frac{\partial v}{\partial y} + w\frac{\partial v}{\partial z},\\
a_z &= \frac{\partial w}{\partial t} + u\frac{\partial w}{\partial x} + v\frac{\partial w}{\partial y} + w\frac{\partial w}{\partial z}.
\end{aligned}$$

Memory: the component being differentiated stays the same along a line; the multipliers u, v, w go with x, y, z in order.

Along a streamline. A *streamline* is a curve everywhere tangent to the velocity. Measure distance s along it and let V be the speed. In a steady flow the acceleration splits into a part along the streamline and a part toward the centre of its curvature:

$$a_s = V\frac{dV}{ds}, \qquad a_n = \frac{V^2}{R_c},$$

where R_c is the local radius of curvature of the streamline. The first is the convective term written along the path; the second is the centripetal acceleration of circular motion. On a straight streamline, such as the centreline of a symmetric nozzle, only a_s survives. A particle on a curved streamline at constant speed still accelerates. (With unsteadiness, add $\partial V/\partial t$ to a_s.)

Cylindrical coordinates. With $\mathbf{V} = v_r\,\mathbf{e}_r + v_\theta\,\mathbf{e}_\theta + v_z\,\mathbf{e}_z$, the operator is $D/Dt = \partial_t + v_r\,\partial_r + (v_\theta/r)\,\partial_\theta + v_z\,\partial_z$; the $1/r$ appears because a step $d\theta$ is a distance $r\,d\theta$. The unit vectors $\mathbf{e}_r$ and $\mathbf{e}_\theta$ turn as a particle moves around, and that turning adds two terms:

$$a_r = \frac{Dv_r}{Dt} - \frac{v_\theta^2}{r}, \qquad a_\theta = \frac{Dv_\theta}{Dt} + \frac{v_r v_\theta}{r}, \qquad a_z = \frac{Dv_z}{Dt}.$$

The $-v_\theta^2/r$ is the centripetal acceleration of anything going round. The $v_r v_\theta/r$ is the same term that makes a skater spin faster when pulling in their arms. These reappear in the cylindrical Navier–Stokes equations of Section 13.

First moves. Write the three (or two) component lines before substituting anything. Check for t in the field: none means the local term is zero, say so. Compute the nine (or four) partial derivatives as a small table, then substitute the point at the end, not the start. If the question asks for a magnitude, give $|\mathbf{a}| = \sqrt{a_x^2 + a_y^2 + a_z^2}$ and its direction. If it asks for acceleration in g's, divide by $9.81\,\mathrm{m/s^2}$. On a centreline or symmetry line, check whether the cross-velocity is zero there; it usually is, and the problem collapses to $a_s = u\,du/dx$.

Worked examples

Example 11.1 (A squeeze-film press)**.** A flat platen is pressed down onto a thin layer of adhesive spread on a fixed base. The layer is a long strip, so the flow is two-dimensional: x along the strip from its centre, y up from the base. The platen moves down at constant speed $W = 2\,\mathrm{mm/s}$, so the gap is $h(t) = h_0 - Wt$. A plug-flow model of the squeezed-out adhesive is

$$u = \frac{Wx}{h(t)}, \qquad v = -\frac{Wy}{h(t)}.$$

Find the acceleration of the adhesive at $x = 20\,\mathrm{mm}$ at the instant the gap is $h = 0.5\,\mathrm{mm}$.

Assumptions. The given field is taken as exact; it ignores viscosity (it slips at both plates), which is acceptable for a kinematics question. The field depends on t through h, so the flow is unsteady and the local term must be kept.

Check the field first. At $y = 0$, $v = 0$: no flow through the base. At $y = h$, $v = -W$: the fluid moves down with the platen. Good.

Derivatives. Since $dh/dt = -W$, $\partial(1/h)/\partial t = W/h^2$. So

$$\frac{\partial u}{\partial t} = \frac{W^2 x}{h^2}, \quad \frac{\partial u}{\partial x} = \frac{W}{h}, \quad \frac{\partial u}{\partial y} = 0, \qquad \frac{\partial v}{\partial t} = -\frac{W^2 y}{h^2}, \quad \frac{\partial v}{\partial x} = 0, \quad \frac{\partial v}{\partial y} = -\frac{W}{h}.$$

Components.

$$a_x = \frac{W^2 x}{h^2} + \frac{Wx}{h}\cdot\frac{W}{h} + 0 = \frac{2W^2 x}{h^2},$$
$$a_y = -\frac{W^2 y}{h^2} + 0 + \Big(-\frac{Wy}{h}\Big)\Big(-\frac{W}{h}\Big) = 0.$$

Numbers. $a_x = 2(0.002)^2(0.02)/(0.0005)^2 = 0.64\,\mathrm{m/s^2}$, directed outward along the strip, and $a_y = 0$.

Sanity check. In x the local and convective parts are equal and add: the fluid is pushed out faster both because the gap is shrinking (time) and because it moves outward to where the field is faster (space). In y they cancel exactly, which makes sense: a particle at height y keeps the same fraction y/h of the gap, so its vertical velocity is $-Wy/h$ with y/h fixed, a constant. The acceleration grows as $1/h^2$, which is why the last fraction of a millimetre of squeezing is violent.

Example 11.2 (A wind-tunnel contraction, on and off the centreline)**.** A steady model of the flow in a two-dimensional contraction is

$$u = U_0\Big(1 + \frac{x^2}{L^2}\Big), \qquad v = -\frac{2U_0 xy}{L^2},$$

with $U_0 = 5\,\mathrm{m/s}$ and $L = 0.4\,\mathrm{m}$. Find the acceleration (a) on the centreline $y = 0$ at $x = 0.2\,\mathrm{m}$, (b) at $(0.2, 0.1)\,\mathrm{m}$, and (c) at $(0, 0.1)\,\mathrm{m}$.

Assumptions. Steady: no t appears, so $\partial/\partial t = 0$. Two-dimensional: $w = 0$ and nothing depends on z.

Derivatives. $\partial u/\partial x = 2U_0 x/L^2$, $\partial u/\partial y = 0$, $\partial v/\partial x = -2U_0 y/L^2$, $\partial v/\partial y = -2U_0 x/L^2$.

Components.

$$a_x = u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} = U_0\Big(1 + \frac{x^2}{L^2}\Big)\frac{2U_0 x}{L^2} = \frac{2U_0^2 x}{L^2}\Big(1 + \frac{x^2}{L^2}\Big),$$
$$a_y = u\frac{\partial v}{\partial x} + v\frac{\partial v}{\partial y} = -\frac{2U_0^2 y}{L^2}\Big(1 + \frac{x^2}{L^2}\Big) + \frac{4U_0^2 x^2 y}{L^4} = -\frac{2U_0^2 y}{L^2}\Big(1 - \frac{x^2}{L^2}\Big).$$

With $U_0^2/L^2 = 25/0.16 = 156.25\,\mathrm{s^{-2}}$:

(a) $y = 0$: $a_x = 2(156.25)(0.2)(1.25) = 78.1\,\mathrm{m/s^2}$, $a_y = 0$. The centreline is a straight streamline ($v = 0$ there), so this is $a_s = u\,du/dx = (6.25)(6.25) = 78.1\,\mathrm{m/s^2}$, the same.

(b) $a_x = 78.1\,\mathrm{m/s^2}$ (it does not depend on y), $a_y = -2(156.25)(0.1)(0.75) = -23.4\,\mathrm{m/s^2}$, so $|\mathbf{a}| = 81.6\,\mathrm{m/s^2}$, about $8.3g$.

(c) $u = 5\,\mathrm{m/s}$, $v = 0$, $a_x = 0$, but $a_y = -2(156.25)(0.1) = -31.3\,\mathrm{m/s^2}$.

Sanity check. Part (c) is the lesson. At $x = 0$ the particle is moving purely in x at a speed that is momentarily not changing, yet it accelerates toward the centreline: the streamlines converge into the contraction, so they are curved there, and a_y is the centripetal V^2/R_c. Solving $31.25 = 25/R_c$ gives a radius of curvature of $0.8\,\mathrm{m}$ for the streamline through that point. Steady, and accelerating at three g.

Trap. Setting the acceleration to zero because the flow is steady. Steady kills only $\partial\mathbf{V}/\partial t$. The reverse error: treating $\partial u/\partial t$ alone as the acceleration of an unsteady flow. Substituting the point's coordinates before differentiating, which turns every derivative into zero. Dropping the $v\,\partial u/\partial y$ term because "the flow is mainly in x"; it is often the one that matters off the axis. In cylindrical coordinates, forgetting $-v_\theta^2/r$, which is the entire acceleration of a particle in a vortex.

Ladder.

1. White P4.2. A nozzle with linearly rising speed; the acceleration at entrance and exit. One-dimensional convective acceleration.
2. White P4.4. A 2-D converging nozzle field; the accelerations and where the largest is. Both components of the material derivative.

3. White P4.10. A linear plane field; the acceleration and its direction. Reading the acceleration vector geometrically.
4. White P4.8. Unsteady duct flow after a valve opens; local plus convective acceleration. When the local term matters.

Answers are in the back of White; settle each call the same sitting.

Where it hides. The left side of every momentum equation in this chapter is $\rho\, D\mathbf{V}/Dt$, so this section is the first line of Section 13, and the exact flows of Section 16 are exactly the cases where it vanishes. The material derivative of density appears in continuity (Section 12). The rigid-body line on the crib sheet, $\nabla P = \rho(\mathbf{g} - \mathbf{a})$, is the Navier–Stokes equation for a flow with no shear, with this $\mathbf{a}$. The convective term $u\,\partial u/\partial x$ is what scales as $\rho V^2/L$ in Chapter 5 and puts the density into the Reynolds number. Integrated along a streamline, $a_s = V\,dV/ds$ is where Bernoulli's $V^2/2$ comes from.

12 Differential continuity

Mass is neither created nor destroyed. Chapter 3 wrote that for a finite control volume: what flows in, minus what flows out, equals what accumulates inside. Apply the same sentence to a box so small that the velocity and density barely change across it, and the result is one partial differential equation that every real flow field must satisfy at every point. Its two exam uses are blunt: it finds a velocity component you were not given, and it rejects a proposed velocity field that could not exist.

Recognise it. A velocity field with one component missing, "$v = ?$", "find $f(y)$", "what must w be". "Is this flow physically possible?" or "does this field satisfy conservation of mass?". A flow between walls, in a pipe, or out between two discs where the velocity profile is proposed and one asks whether it is consistent. A gas being compressed or expanded uniformly, where the density changes in time. It is also the first step of every exact-flow problem (Section 16), where it shows which velocity components can vary in which direction.

On the crib sheet. Only the two-dimensional incompressible form is printed, in section 13: $\partial u/\partial x + \partial v/\partial y = 0$. The general compressible form, the three-dimensional form and the cylindrical form are not printed. The control-volume mass balance of Chapter 3 is printed (section 3), and the differential form is that balance on a small box, which is how to rebuild it below.

The model

The box. Take a fixed box with corner at (x, y, z) and sides dx, dy, dz. Fluid crosses its six faces. The mass flow through a face is density times the velocity component normal to the face times the area, so through the left face (at x, area $dy\,dz$) the mass inflow per unit time is $\rho u\,dy\,dz$, evaluated at x.

Through the right face, at $x + dx$, the quantity ρu has changed slightly. A first-order Taylor expansion (keep the first derivative, drop the terms in dx^2, which vanish faster as the box shrinks) gives the outflow

$$\left(\rho u + \frac{\partial(\rho u)}{\partial x}\,dx\right) dy\,dz.$$

Net outflow in the x direction is right minus left: $\partial(\rho u)/\partial x \; dx\, dy\, dz$. The y and z face pairs give the same form with ρv and ρw.

The mass inside is $\rho\, dx\, dy\, dz$, and the box is fixed, so it changes at the rate $\partial\rho/\partial t \; dx\, dy\, dz$. Mass conservation says accumulation plus net outflow is zero. Divide by the volume $dx\, dy\, dz$:

$$\frac{\partial \rho}{\partial t} + \frac{\partial(\rho u)}{\partial x} + \frac{\partial(\rho v)}{\partial y} + \frac{\partial(\rho w)}{\partial z} = 0, \qquad \text{or} \qquad \frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{V}) = 0. \tag{2}$$

The *divergence* of a vector field, $\nabla \cdot \mathbf{F} = \partial F_x/\partial x + \partial F_y/\partial y + \partial F_z/\partial z$, is exactly what the box computed: the net outflow of $\mathbf{F}$ per unit volume at a point. So the equation reads: the density at a point drops at the rate that mass flux diverges away from it.

Incompressible form. Expand the products with the product rule and regroup:

$$\Big(\frac{\partial \rho}{\partial t} + u\frac{\partial \rho}{\partial x} + v\frac{\partial \rho}{\partial y} + w\frac{\partial \rho}{\partial z}\Big) + \rho\Big(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z}\Big) = \frac{D\rho}{Dt} + \rho\, \nabla \cdot \mathbf{V} = 0.$$

The bracket is the material derivative of density (Section 11): how fast a particle's density changes as it moves. A fluid is treated as *incompressible* when that is zero, $D\rho/Dt = 0$: each particle keeps its density. For liquids, and for gases below a Mach number of about 0.3, this is an excellent approximation. Then, dividing by ρ,

$$\nabla \cdot \mathbf{V} = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0. \tag{3}$$

Read it as a volume statement: a particle's volume does not change, so whatever stretching it suffers in one direction ($\partial u/\partial x > 0$) is paid for by squeezing in the others. Note that (3) has no ρ and no t in it: it holds for unsteady incompressible flow too.

Cylindrical form. In (r, θ, z), the box is a wedge-shaped piece of an annulus. Its outer face at $r + dr$ is larger than its inner face at r, because the area of a face at radius r is $r\, d\theta\, dz$. So the radial flux that is conserved is $r\rho v_r$, not ρv_r; and a step $d\theta$ is a length $r\, d\theta$, which puts $1/r$ on the θ term:

$$\frac{\partial \rho}{\partial t} + \frac{1}{r}\frac{\partial(r\rho v_r)}{\partial r} + \frac{1}{r}\frac{\partial(\rho v_\theta)}{\partial \theta} + \frac{\partial(\rho v_z)}{\partial z} = 0. \tag{4}$$

Incompressible: $\frac{1}{r}\frac{\partial(r v_r)}{\partial r} + \frac{1}{r}\frac{\partial v_\theta}{\partial \theta} + \frac{\partial v_z}{\partial z} = 0$. Memory hook: a purely radial outflow from a line source spreads over a circumference $2\pi r$, so $v_r \propto 1/r$; the equation must make $r v_r$ constant, so the r sits inside the derivative.

Finding a missing component. Suppose $u(x, y)$ is known in a 2-D incompressible flow. Then $\partial v/\partial y = -\partial u/\partial x$, and integrating in y *at fixed* x gives

$$v(x, y) = -\int \frac{\partial u}{\partial x}\, dy + f(x).$$

The "constant" of integration is a function $f(x)$, because anything depending only on x has zero y-derivative and would vanish from the equation. It is fixed by a boundary condition, typically $v = 0$ on a wall at $y = 0$ (Section 14). If no condition is given, the honest answer contains $f(x)$, and the simplest choice is $f = 0$; say so. Integrating with a definite integral from the wall, $v = -\int_0^y \partial u/\partial x\, dy'$, builds the wall condition in.

Checking a proposed field. A field is *physically possible* (as far as mass goes) when it satisfies (2); for an incompressible fluid, when its divergence is zero at every point, not only at one. Compute the divergence as a function of position and look at it. Zero identically: possible. Anything else: impossible for an incompressible fluid, though it might describe a compressible flow with the right $\rho(x, y, z, t)$. Satisfying continuity is necessary, not sufficient: the field must also satisfy momentum (Section 13) and the boundary conditions.

First moves. Decide compressible or incompressible and write the matching equation, in the coordinates of the problem. Mark which velocity components are zero and which derivatives vanish (2-D: no z; axisymmetric: no θ; steady: no t). For a missing component, integrate with the unknown function of the remaining variables written explicitly, then name the boundary condition that fixes it. For "is it possible", compute the divergence as a function, not at a point, and state the conclusion in words. Check units: every term is s^{-1} (incompressible) or $\mathrm{kg/(m^3\,s)}$.

Worked examples

Example 12.1 (The upward velocity in a growing wall layer)**.** Near a flat plate, the streamwise velocity in a thin layer is modelled as

$$u = U\Big[2\frac{y}{\delta} - \Big(\frac{y}{\delta}\Big)^2\Big], \qquad 0 \le y \le \delta(x), \qquad \delta = c\sqrt{x},$$

with $U = 3\,\mathrm{m/s}$ and $\delta = 5\,\mathrm{mm}$ at $x = 0.5\,\mathrm{m}$. Find $v(x, y)$, and its value at the edge of the layer at $x = 0.5\,\mathrm{m}$.

Assumptions. Steady, two-dimensional, incompressible (water or low-speed air). The wall is solid, so $v(x, 0) = 0$.

Continuity. $\partial v/\partial y = -\partial u/\partial x$. Since u depends on x only through δ, use $d\delta/dx = c/(2\sqrt{x}) = \delta/(2x)$:

$$\frac{\partial u}{\partial x} = U\Big[-\frac{2y}{\delta^2} + \frac{2y^2}{\delta^3}\Big]\frac{d\delta}{dx} = -\frac{U}{x}\Big[\frac{y}{\delta} - \frac{y^2}{\delta^2}\Big].$$

Integrate from the wall, which applies $v(x, 0) = 0$ and sets $f(x) = 0$:

$$v = \int_0^y \frac{U}{x}\Big[\frac{y'}{\delta} - \frac{y'^2}{\delta^2}\Big]dy' = \frac{U}{x}\Big[\frac{y^2}{2\delta} - \frac{y^3}{3\delta^2}\Big].$$

At $y = \delta$: $v = U\delta/(6x) = (3)(0.005)/(6 \times 0.5) = 0.0050\,\mathrm{m/s}$, that is, $5\,\mathrm{mm/s}$ upward.

Sanity check. The layer slows the fluid near the wall and thickens downstream, so fluid it has slowed must be displaced outward: $v > 0$, as found. Its size relative to U is $\delta/(6x) = 1/600$, as small as the layer is thin. Independently, the flow rate inside the layer is $\int_0^\delta u\,dy = \frac{2}{3}U\delta$, and a mass balance on a slice dx of the layer gives the same edge velocity.

Example 12.2 (Radial flow out of a disc air bearing)**.** Air at low speed is fed through the centre of a thrust bearing and flows radially outward between two parallel discs a distance $2h$ apart, $-h \le z \le h$. A designer proposes

$$v_r = \frac{3Q}{8\pi r h}\Big(1 - \frac{z^2}{h^2}\Big), \qquad v_\theta = v_z = 0.$$

(a) Is this physically possible for an incompressible fluid? (b) Does it carry the volume flow Q? (c) A colleague proposes $v_r = C(1 - z^2/h^2)$ instead. Check it. (d) For $Q = 2\,\mathrm{L/min}$ and $h = 0.1\,\mathrm{mm}$, find the largest velocity at $r = 10\,\mathrm{mm}$.

Assumptions. Incompressible (the pressure changes are small compared with absolute pressure); steady; axisymmetric (no θ dependence).

(a) With $v_\theta = v_z = 0$, continuity is $\frac{1}{r}\partial(rv_r)/\partial r = 0$. Here $rv_r = \frac{3Q}{8\pi h}(1 - z^2/h^2)$ does not depend on r, so its r-derivative is zero everywhere. Possible.

(b) Flow through a cylinder of radius r: $\int_{-h}^{h} v_r \, 2\pi r \, dz = \frac{3Q}{4h}\int_{-h}^{h}(1-z^2/h^2)\,dz = \frac{3Q}{4h}\cdot\frac{4h}{3} = Q$. Yes, at every radius, as it must.

(c) $\frac{1}{r}\partial(rv_r)/\partial r = \frac{C}{r}(1 - z^2/h^2) \neq 0$. Impossible: the same velocity spread over a larger circumference would carry more flow at larger r, created from nothing.

(d) $Q = 2\times10^{-3}/60 = 3.33\times10^{-5}\,\mathrm{m^3/s}$. The maximum is at $z = 0$: $v_r = 3(3.33\times10^{-5})/(8\pi\times 0.01\times10^{-4}) = 3.98\,\mathrm{m/s}$. The mean over the gap is $Q/(2\pi r\cdot 2h) = 2.65\,\mathrm{m/s}$, and the ratio $3/2$ is the parabolic profile's.

Sanity check. The proposal in (a) is the plane-channel parabola of Section 16 with an amplitude falling as $1/r$, which is what both the flow rate and continuity demand.

Example 12.3 (Compression stroke)**.** A piston moves toward the closed end of a cylinder at $V_p = 10\,\mathrm{m/s}$. At an instant when the gas column is $L = 80\,\mathrm{mm}$ long and its density is $1.2\,\mathrm{kg/m^3}$, assume the gas velocity varies linearly from zero at the closed end ($x = 0$) to $-V_p$ at the piston ($x = L$), and that the density is uniform in space. Find $\partial\rho/\partial t$.

Assumptions. One-dimensional; $u = -V_p x/L$; $\rho = \rho(t)$ only. The gas is compressible, so the full equation (2) is needed.

Continuity. With $\partial\rho/\partial x = 0$, $\partial(\rho u)/\partial x = \rho\,\partial u/\partial x = -\rho V_p/L$. So

$$\frac{\partial\rho}{\partial t} = \frac{\rho V_p}{L} = \frac{1.2\times10}{0.08} = 150\,\mathrm{kg/(m^3\,s)}.$$

Sanity check. The mass in the cylinder, ρLA, must stay fixed. Then $d(\rho L)/dt = 0$ gives $L\,d\rho/dt = -\rho\,dL/dt = \rho V_p$, the same answer from the control-volume balance. Here $\nabla\cdot\mathbf{V} = -125\,\mathrm{s^{-1}} \neq 0$, correctly: this field would be impossible for a liquid, and is fine for a gas.

Trap. Writing the integration constant as a number when it is a function of the other variables. Using the Cartesian $\partial v_r/\partial r$ in place of $\frac{1}{r}\partial(rv_r)/\partial r$, which rejects the source flow $v_r = C/r$ that is perfectly possible. Checking the divergence at one point and declaring the field possible. Using the incompressible form for a gas whose density changes in time. Forgetting the v_θ term when a field has swirl and θ-dependence.

Ladder.

1. White P4.20. An exponential profile; the other component from continuity. Integrating continuity with a constant of integration.
2. White P4.14. The most general purely swirling incompressible motion. Polar continuity.
3. White P4.22. Axisymmetric flow with axial velocity linear in z; the radial velocity. The integration constant is a function.
4. White P4.18. A piston compressing a gas with a linear velocity; the density history. Compressible continuity.
5. White P4.24. A channel with porous walls; the constant fixed by the suction. Continuity plus a boundary condition.

Answers are in the back of White; settle each call the same sitting.

Where it hides. Every exact solution in Section 16 starts here: "fully developed" plus continuity forces the cross-stream velocity to zero, which is what makes the Navier–Stokes equations solvable. The crib sheet's $\sum Q_{\text{node}} = 0$ for pipe networks and the Chapter 3 mass balance are this equation integrated over a volume. The boundary-layer displacement thickness on the sheet (section 8) is the outward push found in the first example. The stream function of Section 4.7 is a device that satisfies the 2-D incompressible equation automatically; it is not on this midterm.

13 The Navier-Stokes equations

The Navier–Stokes equations are Newton's second law, $m\mathbf{a} = \sum \mathbf{F}$, written for a tiny particle of fluid and divided by its volume. Mass becomes density, acceleration is the material derivative of Section 11, and the forces are of three kinds: pressure pushing on the faces, gravity pulling on the volume, and viscous friction between neighbouring layers that move at different speeds. Nothing deeper is going on. What makes the equations look hard is only that the friction term needs a law connecting stress to how the fluid deforms, and that cylindrical coordinates add a few terms. This section builds them so that they can be written from memory, then uses them in the most common exam question: given a velocity field, find the pressure.

Recognise it. "Find the pressure field", "find $\partial p/\partial x$", "what pressure gradient drives this flow", "does this velocity field satisfy the Navier–Stokes equations". Anything that names viscosity together with a velocity profile. A rotating flow where you need $p(r)$. "Neglect viscosity" or "inviscid" with a velocity field given: that is Euler's equation, the same thing with $\mu = 0$. Every problem in Section 16 uses this section's equations.

On the crib sheet. Not printed: no material derivative, no Navier–Stokes equation in any form, no Euler equation. The sheet prints the one-dimensional shear law $\tau = \mu\, du/dy$ (section 1), the angular deformation rate $\dot{\gamma}_{xy} = \partial u/\partial y + \partial v/\partial x$ (section 13), the rigid-body result $\nabla P = \rho(\mathbf{g} - \mathbf{a})$ (section 2), and the laminar pipe flow rate (section 6). The course has said the equations need not be memorised; it is possible the paper prints them, but plan on writing them yourself. The recipe at the end of the model takes under two minutes.

The model

Newton's law on a particle. Take a fluid particle of volume $dx\,dy\,dz$ and mass $\rho\,dx\,dy\,dz$. Its acceleration is $D\mathbf{V}/Dt$. Newton's law per unit volume is

$$\rho \frac{D\mathbf{V}}{Dt} = \mathbf{f}_{\text{body}} + \mathbf{f}_{\text{surface}},$$

where $\mathbf{f}$ means force per unit volume. The only body force here is gravity, $\mathbf{f}_{\text{body}} = \rho\mathbf{g}$, with $\mathbf{g}$ pointing down: $\mathbf{g} = -g\,\mathbf{k}$ if z is up, or $g\sin\theta$ down a slope at angle θ.

Stresses on the faces. A *stress* is a force per unit area on a surface. On each face of the box the neighbouring fluid pushes and drags. Name the stresses with two subscripts: σ_{ij} acts on a face whose outward normal points in direction i, and pushes in direction j. When $i = j$ it is a *normal stress* (push or pull straight on the face), written σ_{xx}; when $i \neq j$ it is a *shear stress* (drag along the face), written τ_{yx} and so on.

The x-force on the box comes from the three face pairs. On the pair of faces normal to x, the stress σ_{xx} acts at $x+dx$ and the opposite stress at x; by the same Taylor expansion as in Section 12, the net force is $\partial\sigma_{xx}/\partial x\; dx\,dy\,dz$. The y-faces contribute $\partial\tau_{yx}/\partial y$ and the z-faces $\partial\tau_{zx}/\partial z$. So the net surface force per volume in x is

$$f_{x,\text{surface}} = \frac{\partial\sigma_{xx}}{\partial x} + \frac{\partial\tau_{yx}}{\partial y} + \frac{\partial\tau_{zx}}{\partial z}.$$

Note what this says: a uniform stress produces no net force, because it pushes equally on opposite faces. Only differences across the particle accelerate it. That is why a pressure *gradient*, not a pressure, drives a flow.

The Newtonian fluid. A fluid at rest carries only pressure, which pushes inward on every face equally: $\sigma_{xx}=\sigma_{yy}=\sigma_{zz}=-p$, shear zero. When it moves, extra stresses appear that resist deformation. A *Newtonian* fluid is one in which those extra stresses are proportional to the *rate* of deformation, with one constant of proportionality, the viscosity μ. Water, air and oils are Newtonian; paint, blood and ketchup are not (the power law on the crib sheet is for them). The law is

$$\tau_{xy}=\tau_{yx}=\mu\Big(\frac{\partial u}{\partial y}+\frac{\partial v}{\partial x}\Big), \qquad \sigma_{xx}=-p+2\mu\frac{\partial u}{\partial x},$$

and the same with (y,z) and (z,x). The shear stress is μ times the angular deformation rate printed on the crib sheet: the rate at which a right angle drawn in the fluid closes up. For a plain shear flow $u(y)$ it is the sheet's $\tau=\mu\,du/dy$. The normal part says that stretching a particle ($\partial u/\partial x>0$) adds a tension. The factor 2 is not a separate assumption: a shear seen in axes turned 45° is a stretch along one diagonal and a squeeze along the other, and the law must give the same stresses in both sets of axes, which fixes the 2.

Substitute. Put the Newtonian stresses into the x-surface force and take μ constant:

$$f_{x,\text{surface}} = -\frac{\partial p}{\partial x} + \mu\Big(2\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}+\frac{\partial^2 v}{\partial x\,\partial y}+\frac{\partial^2 u}{\partial z^2}+\frac{\partial^2 w}{\partial x\,\partial z}\Big).$$

Split one $\partial^2 u/\partial x^2$ off the first term and regroup:

$$f_{x,\text{surface}} = -\frac{\partial p}{\partial x} + \mu\Big(\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}+\frac{\partial^2 u}{\partial z^2}\Big) + \mu\frac{\partial}{\partial x}\Big(\frac{\partial u}{\partial x}+\frac{\partial v}{\partial y}+\frac{\partial w}{\partial z}\Big).$$

The last bracket is $\nabla\cdot\mathbf{V}$, which is zero for an incompressible fluid. What remains uses the *Laplacian*, $\nabla^2 u=\partial^2u/\partial x^2+\partial^2u/\partial y^2+\partial^2u/\partial z^2$, the sum of the second derivatives. In plain words it measures how much u at a point differs from the average of u around it: positive where u is lower than its surroundings. Viscosity drags each particle toward the velocity of its neighbours, and $\mu\nabla^2 u$ is that drag per unit volume.

Theorem 13.1 (Navier–Stokes, incompressible, constant viscosity)**.**

$$\rho\frac{D\mathbf{V}}{Dt} = -\nabla p + \rho\mathbf{g} + \mu\nabla^2\mathbf{V}, \qquad \nabla\cdot\mathbf{V}=0.$$

In Cartesian components,

$$\rho\Big(\frac{\partial u}{\partial t}+u\frac{\partial u}{\partial x}+v\frac{\partial u}{\partial y}+w\frac{\partial u}{\partial z}\Big)=-\frac{\partial p}{\partial x}+\rho g_x+\mu\Big(\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}+\frac{\partial^2 u}{\partial z^2}\Big),$$
$$\rho\Big(\frac{\partial v}{\partial t}+u\frac{\partial v}{\partial x}+v\frac{\partial v}{\partial y}+w\frac{\partial v}{\partial z}\Big)=-\frac{\partial p}{\partial y}+\rho g_y+\mu\Big(\frac{\partial^2 v}{\partial x^2}+\frac{\partial^2 v}{\partial y^2}+\frac{\partial^2 v}{\partial z^2}\Big),$$
$$\rho\Big(\frac{\partial w}{\partial t}+u\frac{\partial w}{\partial x}+v\frac{\partial w}{\partial y}+w\frac{\partial w}{\partial z}\Big)=-\frac{\partial p}{\partial z}+\rho g_z+\mu\Big(\frac{\partial^2 w}{\partial x^2}+\frac{\partial^2 w}{\partial y^2}+\frac{\partial^2 w}{\partial z^2}\Big).$$

Four equations (three momentum, one continuity) for four unknowns (u, v, w, p). They are nonlinear because of the convective terms $u\,\partial u/\partial x$, which is why few exact solutions exist and why the ones that do are the flows where those terms vanish.

Euler's equation. Set $\mu = 0$: $\rho\, D\mathbf{V}/Dt = -\nabla p + \rho\mathbf{g}$. This is the frictionless case, good away from walls in high-Reynolds-number flows. Integrated along a streamline it gives Bernoulli. With $\mathbf{a} = D\mathbf{V}/Dt$ uniform it is the crib sheet's rigid-body line $\nabla P = \rho(\mathbf{g}-\mathbf{a})$. One fact connects the two: in an incompressible flow with no vorticity (Section 15), $\nabla^2\mathbf{V} = 0$ identically, so the viscous term vanishes even when $\mu \neq 0$, and Euler and Navier–Stokes give the same pressure.

Cylindrical forms. For pipes, annuli and rotating flows, use (r, θ, z) with velocities (v_r, v_θ, v_z). Write

$$\frac{D}{Dt}=\frac{\partial}{\partial t}+v_r\frac{\partial}{\partial r}+\frac{v_\theta}{r}\frac{\partial}{\partial\theta}+v_z\frac{\partial}{\partial z},\qquad \nabla^2=\frac{1}{r}\frac{\partial}{\partial r}\Big(r\frac{\partial}{\partial r}\Big)+\frac{1}{r^2}\frac{\partial^2}{\partial\theta^2}+\frac{\partial^2}{\partial z^2}.$$

Then

$$r:\quad \rho\Big(\frac{Dv_r}{Dt}-\frac{v_\theta^2}{r}\Big)=-\frac{\partial p}{\partial r}+\rho g_r+\mu\Big(\nabla^2 v_r-\frac{v_r}{r^2}-\frac{2}{r^2}\frac{\partial v_\theta}{\partial\theta}\Big),$$
$$\theta:\quad \rho\Big(\frac{Dv_\theta}{Dt}+\frac{v_r v_\theta}{r}\Big)=-\frac{1}{r}\frac{\partial p}{\partial\theta}+\rho g_\theta+\mu\Big(\nabla^2 v_\theta-\frac{v_\theta}{r^2}+\frac{2}{r^2}\frac{\partial v_r}{\partial\theta}\Big),$$
$$z:\quad \rho\frac{Dv_z}{Dt}=-\frac{\partial p}{\partial z}+\rho g_z+\mu\nabla^2 v_z.$$

Where the extra terms come from: the unit vectors $\mathbf{e}_r$ and $\mathbf{e}_\theta$ rotate as a particle moves around the axis, so differentiating $v_r\mathbf{e}_r + v_\theta\mathbf{e}_\theta$ produces terms the components alone miss. On the left they are the centripetal $-v_\theta^2/r$ and the angular-momentum term $v_r v_\theta/r$ of Section 11. On the right they are the corrections that stop the viscous term from charging friction to motions that do not deform the fluid. The z equation has none, because $\mathbf{e}_z$ never turns.

Memory hook. Two motions involve no deformation of the fluid, so their viscous term must be zero, and the extra terms are exactly what makes it so. Rigid rotation, $v_\theta = \Omega r$: $\nabla^2(\Omega r) = \Omega/r$, and subtracting $v_\theta/r^2 = \Omega/r$ gives zero. Radial outflow from a line source, $v_r = C/r$: $\nabla^2(C/r) = C/r^3$, and subtracting v_r/r^2 gives zero. If you forget the sign or the power, check it on these two. The $\partial/\partial\theta$ cross terms vanish in every axisymmetric problem on this midterm.

For axisymmetric flows ($\partial/\partial\theta = 0$) the viscous terms combine into a form that integrates directly:

$$\nabla^2 v_\theta-\frac{v_\theta}{r^2}=\frac{\partial}{\partial r}\Big[\frac{1}{r}\frac{\partial(rv_\theta)}{\partial r}\Big]+\frac{\partial^2 v_\theta}{\partial z^2},\qquad \nabla^2 v_z=\frac{1}{r}\frac{\partial}{\partial r}\Big(r\frac{\partial v_z}{\partial r}\Big)+\frac{\partial^2 v_z}{\partial z^2}.$$

Pressure from a given velocity field. Rearrange each component as a formula for the pressure gradient:

$$\frac{\partial p}{\partial x} = -\rho a_x + \rho g_x + \mu \nabla^2 u, \qquad \frac{\partial p}{\partial y} = -\rho a_y + \rho g_y + \mu \nabla^2 v, \qquad \frac{\partial p}{\partial z} = \cdots$$

Then:

1. Check continuity. If the divergence is not zero, stop: the field is not an incompressible flow.
2. Compute a_x, a_y, a_z (Section 11) and the Laplacians of each component.
3. Write $\partial p/\partial x$, $\partial p/\partial y$, $\partial p/\partial z$.
4. Check *compatibility*: $\partial/\partial y$ of $\partial p/\partial x$ must equal $\partial/\partial x$ of $\partial p/\partial y$, because mixed partial derivatives of a smooth p are equal. If they differ, no pressure field exists, so the given velocity field does not satisfy Navier–Stokes.
5. Integrate $\partial p/\partial x$ in x, adding an unknown $f(y, z)$. Differentiate the result in y, set it equal to $\partial p/\partial y$, and solve for f; repeat for z.
6. Fix the remaining constant with the pressure given at one point.

Writing it down from nothing, in under two minutes.

1. Say the sentence: density times particle acceleration equals pressure-gradient force plus gravity plus viscous force. $\rho\, D\mathbf{V}/Dt = -\nabla p + \rho\mathbf{g} + \mu\nabla^2\mathbf{V}$.
2. Left side, x line: $\partial u/\partial t$ plus u, v, w times $\partial u/\partial x, \partial u/\partial y, \partial u/\partial z$.
3. Right side, x line: $-\partial p/\partial x + \rho g_x + \mu$ times the three second derivatives of u. Copy for v and w.
4. Check the sign of the pressure term: flow goes from high to low pressure, so with everything else zero, $\partial p/\partial x < 0$ must push fluid toward $+x$.
5. Cylindrical: the z line is the Cartesian one with the cylindrical D/Dt and ∇^2. For r and θ add $-v_\theta^2/r$ and $+v_r v_\theta/r$ on the left and $-v_r/r^2$ or $-v_\theta/r^2$ inside the viscous bracket on the right. That is the whole correction when nothing depends on θ; otherwise add the $\pm(2/r^2)\,\partial/\partial\theta$ terms of the full forms above. Check on rigid rotation.

First moves. State incompressible, constant μ, and the direction of gravity in your axes; write g_x, g_y, g_z explicitly. Write the full component equations once, then strike terms with a reason under each. For a pressure question, follow the six steps above and do the compatibility check before integrating. Give the pressure as $p(x, y, z)$ with the constant fixed, then the number asked for, in Pa. If the field has no t and is linear in x, y, the viscous terms vanish; say so.

Worked examples

Example 13.2 (Pressure in a corner flow)**.** Water ($\rho = 1000\,\mathrm{kg/m^3}$) flows in the horizontal plane inside a 60° corner with

$$u = a(x^2 - y^2), \qquad v = -2axy, \qquad w = 0, \qquad a = 0.5\,\mathrm{m^{-1}s^{-1}}.$$

The pressure at the corner $(0, 0)$ is p_0. Find $p(x, y)$ and the pressure difference $p - p_0$ at $(2, 1)\,\mathrm{m}$.

Assumptions. Steady, incompressible, constant μ; the plane is horizontal, so gravity acts only in z and p has a hydrostatic part in z that is the same at every (x, y) and can be left out.

Walls. On $y = 0$, $v = 0$. On the line $y = \sqrt{3}\,x$, $v/u = -2\sqrt{3}x^2/(x^2 - 3x^2) = \sqrt{3}$, so the flow is along the line. Both are streamlines, and the field fits a 60° corner.

Continuity. $\partial u/\partial x + \partial v/\partial y = 2ax - 2ax = 0$. Possible.

Acceleration. $a_x = u\,\partial u/\partial x + v\,\partial u/\partial y = a(x^2 - y^2)(2ax) + (-2axy)(-2ay) = 2a^2x(x^2 + y^2)$. Likewise $a_y = 2a^2y(x^2 + y^2)$.

Viscous terms. $\nabla^2 u = 2a - 2a = 0$ and $\nabla^2 v = 0$. Viscosity does not affect the pressure in this flow, even though the fluid is viscous.

Pressure gradient. $\partial p/\partial x = -2\rho a^2 x(x^2 + y^2)$, $\partial p/\partial y = -2\rho a^2 y(x^2 + y^2)$. Compatibility: $\partial/\partial y$ of the first is $-4\rho a^2 xy$, and $\partial/\partial x$ of the second is $-4\rho a^2 xy$. Equal, so a pressure exists.

Integrate. In x: $p = -\rho a^2\left(\frac{x^4}{2} + x^2y^2\right) + f(y)$. Then $\partial p/\partial y = -2\rho a^2 x^2 y + f'(y)$, which must equal $-2\rho a^2 x^2 y - 2\rho a^2 y^3$, so $f' = -2\rho a^2 y^3$ and $f = -\rho a^2 y^4/2 + p_0$. Together:

$$p = p_0 - \frac{\rho a^2}{2}(x^2 + y^2)^2.$$

At $(2, 1)$: $p - p_0 = -\frac{1000(0.25)}{2}(25) = -3125\,\text{Pa}$.

Sanity check. The speed squared is $u^2 + v^2 = a^2(x^2 + y^2)^2$, so the answer is $p_0 - \frac{1}{2}\rho V^2$, Bernoulli from a stagnation point at the corner, as it must be for a flow with no viscous force. At $(2, 1)$, $V = 2.5\,\text{m/s}$ and $\frac{1}{2}\rho V^2 = 3125\,\text{Pa}$.

Example 13.3 (A vertical slot: what the gauge reads)**.** A water–glycol mixture ($\rho = 1100\,\text{kg/m}^3$, $\mu = 0.010\,\text{Pa s}$) flows upward between two vertical plates $h = 2\,\text{mm}$ apart. A measured profile fits $u = 6\bar{U}(y/h)(1 - y/h)$, $v = w = 0$, with mean velocity $\bar{U} = 0.05\,\text{m/s}$; x is vertical (up) and y across the gap. Find the pressure gradient, the pressure difference between two taps $0.5\,\text{m}$ apart vertically, and how much of it is friction.

Assumptions. Steady, incompressible, constant μ, plates wide (no z dependence). $g_x = -g$, $g_y = 0$.

Continuity. $\partial u/\partial x = 0$ since $u = u(y)$; with $v = 0$ it is satisfied.

x-momentum, term by term. $\partial u/\partial t = 0$ (steady); $u\,\partial u/\partial x = 0$ (u does not change with x); $v\,\partial u/\partial y = 0$ ($v = 0$); $w\,\partial u/\partial z = 0$. So the left side is zero and

$$0 = -\frac{\partial p}{\partial x} - \rho g + \mu\frac{d^2u}{dy^2}, \qquad \frac{d^2u}{dy^2} = -\frac{12\bar{U}}{h^2} = -1.5 \times 10^5\,\text{s}^{-1}\text{m}^{-1}.$$

So $\partial p/\partial x = -\rho g - 12\mu\bar{U}/h^2 = -10\,791 - 1500 = -12\,291\,\text{Pa/m}$.

y-momentum. Every term is zero except $-\partial p/\partial y$, so $\partial p/\partial y = 0$: the pressure is uniform across the gap.

Answer. The lower tap reads $12\,291 \times 0.5 = 6150\,\text{Pa}$ more than the upper one. Of this, $\rho g(0.5) = 5400\,\text{Pa}$ is just the weight of the column and $750\,\text{Pa}$ is friction.

Sanity check. With no flow the gauge would read the hydrostatic $5400\,\text{Pa}$; flow upward needs extra, and the extra is positive. The wall shear is $\mu\,du/dy|_0 = 6\mu\bar{U}/h = 1.5\,\text{Pa}$ on each plate; two plates over $0.5\,\text{m}$ of height give $1.5\,\text{N}$ per metre of width, and the friction pressure difference times the gap, $750 \times 0.002 = 1.5\,\text{N}$ per metre, balances it.

Example 13.4 (A drain vortex)**.** Water swirls into a bathtub drain. Away from the drain core the swirl fits $v_\theta = K/r$ with $K = 0.02\,\text{m}^2/\text{s}$, v_r and v_z negligible. Far from

the drain the free surface is flat at atmospheric pressure. Find how far the surface dips at $r = 10\,\text{mm}$.

Assumptions. Steady, axisymmetric, incompressible; z up, so $g_r = 0$, $g_z = -g$; radial and vertical velocities neglected.

θ-equation. The left side $v_r v_\theta / r$ and Dv_θ/Dt vanish (no v_r, no t, no θ dependence); $\partial p/\partial\theta = 0$ by symmetry. Viscous: $\partial_r[\frac{1}{r}\partial_r(r v_\theta)] = \partial_r[\frac{1}{r}\partial_r K] = 0$. Satisfied identically.

r-equation. Only the centripetal term and the pressure survive: $-\rho v_\theta^2/r = -\partial p/\partial r$ (the viscous term is the same zero). So $\partial p/\partial r = \rho K^2/r^3$.

z-equation. $\partial p/\partial z = -\rho g$, hydrostatic.

Integrate. $p = -\rho K^2/(2r^2) - \rho g z + C$. On the free surface $p = p_{\text{atm}}$; far away the surface is at $z = 0$, so $C = p_{\text{atm}}$. Then the surface is $z_s = -K^2/(2gr^2)$. At $r = 0.01\,\text{m}$: $z_s = -(0.02)^2/(2 \times 9.81 \times 10^{-4}) = -0.204\,\text{m}$.

Sanity check. The speed there is $2\,\text{m/s}$, and $V^2/2g = 0.204\,\text{m}$: the dip is the velocity head, Bernoulli along the surface, because this flow has no vorticity away from the axis (Section 15). A $20\,\text{cm}$ dip at $1\,\text{cm}$ means the core is air: the model fails before $r = 0$, where it predicts infinite speed.

Trap. Sign of the pressure term: it is $-\nabla p$ on the right. Forgetting gravity, or putting it in the wrong component when x is vertical or along a slope. Dropping $\mu\nabla^2\mathbf{V}$ because "the flow is inviscid" when the problem gives μ. Integrating $\partial p/\partial x$ and adding a constant instead of $f(y)$. Skipping the compatibility check and reporting a pressure for a field that has none. In cylindrical coordinates, using the Cartesian Laplacian $\partial^2/\partial r^2$, which misses $\frac{1}{r}\partial/\partial r$, or forgetting $-v_\theta^2/r$.

Ladder.

1. White P4.28. A linear plane field: continuity, Navier–Stokes, the pressure field. Pressure from a given velocity field.
2. White P4.30. A nozzle field: the same three questions. The same, with convective terms that survive.
3. White P4.36. A given parabolic film profile on an incline; the constant and the flux. Checking a profile against the equations.
4. White P4.32. Which swirl profiles solve the equations exactly, and what they mean. The θ equation in cylindrical coordinates.

Answers are in the back of White; settle each call the same sitting.

Where it hides. Every result in Section 16, including the crib sheet's laminar $Q = \pi D^4 \Delta P/(128\mu L)$ and $f = 64/\text{Re}$, is this equation with most terms struck. Making it dimensionless in Chapter 5 divides the convective term by the viscous one and produces the Reynolds number; the pressure term gives the Euler number and gravity the Froude number, all on the sheet. The boundary-layer equations behind the sheet's flat-plate correlations are this equation with the small terms dropped. The rigid-body and hydrostatic lines of section 2 are the case $\mathbf{V}$ uniform or zero.

14 Boundary conditions

The Navier–Stokes equations are the same for a pipe, a bearing and a river. What makes the answers different is what happens at the edges: which surfaces are solid, which move, which are open to air. Mathematically, integrating a second-order

equation leaves two unknown constants per direction, and each needs one fact about the boundary to fix it. In an exam, the boundary conditions are where most of the marks for an exact solution sit, because they are where the physics of the particular device enters. This section lists them, says why each holds, and shows the one condition that is easy to forget: at the centreline of a pipe.

Recognise it. Any problem that integrates a velocity profile, which is every problem in Section 16. "Determine the constants in the profile". A proposed profile, with the question whether it satisfies the conditions at the walls. A moving belt, plate, shaft, piston or cylinder. A free surface (a film, a puddle, a layer open to air). Two immiscible liquids in layers. A symmetric channel where you are told, or may choose, to solve only half.

On the crib sheet. One line, in section 13: "No slip at a wall: $V_{\text{fluid}} = V_{\text{wall}}$; use the stated profile form and wall conditions to determine constants." That is the most important condition. No-penetration, the free-surface conditions, the interface conditions, symmetry and the centreline condition are not printed.

The model

Counting. A momentum equation like $\mu\, d^2u/dy^2 = dp/dx$ is second order in y: integrating twice gives $u = \frac{1}{2\mu}\frac{dp}{dx}y^2 + C_1 y + C_2$, so it needs two conditions in y, usually one at each boundary. In cylindrical coordinates the same is true in r. A first-order equation such as continuity needs one. Before applying conditions, count constants and count conditions; they must match.

Solid walls: no slip and no penetration. At a solid surface, the fluid touching it moves with it.

- *No slip*: the tangential velocity of the fluid equals the tangential velocity of the wall. A fixed wall: $u = 0$. A plate sliding at U: $u = U$. A shaft of radius a spinning at Ω: $v_\theta = \Omega a$ at $r = a$. A rod pulled axially at U: $v_z = U$ at its surface.
- *No penetration*: the normal velocity of the fluid equals the normal velocity of the wall, because fluid cannot pass through a solid. A fixed wall: $v = 0$. A wall moving toward the fluid at W, like the platen in Section 11: $v = -W$. A porous wall with suction: v equals the given suction velocity.

Together: $\mathbf{V}_{\text{fluid}} = \mathbf{V}_{\text{wall}}$ at the wall. No penetration is plain geometry. No slip is an experimental fact for ordinary liquids and gases: molecules at the surface are trapped momentarily and leave with the wall's average velocity. It fails only for rarefied gases and some exotic surfaces (the "slip flows" at the end of White's problem list, not on this midterm). Inviscid (Euler) theory can keep only no penetration, because without viscosity the equations are lower order and there is no way to satisfy both.

Inlet and outlet. Where fluid enters or leaves the domain, the velocity profile or the pressure is usually given. In a long duct, exact solutions sidestep this by assuming *fully developed* flow: far enough from the inlet that the profile no longer changes along the duct. Then the inlet and outlet enter only through the pressure difference between them, Δp over the length L, which sets $dp/dx = -\Delta p/L$.

Free surface. A surface between a liquid and a gas that is free to move.

- *Pressure*: the liquid pressure equals the gas pressure, usually atmospheric. (Surface tension adds Υ/R on a curved surface; for a flat film it is zero.)
- *Shear*: the shear stress in the liquid equals the shear stress applied by the gas. Air is about 50 times less viscous than water, so unless a wind is stated, take it as zero: $\mu\,\partial u/\partial y = 0$, so $\partial u/\partial y = 0$ at the surface. The velocity profile meets a free surface with zero slope.
- *Kinematic*: no fluid crosses the surface. For a flat steady film this is $v = 0$ there.

Note what is *not* imposed: the velocity of the surface. It is an output.

Liquid–liquid interface. At the flat interface between two immiscible fluids 1 and 2:

- the velocities are equal, $u_1 = u_2$ (no slip between the fluids);
- the shear stresses are equal, $\mu_1\,\partial u_1/\partial y = \mu_2\,\partial u_2/\partial y$, by Newton's third law on a sheet of zero mass;
- the pressures are equal (flat interface, no surface tension jump).

Because the shear stresses match and the viscosities differ, the velocity gradients jump: the less viscous fluid shears more. A free surface is the special case where fluid 2 has negligible viscosity.

Symmetry planes. If the geometry and the driving are symmetric about the plane $y = 0$, the velocity is an even function of y, so $\partial u/\partial y = 0$ on the plane. That condition lets you solve the half-channel, $0 \le y \le h$, with no slip at $y = h$ and zero slope at $y = 0$, in place of no slip at both walls. The two give the same answer; use whichever is less algebra, and state it.

The centreline in cylindrical coordinates. In a pipe or around any axis, integrating the z momentum equation gives

$$\frac{1}{r}\frac{d}{dr}\Big(r\frac{dv_z}{dr}\Big) = \frac{1}{\mu}\frac{dp}{dz} \quad\Longrightarrow\quad v_z = \frac{1}{4\mu}\frac{dp}{dz}r^2 + C_1\ln r + C_2.$$

A pipe has only one wall, at $r = R$, but two constants. The second condition comes from the axis: at $r = 0$ the fluid is ordinary fluid, so its velocity is finite. Since $\ln r \to -\infty$ as $r \to 0$, the only way to keep v_z finite is $C_1 = 0$. Equivalent statements: the shear stress $\mu\,dv_z/dr$ is finite at the axis (the $\ln r$ term would give C_1/r), or $dv_z/dr = 0$ at $r = 0$ by symmetry. This is the condition that finishes Hagen–Poiseuille flow.

The condition holds *only if the axis is in the fluid*. In an annulus, $a \le r \le b$, the axis is inside the solid core, $\ln r$ is perfectly finite over the domain, and C_1 is fixed by the second wall. Killing the $\ln r$ term in an annulus is a wrong answer. The same applies to B/r in the swirl solution $v_\theta = Ar + B/r$: zero if the fluid fills the axis, finite and essential between two cylinders.

First moves. Draw the cross-section, label every boundary, and write one line per boundary: which condition, and why ("fixed wall, no slip", "free surface, no shear from air", "axis in fluid, finite velocity"). Count the constants after integrating and check the count matches. Use the wall's velocity in the fluid's coordinate direction, with its sign. After solving, substitute the profile back into every condition; it takes thirty seconds and catches most algebra slips.

Worked examples

Example 14.1 (An oil layer over water, dragged by a belt)**.** In a skimmer, a belt moving at $U = 0.5\,\mathrm{m/s}$ runs along the top of a layer of oil ($\mu_2 = 0.1\,\mathrm{Pa\,s}$, thickness $h_2 = 1\,\mathrm{mm}$) that floats on water ($\mu_1 = 0.001\,\mathrm{Pa\,s}$, thickness $h_1 = 2\,\mathrm{mm}$) above a fixed floor. Find the interface velocity and the shear stress on the belt.

Assumptions. Steady, fully developed, wide, no pressure gradient along the flow, gravity normal to the flow. Then each layer obeys $\mu\, d^2u/dy^2 = 0$, so each profile is a straight line, with two constants per layer: four in all.

Conditions (y up from the floor). Floor, no slip: $u_1(0) = 0$. Interface, equal velocity: $u_1(h_1) = u_2(h_1) = u_i$. Interface, equal shear: $\mu_1 u_1' = \mu_2 u_2'$. Belt, no slip: $u_2(h_1 + h_2) = U$. Four conditions.

Solve. With straight lines, $u_1' = u_i/h_1$ and $u_2' = (U - u_i)/h_2$. Shear continuity:

$$\frac{\mu_1 u_i}{h_1} = \frac{\mu_2 (U - u_i)}{h_2} \quad \Longrightarrow \quad u_i = U\,\frac{\mu_2/h_2}{\mu_1/h_1 + \mu_2/h_2} = 0.5\,\frac{100}{0.5 + 100} = 0.4975\,\mathrm{m/s}.$$

The shear is the same in both layers: $\tau = \mu_1 u_i/h_1 = 0.001(0.4975)/0.002 = 0.249\,\mathrm{Pa}$. The belt must pull with $0.249\,\mathrm{N}$ per square metre of contact.

Sanity check. The oil is a hundred times more viscous and half as thick, so it is far stiffer in shear and moves almost as a block with the belt; nearly all the velocity change happens in the water, whose shear rate is $249\,\mathrm{s^{-1}}$ against $2.5\,\mathrm{s^{-1}}$ in the oil. The layers act like two resistors in series, h/μ each, with the same "current" τ; the formula is the voltage divider.

Example 14.2 (A push–pull cable in its sheath)**.** The inner wire of a control cable, radius $a = 1.00\,\mathrm{mm}$, slides axially at $U = 0.3\,\mathrm{m/s}$ inside a fixed sheath of bore radius $b = 1.05\,\mathrm{mm}$, over a length $L = 0.5\,\mathrm{m}$. The gap is filled with oil, $\mu = 0.3\,\mathrm{Pa\,s}$. Find the velocity profile and the viscous force resisting the wire.

Assumptions. Steady, incompressible, fully developed ($v_z = v_z(r)$), axisymmetric, no swirl, no imposed pressure gradient (the ends are open to the same pressure), gravity neglected (horizontal, or its effect is hydrostatic only).

Equation. The z-momentum equation reduces to $\frac{1}{r}\frac{d}{dr}\big(r\,\frac{dv_z}{dr}\big) = 0$, so $v_z = C_1 \ln r + C_2$.

Conditions. Wire, no slip: $v_z(a) = U$. Sheath, no slip: $v_z(b) = 0$. The axis is inside the wire, not in the fluid, so the $\ln r$ term stays.

Solve. $C_2 = -C_1 \ln b$, then $U = C_1 \ln(a/b)$:

$$v_z = U\,\frac{\ln(r/b)}{\ln(a/b)}.$$

Force. The shear stress is $\tau_{rz} = \mu\, dv_z/dr = \mu U/(r \ln(a/b))$, negative because $\ln(a/b) < 0$: it opposes the wire. At the wire, $|\tau| = \mu U/(a \ln(b/a))$, and the force is $|\tau| \cdot 2\pi a L$:

$$F = \frac{2\pi\mu U L}{\ln(b/a)} = \frac{2\pi(0.3)(0.3)(0.5)}{\ln 1.05} = \frac{0.2827}{0.04879} = 5.80\,\mathrm{N}.$$

Sanity check. Substituting $r = a$ and $r = b$ returns U and 0. The gap is thin ($c = 0.05\,\mathrm{mm} \ll a$), so it should behave like plane Couette flow: $\tau \approx \mu U/c = 1800\,\mathrm{Pa}$ on area $2\pi a L$ gives $5.65\,\mathrm{N}$, within 3%. Dropping the $\ln r$ term, as one would in a pipe, would leave $v_z = C_2$ constant, unable to meet both walls.

Example 14.3 (Wind on a thin puddle)**.** Wind exerts a steady shear stress $\tau_s = 0.02\,\text{Pa}$ on a water layer $h = 2\,\text{mm}$ deep on a level car park. Far from the edges, and ignoring the slight tilt of the surface that eventually builds up a back-pressure, find the surface velocity.

Assumptions. Steady, fully developed, no pressure gradient, laminar, $\mu = 0.001\,\text{Pa s}$. Then $d^2u/dy^2 = 0$ and $u = C_1 y + C_2$.

Conditions. Ground, no slip: $u(0) = 0$, so $C_2 = 0$. Free surface with an applied shear: $\mu\, du/dy = \tau_s$ at $y = h$, so $C_1 = \tau_s/\mu$. Here the free-surface shear is not zero, because the air is the driver.

Answer. $u = \tau_s y/\mu$, and the surface moves at $u(h) = \tau_s h/\mu = 0.02(0.002)/0.001 = 0.04\,\text{m/s}$.

Sanity check. The shear is uniform through the layer, so the profile is linear; with $\tau_s = 0$ it gives $u = 0$, a still puddle, as it should. In a closed basin the surface tilts until a pressure gradient pushes water back along the bottom; that adds a parabola and is the Couette-with-pressure-gradient case of Section 16.

Trap. Applying "finite at $r = 0$" in an annulus. Putting $u = 0$ at a free surface, which is no slip at a place with no wall. Using equal velocity gradients at an interface in place of equal shear stresses. Forgetting the wall's own velocity when the wall moves, or its sign when it moves against the flow. Fewer conditions than constants, discovered only after an hour of algebra: count first.

Ladder.

1. White P4.24. A channel with porous walls. A nonzero normal velocity at a wall.
2. White P4.80. A film draining down a vertical wall. No slip at the wall, no shear at the free surface.
3. White P4.88. A rod sliding inside a fixed tube. A moving wall in cylindrical coordinates.
4. White P4.94. A cylinder spinning in fluid at rest far away. A condition at infinity.

Answers are in the back of White; settle each call the same sitting.

Where it hides. Every profile in Section 16 is a general solution plus these conditions. The no-slip condition is the reason boundary layers exist (crib sheet section 8): the wall forces the velocity to zero, and the layer is where the flow adjusts. The finite-at-the-axis argument returns wherever cylindrical coordinates do, including heat conduction in a rod. The interface condition is behind the two-fluid manometers of Chapter 2 only in the static form, equal pressure; in motion the shear matches too.

15 Vorticity and irrotationality

A small particle of fluid carried along by a flow does three things at once: it moves, it changes shape, and it spins. The spin is measured by the vorticity. It is the property that separates flows where viscosity matters from flows where it can be ignored, and its surprises are worth learning early: fluid going round in circles need not spin, and fluid moving in straight lines can. This section defines the spin of a particle from two small line segments, gives the formula, and splits the motion near a point into its pieces.

Recognise it. "Find the vorticity", "the angular velocity of a fluid element", "the rate of rotation", "is this flow irrotational?", "is the flow rotational?". Questions asking for the strain rates or the rate of angular deformation of a given field. A vortex, swirl or stirred tank, and the question whether the fluid particles rotate. A shear flow, with the question whether particles moving in straight lines rotate.

On the crib sheet. Printed, in section 13: the vorticity $\zeta = \nabla \times \mathbf{V}$ with $\zeta_z = \partial v/\partial x - \partial u/\partial y$, the element rotation $\Omega = \frac{1}{2}\zeta$ with $\Omega_z > 0$ counterclockwise, the normal strain rates $\dot\epsilon_{xx} = \partial u/\partial x$, $\dot\epsilon_{yy} = \partial v/\partial y$, and the angular deformation $\dot\gamma_{xy} = \partial u/\partial y + \partial v/\partial x$. This is the best-covered section of the chapter. Not printed: the x and y components of the curl, and the cylindrical form, both below.

The model

The spin of a particle. Draw a small cross on the fluid at a point: a segment dx along x and a segment dy along y. In a time dt the right end of the x-segment moves up, relative to its left end, by $(\partial v/\partial x)\,dx\,dt$, so the segment turns counterclockwise through the angle $(\partial v/\partial x)\,dt$. The top end of the y-segment moves right, relative to its bottom, by $(\partial u/\partial y)\,dy\,dt$, so that segment turns *clockwise* through $(\partial u/\partial y)\,dt$. The two arms may turn by different amounts, which is deformation. Their average counterclockwise turning rate is the particle's angular velocity:

$$\Omega_z = \frac{1}{2}\Big(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}\Big).$$

The difference of their turning rates is the rate at which the right angle closes: $\dot\gamma_{xy} = \partial u/\partial y + \partial v/\partial x$, the crib sheet's angular deformation.

Vorticity, the curl. The *vorticity* is twice that angular velocity, $\zeta = 2\Omega$. The factor 2 is a convention that makes it equal to the *curl* of the velocity, $\nabla \times \mathbf{V}$, which is the vector built from the cross-derivatives:

$$\zeta = \nabla \times \mathbf{V} = \Big(\frac{\partial w}{\partial y} - \frac{\partial v}{\partial z}\Big)\mathbf{i} + \Big(\frac{\partial u}{\partial z} - \frac{\partial w}{\partial x}\Big)\mathbf{j} + \Big(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}\Big)\mathbf{k}.$$

Memory: the z component is the 2-D formula; the others follow by cycling $x \to y \to z \to x$ and $u \to v \to w \to u$. In words, the curl at a point measures how much the velocity circulates around it: put a tiny paddle wheel with its axle along $\mathbf{k}$ at the point, and it turns at $\Omega_z = \zeta_z/2$. In cylindrical coordinates the axial component is

$$\zeta_z = \frac{1}{r}\frac{\partial (r v_\theta)}{\partial r} - \frac{1}{r}\frac{\partial v_r}{\partial \theta},$$

with the same r-inside-the-derivative pattern as cylindrical continuity.

The motion near a point, split. Near a point, the velocity differs from its value at the point by the velocity gradients times the displacement. In 2-D there are four gradients. Rewrite the two cross-derivatives as

$$\frac{\partial u}{\partial y} = \tfrac{1}{2}\dot\gamma_{xy} - \Omega_z, \qquad \frac{\partial v}{\partial x} = \tfrac{1}{2}\dot\gamma_{xy} + \Omega_z.$$

So every small motion is a sum of four parts: *translation* (the velocity at the point), *stretching* along x and y ($\dot\epsilon_{xx}$, $\dot\epsilon_{yy}$), *shear* ($\dot\gamma_{xy}$), and *rigid rotation* (Ω_z). The first three

are the *strain rate*, the rate at which shape and size change. For an incompressible fluid $\dot{\epsilon}_{xx}+\dot{\epsilon}_{yy}+\dot{\epsilon}_{zz}=0$, which is continuity again. The Newtonian stress of Section 13 depends only on the strain rate, never on the rotation: spinning a particle without deforming it costs no friction.

Irrotational flow. A flow is *irrotational* where $\zeta = 0$, and *rotational* where it is not. Two consequences matter now. First, for an incompressible irrotational flow, $\nabla^2\mathbf{V} = 0$, so the viscous term in Navier–Stokes vanishes even though the fluid is viscous, and the pressure follows from Bernoulli between any two points, not just along a streamline. Second, vorticity is created only by viscous shear at walls (and by some body forces); a flow that starts from rest and stays away from walls stays irrotational. That is why the region outside a thin boundary layer is modelled as irrotational. An irrotational flow can be written as the gradient of a scalar, the velocity potential of Section 4.9; that is not on this midterm.

Three flows that test the idea.

- *Rigid rotation*, $v_\theta = \omega_0 r$, as in a tank spun up to steady speed: $\zeta_z = \frac{1}{r}\frac{d}{dr}(\omega_0 r^2) = 2\omega_0$. Every particle spins at the tank's angular speed, as a painted cross on the fluid would show.
- *Free vortex*, $v_\theta = K/r$, as around a drain: $\zeta_z = \frac{1}{r}\frac{d}{dr}(K) = 0$ for $r > 0$. Particles go round in circles but do not spin: a paddle wheel carried round keeps its orientation, because the faster inner fluid turns it one way exactly as much as the curvature of the path turns it the other.
- *Couette flow*, $u = Uy/h$ between a fixed and a sliding plate: $\zeta_z = -\partial u/\partial y = -U/h$. The streamlines are straight, yet every particle spins clockwise at $U/2h$, because its top moves faster than its bottom. Every viscous flow next to a wall is rotational.

Whether a flow is rotational has nothing to do with whether its streamlines are curved.

First moves. Check continuity first, if the question allows it; an impossible field makes the rest moot. Write the four (2-D) or nine (3-D) velocity gradients as a small table. Then $\zeta_z = \partial v/\partial x - \partial u/\partial y$ and the other components by cycling. "Irrotational" needs every component zero at every point: compute them as functions. Give the element angular velocity as $\Omega = \zeta/2$ with a direction (counterclockwise for positive ζ_z, with z out of the page). For a swirl, use the cylindrical formula; never the Cartesian one with r substituted for x.

Worked examples

Example 15.1 (A PIV snapshot)**.** Particle-image velocimetry in a mixing chamber gives, near a point taken as the origin, the linear fit

$$u = 3x + 4y, \qquad v = 2x - 3y \qquad (u, v \text{ in m/s}, x, y \text{ in m}).$$

(a) Is the fit consistent with an incompressible flow? (b) Find the vorticity, the element rotation, the strain rates and the angular deformation. (c) What change to the coefficient 4 would make the flow irrotational? (d) Is $\mathbf{V} = yz\,\mathbf{i} + xz\,\mathbf{j} + xy\,\mathbf{k}$ irrotational?

Gradients (all in s^{-1}). $\partial u/\partial x = 3$, $\partial u/\partial y = 4$, $\partial v/\partial x = 2$, $\partial v/\partial y = -3$.

(a) $\partial u/\partial x + \partial v/\partial y = 3 - 3 = 0$. Consistent.

(b) $\zeta_z = 2 - 4 = -2\,\mathrm{s}^{-1}$. $\Omega_z = -1\,\mathrm{rad/s}$: particles spin clockwise at $1\,\mathrm{rad/s}$. $\dot{\epsilon}_{xx} = 3\,\mathrm{s}^{-1}$ (stretching along x), $\dot{\epsilon}_{yy} = -3\,\mathrm{s}^{-1}$ (squeezing along y), $\dot{\gamma}_{xy} = 4 + 2 = 6\,\mathrm{s}^{-1}$.

(c) Replace 4 by b; $\zeta_z = 2 - b = 0$ needs $b = 2$. Continuity does not involve b, so the result is still incompressible.

(d) $\zeta_x = \partial(xy)/\partial y - \partial(xz)/\partial z = x - x = 0$; $\zeta_y = \partial(yz)/\partial z - \partial(xy)/\partial x = y - y = 0$; $\zeta_z = \partial(xz)/\partial x - \partial(yz)/\partial y = z - z = 0$. Irrotational everywhere. (Its divergence is $0 + 0 + 0$, so it is also incompressible.)

Sanity check. Using the split, $\partial u/\partial y = \frac{1}{2}(6) - (-1) = 4$ and $\partial v/\partial x = \frac{1}{2}(6) + (-1) = 2$, which recovers the data. The shear is three times the rotation here: the particles deform much faster than they spin.

Example 15.2 (Vorticity across a channel)**.** Oil flows between parallel plates at $y = \pm h$, $h = 5\,\mathrm{mm}$, with the laminar profile $u = u_m(1 - y^2/h^2)$, $u_m = 0.3\,\mathrm{m/s}$, $v = 0$. Find the vorticity distribution and the spin of a particle at each wall and on the centreline.

Vorticity. $\zeta_z = \partial v/\partial x - \partial u/\partial y = 0 + 2u_m y/h^2$. It is linear in y, zero on the centreline, largest at the walls.

Values. At $y = +h$: $\zeta_z = 2u_m/h = 2(0.3)/0.005 = 120\,\mathrm{s}^{-1}$, so $\Omega_z = 60\,\mathrm{rad/s}$ counterclockwise. At $y = -h$: $60\,\mathrm{rad/s}$ clockwise. At $y = 0$: no spin.

Sanity check. Near the top wall the fluid below is faster than the fluid above, so a paddle wheel there has its bottom pushed forward and its top held back: counterclockwise, as found. The centreline is a symmetry plane with zero slope (Section 14), so no spin there. The flow is rotational everywhere else even though every streamline is straight. And the wall vorticity is the wall shear divided by μ, which is how the wall makes vorticity.

Example 15.3 (A stirred tank)**.** A small impeller makes a swirl modelled as rigid rotation for $r \le R$ and a free vortex outside: $v_\theta = \omega_s r$ for $r \le R$, $v_\theta = \omega_s R^2/r$ for $r > R$, with $\omega_s = 10\,\mathrm{rad/s}$ and $R = 20\,\mathrm{mm}$. Find the vorticity in each region, the speed at $r = 50\,\mathrm{mm}$, and check the circulation.

Vorticity, cylindrical, with $v_r = 0$. Core: $\zeta_z = \frac{1}{r}\frac{d}{dr}(\omega_s r^2) = 2\omega_s = 20\,\mathrm{s}^{-1}$. Outside: $\zeta_z = \frac{1}{r}\frac{d}{dr}(\omega_s R^2) = 0$. The velocity is continuous at $r = R$, where both give $\omega_s R = 0.2\,\mathrm{m/s}$, the fastest point.

At $50\,\mathrm{mm}$: $v_\theta = 10(0.02)^2/0.05 = 0.08\,\mathrm{m/s}$.

Circulation. The circulation round a closed loop is $\Gamma = \oint \mathbf{V} \cdot d\mathbf{s}$, the velocity along the loop summed round it. Stokes' theorem says it equals the total vorticity passing through the loop. For a circle of any radius $r > R$: $\Gamma = 2\pi r\, v_\theta = 2\pi\omega_s R^2 = 0.0251\,\mathrm{m^2/s}$, independent of r. The vorticity through it is $\zeta_z \cdot \pi R^2 = 20\pi(0.02)^2 = 0.0251\,\mathrm{m^2/s}$. They agree.

Sanity check. The circulation outside does not grow with radius because no vorticity lives outside the core. All the "rotation" a large loop sees is concentrated in the small core, which is the picture of a vortex in a real tank: a spinning core and an outer region that circulates without spinning.

Trap. Calling a flow rotational because its streamlines are circles, or irrotational because they are straight. Dropping the factor $\frac{1}{2}$ between vorticity and element rotation, or applying it twice. Sign: $\zeta_z = \partial v/\partial x - \partial u/\partial y$, not the reverse; positive is counterclockwise. Using $\zeta_z = \partial v_\theta/\partial r$ in a swirl, which misses v_θ/r and turns rigid rotation's $2\omega_0$ into ω_0. Checking irrotationality at one point.

Ladder.

1. White P4.28. The linear field again: compute its vorticity yourself (the book's answer does not give it). The curl in two dimensions.
2. White P4.60. A drain vortex with swirl falling as $1/r$; rotational or not, then the surface depth. Irrotational, so Bernoulli across streamlines. The book's answer writes ω where the field's constant K is meant.
3. White P4.94. Flow outside a spinning cylinder. A viscous flow that is irrotational.
4. White P4.32. The two exact swirl profiles. Rigid rotation against the free vortex.

Answers are in the back of White; settle each call the same sitting.

Where it hides. The viscous term of Section 13 is $-\mu\nabla\times\zeta$ for an incompressible fluid, which is why irrotational flows feel no net viscous force and obey Bernoulli everywhere. The exact flows of Section 16 are all rotational, and their wall vorticity is the wall shear over μ. Rigid rotation is the crib sheet's $z=\omega^2r^2/2g$ paraboloid; the free vortex is the outer part of every drain and tornado. The boundary layers of crib sheet section 8 are the thin rotational regions next to walls, with an irrotational flow outside.

16 Exact viscous flows

The Navier–Stokes equations are nonlinear and in general unsolvable by hand. A handful of flows are simple enough that the hard terms vanish exactly, and those are the ones examined: flow between plates, through a pipe, down a slope and between rotating cylinders. Each is solved by the same procedure, and the procedure is the thing being marked. Every term you strike needs a reason, and each reason is a modelling call: a statement about the flow that you are choosing to believe. This section sets out the procedure once, then runs it on six flows. The results are worth knowing, but the derivation is what earns the marks, and it is short enough to redo under time.

Recognise it. Viscous fluid between two walls, one or both moving, with or without a pressure difference: a bearing, a seal, a slideway, a piston clearance, a belt. Flow in a long pipe or tube at low Reynolds number. A liquid film on a wall, open to air. Fluid between a rotating shaft and a housing, or in a cup viscometer. The phrases "fully developed", "laminar", "steady", "derive the velocity profile", "find the flow rate per unit width", "find the torque", "find the force needed to pull the plate".

On the crib sheet. Only the end result for one case is printed: laminar pipe flow, $Q=\pi D^4\Delta P/(128\mu L)$, $f=64/\mathrm{Re}$, valid for $\mathrm{Re}<2300$, with the entry length $L_{e,\mathrm{lam}}\approx 0.06\,\mathrm{Re}\,D$ (section 6). Also printed: $\tau=\mu\,du/dy$ (section 1) and no slip (section 13). Not printed: the velocity profile of any flow, the Couette and channel results, the film, the rotating cylinders, and the Navier–Stokes equations they come from (rebuild those from Section 13). The pipe line on the sheet is a check on your derivation, not a substitute for it.

The model

The vocabulary.

- *Laminar*: the fluid moves in smooth layers, with no random eddies. All exact solutions are laminar; turbulent flow has no exact solution.
- *Parallel flow*: every streamline is a straight line in one direction, say x, so the only nonzero velocity component is u.

- *Fully developed*: the velocity profile no longer changes along the flow, $\partial u/\partial x = 0$. True far from an inlet (beyond the entry length on the sheet).
- *Two-dimensional*: nothing depends on the third coordinate z, because the walls are wide. *Axisymmetric*: nothing depends on θ.

The procedure.

1. *List the assumptions*, each as a sentence: steady; incompressible; constant viscosity; laminar; fully developed; 2-D or axisymmetric; gravity included or not, and in which direction.
2. *Continuity*: show that the assumptions force the cross-stream velocity to zero.
3. *Strike terms* in each Navier–Stokes component, writing the reason under each.
4. *Pressure*: use the cross-stream equation to show p varies across the flow only hydrostatically, then argue dp/dx is a constant.
5. *Integrate* the remaining ordinary differential equation twice.
6. *Boundary conditions* (Section 14): one per constant.
7. *Results*: the profile, then whatever is asked, flow rate, maximum and mean velocity, wall shear, force or torque, then a sanity check.

Steps 2 to 4 for any plane parallel flow. Take walls parallel to x, separated in y, wide in z. Two-dimensional: $w = 0$ and $\partial/\partial z = 0$. Fully developed: $\partial u/\partial x = 0$. Continuity then reads $\partial v/\partial y = 0$, so v does not change across the gap; at a solid wall $v = 0$, so $v = 0$ everywhere. Hence $u = u(y)$ only.

Now the x-momentum equation, term by term:

$$\rho\Big(\underbrace{\frac{\partial u}{\partial t}}_{\text{steady}} + \underbrace{u\frac{\partial u}{\partial x}}_{\text{fully developed}} + \underbrace{v\frac{\partial u}{\partial y}}_{v=0} + \underbrace{w\frac{\partial u}{\partial z}}_{w=0}\Big) = -\frac{\partial p}{\partial x} + \rho g_x + \mu\Big(\underbrace{\frac{\partial^2 u}{\partial x^2}}_{\text{fully developed}} + \frac{d^2u}{dy^2} + \underbrace{\frac{\partial^2 u}{\partial z^2}}_{\text{2-D}}\Big).$$

The whole left side vanishes: this is why these flows are solvable. The fluid does not accelerate, so each layer is in equilibrium between pressure, gravity and the viscous force. The y-equation has every velocity term zero, leaving $\partial p/\partial y = \rho g_y$: across the gap the pressure is hydrostatic only.

Why dp/dx is a constant: the x-equation says $\partial p/\partial x - \rho g_x = \mu\, d^2u/dy^2$. The right side depends only on y. Differentiating $\partial p/\partial y = \rho g_y$ with respect to x shows $\partial p/\partial x$ does not depend on y; so the left side depends only on x. A function of x equal to a function of y for all x and y must be a constant. So in a fully developed flow the pressure falls linearly along the duct, $dp/dx = -\Delta p/L$, and the equation to solve is

$$\mu\frac{d^2u}{dy^2} = \frac{dp}{dx} - \rho g_x. \tag{5}$$

Gravity along the flow and a pressure gradient act the same way; together they are the gradient of the *piezometric pressure* $p + \rho g z_{\text{elev}}$ (the crib sheet's HGL times γ). For horizontal flow, $g_x = 0$.

In what follows $G = -dp/dx > 0$ when the pressure falls in the $+x$ direction. Shear stress on a wall is read off the profile: the fluid exerts on a wall below it the stress $+\mu\, du/dy$ at the wall, and on a wall above it $-\mu\, du/dy$, in the x direction. Force is stress times wetted area; torque is force times radius.

1. Plane Couette flow. Fluid between a fixed plate at $y = 0$ and a plate at $y = c$ sliding at U; no pressure gradient, gravity normal to the plates. Equation (5): $d^2u/dy^2 = 0$, so $u = C_1 y + C_2$. No slip: $u(0) = 0$ gives $C_2 = 0$; $u(c) = U$ gives $C_1 = U/c$.

$$u = U\frac{y}{c}, \qquad q = \int_0^c u\,dy = \frac{Uc}{2}, \qquad \tau = \mu\frac{U}{c} \text{ everywhere.}$$

Here q is the volume flow per unit width, in m^2/s. The shear is the same at every height, which is the crib sheet's $\tau = \mu\,du/dy$ with a constant slope: the setup that defines viscosity.

2. Plane Poiseuille flow. Fixed plates at $y = \pm h$ (gap $c = 2h$), driven by a pressure gradient. Integrating $\mu u'' = -G$ twice: $u = -\frac{G}{2\mu}y^2 + C_1 y + C_2$. No slip at $y = \pm h$, or symmetry ($u'(0) = 0$) plus no slip at $y = h$, gives $C_1 = 0$ and $C_2 = Gh^2/(2\mu)$:

$$u = \frac{G}{2\mu}(h^2 - y^2), \qquad u_{\max} = \frac{Gh^2}{2\mu}, \qquad q = \int_{-h}^{h} u\,dy = \frac{2Gh^3}{3\mu} = \frac{Gc^3}{12\mu}, \qquad \bar{u} = \frac{q}{2h} = \frac{2}{3}u_{\max},$$

and the wall shear $\tau_w = \mu|u'(\pm h)| = Gh$. Check by a force balance on the whole slab of fluid, length L, unit width: pressure force $\Delta p \cdot 2h$ equals shear on two walls $2\tau_w L$, so $\tau_w = h\,\Delta p/L = Gh$. The flow rate goes as the gap cubed: halve a leakage gap and the leak falls eightfold.

3. Couette flow with a pressure gradient. Fixed wall at $y = 0$, wall at $y = c$ moving at U, and a pressure gradient. The equation is linear, so the solution is the sum of the two above (with the plane-Poiseuille parabola rewritten for walls at 0 and c):

$$u = U\frac{y}{c} + \frac{G}{2\mu}\,y(c - y), \qquad q = \frac{Uc}{2} + \frac{Gc^3}{12\mu}.$$

Check: $u(0) = 0$, $u(c) = U$, and $\mu u'' = -G$. The shear on the fixed wall is $\tau_0 = \mu U/c + Gc/2$, and the force per area on the moving wall is $-\mu u'(c) = -\mu U/c + Gc/2$. If the pressure rises in the direction of motion ($G < 0$) strongly enough, $u'(0) < 0$ and fluid near the fixed wall flows backward; this starts when $G = -2\mu U/c^2$. Flow rate zero needs $G = -6\mu U/c^2$: that is a dead-ended slot pumped by a moving wall, the principle of a viscous pump and of a slider bearing's pressure build-up.

4. Hagen–Poiseuille flow in a pipe. A round pipe of radius $R = D/2$, axis along z. Assumptions: steady, incompressible, laminar, fully developed, axisymmetric, no swirl; horizontal (or replace p by the piezometric pressure). Continuity in cylindrical form with $\partial v_z/\partial z = 0$ and $v_\theta = 0$ gives $\frac{1}{r}\frac{d}{dr}(rv_r) = 0$, so rv_r is constant; it is zero at the wall, so $v_r = 0$ everywhere. The z-equation (Section 13) loses its entire left side for the same reasons as above, and

$$\frac{\mu}{r}\frac{d}{dr}\Big(r\frac{dv_z}{dr}\Big) = -G, \qquad G = \frac{\Delta p}{L}.$$

Multiply by r/μ and integrate: $r\,dv_z/dr = -Gr^2/(2\mu) + C_1$. Divide by r and integrate again:

$$v_z = -\frac{Gr^2}{4\mu} + C_1 \ln r + C_2.$$

The axis is in the fluid, so v_z is finite at $r=0$: $C_1=0$. No slip at $r=R$: $C_2 = GR^2/(4\mu)$. So

$$v_z = \frac{G}{4\mu}(R^2 - r^2), \qquad v_{\max} = \frac{GR^2}{4\mu}.$$

Flow rate: add up rings of area $2\pi r\,dr$,

$$Q = \int_0^R v_z\, 2\pi r\, dr = \frac{\pi G R^4}{8\mu} = \frac{\pi D^4 \Delta p}{128 \mu L},$$

which is the crib sheet's line. Mean velocity $V = Q/(\pi R^2) = GR^2/(8\mu) = \tfrac{1}{2} v_{\max}$. Wall shear $\tau_w = \mu |dv_z/dr|_R = GR/2 = D\Delta p/(4L)$; the force balance $\Delta p\, \pi R^2 = \tau_w\, 2\pi R L$ confirms it.

The friction factor. The crib sheet's head loss is $h_L = f\,(L/D)\,V^2/2g$, and for a horizontal pipe of constant diameter the energy equation gives $\Delta p = \rho g h_L$. From $V = GR^2/(8\mu)$, $\Delta p = 32\mu L V/D^2$. Then

$$f = \frac{\Delta p}{\rho g}\cdot\frac{D}{L}\cdot\frac{2g}{V^2} = \frac{2D}{\rho L V^2}\cdot\frac{32\mu L V}{D^2} = \frac{64\mu}{\rho V D} = \frac{64}{\mathrm{Re}}.$$

The other line on the sheet. Both hold only while the flow is laminar, $\mathrm{Re} < 2300$, and fully developed.

5. A film down an inclined plane. A liquid layer of uniform thickness h flows down a wide plate inclined at θ to the horizontal, open to air. Axes: x down the slope, y normal to the plate, out of it. Gravity has components $g_x = g\sin\theta$, $g_y = -g\cos\theta$. Assumptions: steady, laminar, fully developed (uniform h), 2-D, air shear negligible.

The y-equation gives $\partial p/\partial y = -\rho g\cos\theta$. At the surface $p = p_a$, so $p = p_a + \rho g\cos\theta\,(h-y)$. Since h does not change along x, neither does p: $\partial p/\partial x = 0$. The film is driven by gravity alone. Equation (5): $\mu u'' = -\rho g\sin\theta$. Conditions: no slip, $u(0)=0$; free surface, no shear, $u'(h)=0$. Integrating, $\mu u' = \rho g\sin\theta\,(h-y)$ (the constant chosen to make $u'(h)=0$), and again,

$$u = \frac{\rho g\sin\theta}{\mu}\Big(hy - \frac{y^2}{2}\Big), \qquad u_s = \frac{\rho g\sin\theta\, h^2}{2\mu}, \qquad q = \frac{\rho g\sin\theta\, h^3}{3\mu}, \qquad \bar u = \tfrac{2}{3}u_s.$$

Wall shear $\tau_w = \rho g h\sin\theta$: the downslope weight of the film per unit area, as a force balance says it must be. Usually q is known and h is wanted: $h = (3\mu q/(\rho g\sin\theta))^{1/3}$. The profile is half of the channel parabola: the free surface acts like the centreline of a channel twice as deep.

6. Flow between rotating cylinders. Fluid fills the gap between an inner cylinder of radius r_i rotating at Ω_i and a fixed outer cylinder of radius r_o, length L. Assumptions: steady, incompressible, laminar, long cylinders ($\partial/\partial z = 0$), axisymmetric, purely circular flow: $v_r = v_z = 0$, $v_\theta = v_\theta(r)$. Continuity is satisfied identically. The θ-equation keeps only the viscous term (no θ pressure gradient by symmetry, no acceleration along the circle), in its axisymmetric form:

$$0 = \mu\frac{d}{dr}\Big[\frac{1}{r}\frac{d(rv_\theta)}{dr}\Big] \quad\Longrightarrow\quad \frac{1}{r}\frac{d(rv_\theta)}{dr} = 2A \quad\Longrightarrow\quad v_\theta = Ar + \frac{B}{r}.$$

The first part is rigid rotation, the second a free vortex. The axis is not in the fluid, so B/r stays. No slip, $v_\theta(r_i) = \Omega_i r_i$ and $v_\theta(r_o) = 0$, gives

$$v_\theta = \frac{\Omega_i r_i^2}{r_o^2 - r_i^2}\Big(\frac{r_o^2}{r} - r\Big).$$

The r-equation then gives the pressure, $dp/dr = \rho v_\theta^2/r$: higher outward, holding each particle on its circle.

Shear stress and torque. In a swirl the shear stress is $\tau_{r\theta} = \mu\, r\, \frac{d}{dr}(v_\theta/r)$, not $\mu\, dv_\theta/dr$: the extra piece subtracts the rigid-rotation part, which deforms nothing and so carries no stress (Section 15). Here

$$\tau_{r\theta} = -\frac{2\mu\Omega_i r_i^2 r_o^2}{r^2(r_o^2 - r_i^2)}, \qquad T = |\tau_{r\theta}| \cdot 2\pi r L \cdot r = \frac{4\pi\mu\Omega_i L\, r_i^2 r_o^2}{r_o^2 - r_i^2}.$$

The torque does not depend on r: every cylindrical layer of fluid is in rotational equilibrium, so the torque the inner cylinder applies passes undiminished to the outer one. For a narrow gap $c = r_o - r_i \ll r_i$ it reduces to plane Couette: $\tau \approx \mu\Omega_i r_i/c$ on area $2\pi r_i L$ at arm r_i. Measuring T at known Ω_i gives μ; that is the rotating-cylinder viscometer. At high enough inner speed this flow becomes unstable and forms stacked ring vortices, so, like the others, the solution is exact but not always the flow you get.

First moves. Sketch the geometry with axes, gravity and wall velocities. Write the assumption list as numbered sentences, then point to them by number when striking terms. Use continuity to kill the cross-stream velocity. Write the one surviving equation, integrate twice with constants shown, count conditions, apply them, substitute back. Get q or Q by integrating the profile, wall shear from $\mu\, du/dy$ at the wall, force and torque from shear times area (times radius). Finish with a check: a force balance on the whole slab, a limit ($U \to 0$, $G \to 0$, thin gap), the crib sheet's pipe formula, and the Reynolds number to confirm laminar flow.

Worked examples

Example 16.1 (Leakage past a hydraulic piston)**.** A hydraulic piston of diameter $D = 50\,\text{mm}$ and length $L = 30\,\text{mm}$ runs in its cylinder with a radial clearance $c = 20\,\mu\text{m}$. Oil ($\mu = 0.04\,\text{Pa s}$, $\rho = 870\,\text{kg/m}^3$) is at $10\,\text{MPa}$ (gauge) on one side and at 0 on the other. The piston is moving toward the high-pressure side at $V = 0.1\,\text{m/s}$. Find the leakage flow and the viscous force on the piston.

Assumptions. (1) Steady, (2) incompressible, constant μ, (3) laminar (checked below), (4) fully developed along the piston, ignoring end effects, (5) the gap is thin, $c/R = 8 \times 10^{-4}$, so it is unrolled into a plane slot of width πD, (6) gravity negligible across a $20\,\mu\text{m}$ gap.

Set-up. Let x run along the gap from the high-pressure side toward the low, and y from the cylinder wall ($y = 0$, fixed) to the piston ($y = c$, moving at $-V$ since it moves toward high pressure). Then $G = \Delta p/L = 10^7/0.03 = 3.33 \times 10^8\,\text{Pa/m}$, and the solution is case 3 with $U = -V$:

$$u = -V\frac{y}{c} + \frac{G}{2\mu}\, y(c - y), \qquad q = -\frac{Vc}{2} + \frac{Gc^3}{12\mu}.$$

(Continuity, the struck terms and the constant pressure gradient are exactly as in the model, by assumptions 1, 4 and 5.)

Numbers. $Gc^3/(12\mu) = (3.33 \times 10^8)(8 \times 10^{-15})/0.48 = 5.56 \times 10^{-6}\,\text{m}^2/\text{s}$. $Vc/2 = 1.00 \times 10^{-6}\,\text{m}^2/\text{s}$. So $q = 4.56 \times 10^{-6}\,\text{m}^2/\text{s}$ and

$$Q = \pi D q = \pi(0.05)(4.56 \times 10^{-6}) = 7.16 \times 10^{-7}\,\text{m}^3/\text{s} = 0.043\,\text{L/min}.$$

Force. On the piston (the upper wall), per unit area, $-\mu u'(c) = \mu V/c + Gc/2 = 200 + 3333 = 3533\,\text{Pa}$, toward the low-pressure side. Over the area $\pi DL = 4.71 \times 10^{-3}\,\text{m}^2$: $F = 16.7\,\text{N}$.

Sanity check. Reynolds number of the gap flow, $\rho q/\mu = 870(4.56 \times 10^{-6})/0.04 = 0.10$: deeply laminar. The piston's motion drags oil back toward the chamber and cuts the leak by 18% from the $0.052\,\text{L/min}$ it would be at rest; a piston speed of $2q_p/c = 0.56\,\text{m/s}$ would stop it entirely. The viscous force is negligible beside the pressure force on the face, $10^7 \times \pi(0.025)^2 = 19.6\,\text{kN}$. Doubling the clearance would multiply the pressure-driven leak by eight, which is why such clearances are held to microns.

Example 16.2 (A drip-irrigation microtube)**.** A polyethylene microtube of inner diameter $D = 1.0\,\text{mm}$ and length $L = 0.60\,\text{m}$ feeds a plant from a tank whose water surface is $H = 1.5\,\text{m}$ above the tube outlet. Water at $20°\text{C}$: $\rho = 998\,\text{kg/m}^3$, $\mu = 1.00 \times 10^{-3}\,\text{Pa s}$. Estimate the delivery in litres per hour, and check the model.

Assumptions. (1) Steady, (2) incompressible, (3) laminar and fully developed over the whole length (both checked below), (4) the whole head is spent on pipe friction: the inlet loss and the kinetic energy leaving the outlet are neglected at first, (5) the tube is straight; with gravity, the driving pressure is the piezometric difference $\Delta p = \rho g H$.

Model. Hagen–Poiseuille, case 4: $Q = \pi D^4 \Delta p/(128\mu L)$ with $\Delta p = 998(9.81)(1.5) = 14\,690\,\text{Pa}$:

$$Q = \frac{\pi(10^{-3})^4(14\,690)}{128(10^{-3})(0.60)} = 6.01 \times 10^{-7}\,\text{m}^3/\text{s} = 2.16\,\text{L/h}.$$

Profile and shear. $V = Q/(\pi D^2/4) = 0.765\,\text{m/s}$, $v_{\max} = 2V = 1.53\,\text{m/s}$ on the axis, $\tau_w = D\Delta p/(4L) = 6.12\,\text{Pa}$.

Checks. $\text{Re} = \rho V D/\mu = 998(0.765)(0.001)/0.001 = 763 < 2300$: laminar. Then $f = 64/\text{Re} = 0.0838$ and $h_L = f(L/D)V^2/2g = 0.0838(600)(0.0298) = 1.50\,\text{m}$, the head we started with: the crib sheet's two laminar lines agree with each other. Entry length $0.06\,\text{Re}\,D = 46\,\text{mm}$, under 8% of the tube, so "fully developed" is fair. The exit kinetic energy, $\alpha V^2/2g$ with $\alpha = 2$ for a laminar profile, is $0.06\,\text{m}$, 4% of the head; including it in the energy equation and solving the resulting quadratic in V lowers the delivery to $2.08\,\text{L/h}$. So the estimate is 2.1 to $2.2\,\text{L/h}$, a typical emitter rating, and the dominant uncertainty is the tube's actual bore, which enters to the fourth power.

Example 16.3 (Rain sheeting off a solar panel)**.** Rain falls at $20\,\text{mm/h}$ on a panel inclined at $30°$, $2.0\,\text{m}$ long down the slope. Find the thickness of the water film and its surface speed at the lower edge.

Assumptions. (1) Steady rain, (2) all the rain landing on the panel runs off as a film, with the rate taken per unit panel area, (3) laminar, (4) the film thickens slowly along the panel, so at each station it is treated as locally fully developed with the local q (the film model of case 5), (5) no air shear, (6) water at $20°\text{C}$.

Flow per unit width at the lower edge. $q = (20 \times 10^{-3}/3600)(2.0) = 1.11 \times 10^{-5}\,\text{m}^2/\text{s}$.

Thickness. From $q = \rho g \sin\theta\, h^3/(3\mu)$:

$$h = \Big(\frac{3\mu q}{\rho g \sin\theta}\Big)^{1/3} = \Big(\frac{3(10^{-3})(1.11 \times 10^{-5})}{998(9.81)(0.5)}\Big)^{1/3} = 1.90 \times 10^{-4}\,\text{m} = 0.19\,\text{mm}.$$

Surface speed. $u_s = \frac{3}{2}\bar{u} = \frac{3}{2}q/h = 0.088\,\text{m/s}$, which equals $\rho g \sin\theta\, h^2/(2\mu)$, a check on the arithmetic.

Sanity check. The film Reynolds number $q/\nu = 11$ is far below the hundreds at which films turn turbulent, so laminar holds. Real films ripple with long surface waves at

Reynolds numbers this low, so the smooth film is a model of the mean thickness, not a photograph. The wall shear $\rho g h \sin\theta = 0.93\,\text{Pa}$ is the film's weight per area along the slope. Since $h \propto q^{1/3}$, the film is only $2^{1/3} = 1.26$ times thicker at the bottom than halfway down: rain doubles the flow and barely thickens the film.

Trap. Keeping $u\,\partial u/\partial x$ or striking $\mu\,d^2u/dy^2$; the first is zero by full development, the second is the whole physics. Sign of G: the flow goes toward lower pressure, so $u > 0$ needs $dp/dx < 0$. Using $\tau = \mu\,dv_\theta/dr$ in a swirl. Killing the $\ln r$ or B/r term between two cylinders. Taking $\partial p/\partial x \neq 0$ in a free film, or forgetting gravity's component along a slope or a vertical pipe. Using the mean velocity where the maximum is asked, or the gap c where the half-gap h belongs: $q = Gc^3/(12\mu) = 2Gh^3/(3\mu)$. Applying $Q = \pi D^4 \Delta P/(128\mu L)$ without checking Re < 2300.

Ladder.

1. White P4.36. A given film profile on an incline; the constant and the flux. The reduction, checked against an answer.
2. White P4.80. An oil film draining down a vertical wall; the profile. Cartesian, gravity-driven, free surface.
3. White P4.88. A rod sliding inside a fixed tube; the profile. Cylindrical, shear-driven, a $\ln r$ solution.
4. White P4.92. A tank emptying through a laminar pipe; the level history. Poiseuille inside a mass balance.
5. White P4.84. A film draining down the outside of a vertical rod; the profile and the flow. Cylindrical, gravity-driven, free surface. The mock uses it.
6. White P4.82. A cylinder spinning in still fluid; the equation, conditions and steady state. The θ equation with a condition at infinity.

Answers are in the back of White; settle each call the same sitting.

Where it hides. The crib sheet's laminar lines, $Q = \pi D^4 \Delta P/(128\mu L)$ and $f = 64/\text{Re}$, are Hagen–Poiseuille; the left end of the Moody chart is $f = 64/\text{Re}$, and the energy correction $\alpha = 2.0$ for laminar flow comes from integrating its parabola. The hydraulic diameter $D_h = 4A/P_w$ is a way of reusing pipe results in other ducts; for a wide slot it gives $D_h = 2c$. Plane Couette flow is the definition of viscosity in Chapter 1, and the rotating cylinders are how it is measured. In Chapter 5, the Reynolds number is the ratio of the struck convective terms to the kept viscous ones; these solutions hold exactly when that ratio is small enough for the flow to stay laminar. Chapter 6 picks up from Hagen–Poiseuille for turbulent pipe flow, and the flat-plate boundary layer of Chapter 7 is what the Couette profile becomes when the plate is not confined.

17 Dimensional homogeneity

Every term of a physical equation must carry the same dimensions, because adding a pressure to a velocity is as meaningless as adding metres to seconds. That one rule is the whole of this section and the foundation of the rest of chapter 5. It checks algebra in seconds, it tells you the units of a constant in a formula someone else fitted to data, and it exposes the formulas that only work in the units they were written in. The marks on an exam come from writing dimensions down cleanly, one quantity at a time, from its definition.

Recognise it. "Show that the equation is dimensionally consistent (homogeneous)." "What are the dimensions (or SI units) of the constant C?" "Convert this formula to

other units." A formula with a strange exponent such as $V^{1.85}$ or $h^{3/2}$ and a constant with no units stated. A list of quantities with "give the dimensions of each in the $\{MLT\Theta\}$ system". Any derived result you want to check before moving on, which is every derived result.

On the crib sheet. Section 16, the nomenclature table, gives SI units and dimensions for mass, area, diameter, second moment of area, velocity, acceleration, force, torque, pressure, power, specific weight, density, dynamic and kinematic viscosity, shear stress and surface tension. Not on it: flow rate, mass flow, angular velocity, frequency, energy, specific heat or bulk modulus, nor how to get a dimension from a definition when a question introduces a quantity the table lacks. The table below covers those and shows the derivation for each. The sheet uses Υ for surface tension, as White does; many books write σ.

The model

Primary dimensions. A *dimension* is the kind of quantity a number measures (length, time), as opposed to the *unit*, which is the particular yardstick (metre, foot). Mechanics needs three *primary dimensions*: mass M, length L and time T. Heat transfer adds temperature Θ. Every other dimension is built from these by multiplying and dividing, because every other quantity is defined from them by a formula. The notation $\{q\}$ means "the dimensions of q"; for example $\{V\} = LT^{-1}$. Pure numbers, ratios of two lengths, and angles in radians (an arc length divided by a radius) have dimensions $M^0L^0T^0$ and are called *dimensionless*.

The table, derived. Each line takes its definition and substitutes the dimensions already found. Read it as a chain: force needs acceleration, pressure needs force, viscosity needs pressure.

Quantity	Definition used	Dimensions
Velocity V	distance / time	LT^{-1}
Acceleration a	change of velocity / time	LT^{-2}
Force F	$F = ma$	MLT^{-2}
Pressure, stress p, τ	force / area	$ML^{-1}T^{-2}$
Density ρ	mass / volume	ML^{-3}
Specific weight γ	weight / volume $= \rho g$	$ML^{-2}T^{-2}$
Dynamic viscosity μ	$\tau = \mu\, du/dy$, so $\mu = \tau/(du/dy)$; $\{du/dy\} = T^{-1}$	$ML^{-1}T^{-1}$
Kinematic viscosity ν	$\nu = \mu/\rho$	L^2T^{-1}
Surface tension Υ	force / length of the line it pulls along	MT^{-2}
Energy, work E	force $\times$ distance	ML^2T^{-2}
Power P	energy / time, or force $\times$ velocity	ML^2T^{-3}
Torque $\mathcal{T}$	force $\times$ lever arm	ML^2T^{-2}
Volume flow rate Q	volume / time	L^3T^{-1}
Mass flow $\dot{m}$	mass / time $= \rho Q$	MT^{-1}
Angular velocity Ω	angle / time; the radian is dimensionless	T^{-1}
Frequency f	cycles / time	T^{-1}
Specific heat c_p, c_v	energy / (mass $\times$ temperature change)	$L^2T^{-2}\Theta^{-1}$
Gas constant R	$p = \rho RT$, so $R = p/(\rho T)$	$L^2T^{-2}\Theta^{-1}$
Bulk modulus K	$K = -dp/(d\mathcal{V}/\mathcal{V})$; the volume ratio is dimensionless	$ML^{-1}T^{-2}$
Thermal conductivity k	heat flux $= -k\, dT/dx$; heat flux is power/area	$MLT^{-3}\Theta^{-1}$

Three consequences. Torque and energy share dimensions but are different things (force times a perpendicular arm, force times a parallel displacement), so matching dimensions do not prove two quantities the same. Specific heat and the gas constant are velocity squared per degree, which is why $\sqrt{kRT}$ is a speed (the speed of sound; $k = c_p/c_v$ is dimensionless). Bulk modulus is a pressure, so $\sqrt{K/\rho}$ is also a speed. The SI unit of μ, kg/(m s), is the same as Pa s $=$ N s/m^2; the unit of Υ, N/m, is the same as kg/s^2.

The principle. An equation is *dimensionally homogeneous* when every additive term has the same dimensions. Every equation derived from physical laws is homogeneous, because the laws themselves are statements that hold whatever units are used: if an equation held in metres but not in feet, it would be a statement about the metre, not about the fluid. Two rules follow and both are used on exams.

1. Terms added, subtracted or set equal must match. Factors multiplied together just combine their exponents.
2. The argument of a function that is not a power (exp, ln, sin) must be dimensionless, and so is its value. The reason: $e^x = 1 + x + x^2/2 + \cdots$ adds x to x^2, which is only legal if x has no dimensions. A logarithm of a dimensional quantity, such as $\ln(p)$, only appears legitimately as a difference $\ln p_2 - \ln p_1 = \ln(p_2/p_1)$.

Derivatives and integrals carry the dimensions of what they divide or multiply by: $\{\partial u/\partial x\} = \{u\}/L$, $\{\partial^2 u/\partial y^2\} = \{u\}/L^2$, $\{\int p\, dA\} = \{p\}L^2$. Operators such as ∇ count as L^{-1} and D/Dt as T^{-1}.

Checking an equation. Write the dimensions of each term separately, simplify each to a single $M^aL^bT^c$, and compare. Take the steady Bernoulli equation per unit volume, $p + \frac{1}{2}\rho V^2 + \rho g z = \text{const}$. The three terms have dimensions $ML^{-1}T^{-2}$; $ML^{-3} \cdot L^2T^{-2} = ML^{-1}T^{-2}$; and $ML^{-3} \cdot LT^{-2} \cdot L = ML^{-1}T^{-2}$. They match, so each term is a pressure, an energy per unit volume. The factor $\frac{1}{2}$ is invisible to this check: homogeneity says nothing about pure numbers, signs, or whether a term that should be there is missing. It catches a wrong power of a variable, which is the commonest algebra slip.

The units of an empirical constant. An *empirical formula* is one fitted to measurements rather than derived. If someone writes $y = Cx^n$ and states the units of x and y, then C carries whatever units make the equation balance:

$$\{C\} = \frac{\{y\}}{\{x\}^n}.$$

If the formula is a correct physical law, the units of C will reduce to a combination of physical properties the problem should contain (for example $\{C\} = \{\rho\}$ or $\{C\} = \{\mu\}/L$), and C is really those properties times a pure number. If the units come out as something like $\mathrm{kg\,m^{-2.8}\,s^{-0.2}}$, the formula is dimensionally inconsistent: the constant has absorbed fluid properties and geometry that were held fixed in the test.

The pitfalls of inconsistent formulas. Such a formula is not wrong, but it is narrow.

1. *It is tied to its units.* Changing from SI to US units changes the numerical value of C, by a factor that you compute by substituting the unit conversions. Using the SI constant with feet gives a wrong number with no warning.
2. *It is tied to its fluid and geometry.* The properties absorbed into C were those of the test. Using the formula for a different fluid, temperature or size silently assumes those properties did not matter, which is usually false.
3. *It cannot be extrapolated.* A fitted exponent such as 1.8 is the local slope of a curve over the tested range. Outside that range the physics may change (laminar to turbulent, for example) and the slope with it.

The cure, when you know which properties matter, is to rewrite the formula in a homogeneous form with those properties shown and the constant made dimensionless. The first worked example does exactly this.

First moves. Write $\{\cdot\}$ for every symbol before doing anything else, building each from its definition. Reduce each additive term to one $M^aL^bT^c\Theta^d$. For an empirical constant, solve $\{C\} = \{y\}/\{x\}^n$, then state it in base SI units. For a unit conversion, substitute $x_{\text{SI}} = (\text{factor}) \times x_{\text{other}}$ into the formula and collect the number. Say in one line whether the formula is homogeneous; if not, say what it has absorbed. Remember that homogeneity is necessary, not sufficient.

Worked examples

Example 17.1. A supplier's datasheet for a woven-wire strainer in a water line gives the clean pressure drop as

$$\Delta p = 410\,V^{1.8}, \qquad \Delta p \text{ in Pa},\ V \text{ (approach velocity) in m/s},$$

fitted to tests with water at 20°C ($\rho = 998\,\mathrm{kg/m^3}$, $\mu = 1.0\times10^{-3}\,\mathrm{Pa\,s}$). The wire diameter is $d = 0.20\,\mathrm{mm}$. (a) Find the SI units of the constant 410 and say whether the formula is homogeneous. (b) Rewrite the formula for Δp in psi with V in ft/s. (c) Assuming Δp depends only on ρ, μ, V and d for a given weave, write the formula in a dimensionally homogeneous form and find the dimensionless constant. (d) Predict Δp at $V = 1.2\,\mathrm{m/s}$ for a glycol coolant with $\rho = 1040\,\mathrm{kg/m^3}$, $\mu = 3.5\times10^{-3}\,\mathrm{Pa\,s}$, and compare with the datasheet formula used naively.

Solution. (a) $\{C\} = \{\Delta p\}/\{V\}^{1.8}$, so in units C is $\mathrm{Pa/(m/s)^{1.8}} = \mathrm{kg\,m^{-1}s^{-2}\cdot m^{-1.8}s^{1.8}} = \mathrm{kg\,m^{-2.8}\,s^{-0.2}}$. The fractional powers of metres and seconds are the fingerprint of a constant that has silently absorbed fluid properties and a size (part (c) shows which). As written, with a bare number 410, the formula is not homogeneous: it is valid only in Pa and m/s, and only for water at 20°C on this weave.

(b) Substitute $V_{\mathrm{SI}} = 0.3048\,V_{\mathrm{ft/s}}$ and $\Delta p_{\mathrm{Pa}} = 6894.76\,\Delta p_{\mathrm{psi}}$:

$$6894.76\,\Delta p_{\mathrm{psi}} = 410\,(0.3048)^{1.8} V_{\mathrm{ft/s}}^{1.8} \quad\Longrightarrow\quad \Delta p_{\mathrm{psi}} = 0.00701\,V_{\mathrm{ft/s}}^{1.8}.$$

Check at $V = 1\,\mathrm{m/s} = 3.281\,\mathrm{ft/s}$: $0.00701\times3.281^{1.8} = 0.0595\,\mathrm{psi} = 410\,\mathrm{Pa}$, the SI value. The number changed by a factor of about $58\,000$; that is what "tied to its units" means.

(c) A homogeneous relation among $\Delta p, \rho, \mu, V, d$ can be written (by the Pi theorem of the next section, or simply by forming the obvious groups) as $\Delta p/(\rho V^2) = f(\mathrm{Re})$ with $\mathrm{Re} = \rho V d/\mu$. At fixed fluid and wire, $\mathrm{Re}\propto V$. The datasheet says $\Delta p \propto V^{1.8}$, so $f(\mathrm{Re})\,V^2 \propto V^{1.8}$, which forces $f(\mathrm{Re}) = K\,\mathrm{Re}^{-0.2}$ with K a pure number. Therefore

$$\Delta p = K\rho V^2\left(\frac{\rho V d}{\mu}\right)^{-0.2} = K\rho^{0.8}\mu^{0.2}d^{-0.2}V^{1.8}.$$

Dimension check of the right side: $M^{0.8}L^{-2.4}\cdot M^{0.2}L^{-0.2}T^{-0.2}\cdot L^{-0.2}\cdot L^{1.8}T^{-1.8} = ML^{-1}T^{-2}$, a pressure. Matching to the datasheet, $410 = K(998)^{0.8}(1.0\times10^{-3})^{0.2}(2.0\times10^{-4})^{-0.2}$, which gives $K = 1.18$. The dimensional constant 410 was this pure number times $\rho^{0.8}\mu^{0.2}d^{-0.2}$ for water and this wire.

(d) Homogeneous form, glycol: $\Delta p = 1.18\times1040^{0.8}\times(3.5\times10^{-3})^{0.2}\times(2.0\times10^{-4})^{-0.2}\times1.2^{1.8} = 756\,\mathrm{Pa}$. Naive datasheet: $410\times1.2^{1.8} = 569\,\mathrm{Pa}$. The naive use underestimates by about 25%, mostly because the coolant is 3.5 times more viscous.

Assumption to state: the power law fitted for water still holds in the glycol's Reynolds-number range. The water tests at $1\,\mathrm{m/s}$ had $\mathrm{Re} = 998\times1\times2\times10^{-4}/10^{-3}\approx200$; the glycol case has $\mathrm{Re} = 1040\times1.2\times2\times10^{-4}/3.5\times10^{-3}\approx71$, lower than the tested range if the tests stayed near $1\,\mathrm{m/s}$. The prediction is an extrapolation in Re and should be reported as such. □

Sanity check: a more viscous fluid at the same speed should give a larger drop, and the homogeneous form does; the naive form cannot respond to viscosity at all. The exponent 0.2 on μ is small because the flow is mostly inertial ($\Delta p\propto V^{1.8}$ is close to V^2); a purely viscous drop would go as V^1 and μ^1.

Example 17.2. A rotating-disk viscometer has a flat disk of radius R turning at angular velocity Ω a gap h above a fixed plate, with the gap filled by oil of viscosity μ and density ρ. Three students offer formulas for the torque needed to turn it:

$$\text{(A) } \mathcal{T} = \frac{\pi\mu\Omega R^4}{2h}, \qquad \text{(B) } \mathcal{T} = \frac{\pi\mu\Omega R^3}{2h^2}, \qquad \text{(C) } \mathcal{T} = \frac{\pi}{2}\rho\Omega^2R^5.$$

(a) Check each for homogeneity. (b) Of those that pass, which can be the viscous torque, and why? (c) For SAE 30 oil ($\mu = 0.29\,\mathrm{Pa\,s}$), $R = 40\,\mathrm{mm}$, $h = 0.50\,\mathrm{mm}$ at

$300\,\text{rpm}$, find the torque and the power dissipated. (d) The manufacturer instead quotes $\mathcal{T} = 2.53 \times 10^{-4}\, N$ with $\mathcal{T}$ in N m and N in rpm for this oil and disk. Find the units of the constant and check it against your answer.

Solution. (a) The target is $\{\mathcal{T}\} = ML^2T^{-2}$. (A) $\{\mu\Omega R^4/h\} = ML^{-1}T^{-1} \cdot T^{-1} \cdot L^4 \cdot L^{-1} = ML^2T^{-2}$. Homogeneous. (B) $\{\mu\Omega R^3/h^2\} = ML^{-1}T^{-1} \cdot T^{-1} \cdot L^3 \cdot L^{-2} = MT^{-2}$. This is a force per length, not a torque; (B) is wrong. Its error is two lengths short, the signature of a slipped power of R or h. (C) $\{\rho\Omega^2R^5\} = ML^{-3} \cdot T^{-2} \cdot L^5 = ML^2T^{-2}$. Homogeneous.

(b) Only (A). Formula (C) is homogeneous but contains no viscosity, so it predicts the same torque in air and in honey; a viscous torque must vanish when $\mu \to 0$. Homogeneity is necessary, not sufficient. (A) also has the right physics: shear stress $\tau = \mu\Omega r/h$ at radius r (linear velocity profile across the thin gap), acting on a ring of area $2\pi r\, dr$ with arm r, integrates to $\int_0^R \mu\Omega r/h \cdot 2\pi r^2\, dr = \pi\mu\Omega R^4/(2h)$.

(c) $\Omega = 300 \times 2\pi/60 = 31.4\,\text{rad/s}$. Then

$$\mathcal{T} = \frac{\pi(0.29)(31.4)(0.040)^4}{2(5.0 \times 10^{-4})} = 0.0733\,\text{N m}, \qquad P = \mathcal{T}\Omega = 0.0733 \times 31.4 = 2.30\,\text{W}.$$

Units: $\text{Pa s} \cdot \text{s}^{-1} \cdot \text{m}^4/\text{m} = \text{N m}^{-2} \cdot \text{m}^3 = \text{N m}$, and $\text{N m} \cdot \text{rad/s} = \text{W}$.

(d) $\{C\} = \text{N m/rpm}$. Since $1\,\text{rpm} = (2\pi/60)\,\text{rad/s}$ and the radian is dimensionless, $\text{N m/rpm} = (60/2\pi)\,\text{N m s} = 9.55\,\text{N m s}$, so in base units the constant is $2.53 \times 10^{-4} \times 9.55 = 2.42 \times 10^{-3}\,\text{kg m}^2/\text{s}$. Physically it is $\pi\mu R^4/(2h)$ for this oil and disk, which is $\pi(0.29)(0.040)^4/(2 \times 5 \times 10^{-4}) = 2.33 \times 10^{-3}\,\text{kg m}^2/\text{s}$; the quoted value is 4% higher, plausibly an edge correction measured at calibration. At $300\,\text{rpm}$ the quoted formula gives $0.0759\,\text{N m}$, against $0.0733\,\text{N m}$ from (A). □

Sanity check: the torque is fingertip-sized, and $2.3\,\text{W}$ into a few cubic centimetres of oil is why such instruments control temperature.

Trap. Treating torque and energy as the same because they share ML^2T^{-2}. Giving μ the dimensions of ν, or forgetting that $\{\Upsilon\} = MT^{-2}$ has no length in it. Forgetting that rpm and degrees must become rad/s before any formula with Ω is used, even though radians carry no dimension. Using a fitted constant in other units, or for another fluid, without converting. Declaring a formula correct because it is homogeneous: pure numbers, signs and missing terms are invisible to the check. Writing $\ln p$ or e^t with a dimensional argument.

Ladder.

1. White P5.10. The dimensions of four composite expressions. Dimensions from definitions.
2. White P4.20. The dimensions of the constants in a velocity profile. Units of constants in a formula.
3. White P5.12. Five variables, one group; a ratio of familiar groups. Building a group by hand.
4. White P5.8. Build the Archimedes number from four quantities. Same, with a density difference.

Answers are in the back of White; settle each call the same sitting.

Where it hides. Inside every other check on the exam: a control-volume momentum balance where one term has an extra ρ (chapter 3), a Navier–Stokes term with the

wrong number of derivatives (chapter 4), a velocity profile whose constant must carry units of s^{-1} or $m^{-1}s^{-1}$. In the Pi theorem (Section 18), where the table above is the first step. In pipe-flow and pump formulas quoted in US units with g_c or with "Q in gpm", which are inconsistent formulas in disguise.

18 The Pi theorem

A quantity that depends on n variables needs, in general, a table of experiments with $n-1$ independent knobs. The Pi theorem says that dimensional homogeneity alone cuts the number of knobs: the relation can always be rewritten among dimensionless groups, and there are fewer of those than variables, by exactly the number of independent dimensions the variables contain. On the exam it is a procedure worth most of a 15-mark question: list the variables, find how many independent dimensions they really contain, choose the repeating variables, solve small linear systems for the exponents, check, and rename. This section explains why the procedure works, defines the rank of a matrix from scratch, and does each step with its reason.

Recognise it. "Using dimensional analysis" or "using the Pi theorem / Buckingham's method" followed by "X depends on $a, b, c, \dots$". "Find the dimensionless groups", "how many dimensionless parameters govern …", "show that X can be written as $f(\mathrm{Re}, \dots)$". "Rewrite the result so that the unknown appears in only one group." Any list of five or more physical quantities with one singled out as depending on the rest.

On the crib sheet. Section 12: the number of groups is $n-r$, where n is the number of variables and r the rank of the dimensional matrix; repeating variables must be dimensionally independent; and the standard groups $\mathrm{Re} = \rho V L/\mu$, $\mathrm{Fr} = V/\sqrt{gL}$, $\mathrm{Ma} = V/a$, $\mathrm{We} = \rho V^2 L/\Upsilon$, $\mathrm{Eu} = \Delta p/(\rho V^2)$, $\mathrm{St} = \omega L/V$, $\mathrm{Ca} = \mu V/\Upsilon$, the cavitation number and the pressure coefficient. Not on the sheet: how to compute the rank, the procedure itself, the dimensions of variables outside the section 16 table (see Section 17), and how to turn the groups you get into the standard ones.

The model

Dimensionless products. Suppose the variables are $q_1, \dots, q_n$. A *product of powers* is $\Pi = q_1^{a_1} q_2^{a_2} \cdots q_n^{a_n}$ with numbers a_i. Its dimensions are found by adding exponents: if $\{q_j\} = M^{m_j} L^{l_j} T^{t_j}$, then

$$\{\Pi\} = M^{\sum a_j m_j}\, L^{\sum a_j l_j}\, T^{\sum a_j t_j}.$$

So Π is dimensionless exactly when three linear equations hold, one per dimension:

$$\sum_j a_j m_j = 0, \qquad \sum_j a_j l_j = 0, \qquad \sum_j a_j t_j = 0.$$

These are n unknowns a_j in three equations (four if Θ appears). Every result in this section is a statement about how many independent solutions such a system has.

The dimensional matrix. Write the exponents in a table with one row per dimension and one column per variable. That table is the *dimensional matrix*. For a drag force F on a sphere of diameter D in a stream of speed V, density ρ, viscosity μ:

	F	V	D	ρ	μ
M	1	0	0	1	1
L	1	1	1	-3	-1
T	-2	-1	0	0	-1

The three equations above are "each row, weighted by the a_j, sums to zero".

Rank, in plain words. The *rank* of a matrix is the number of its rows that are genuinely different pieces of information: the number left after you throw away every row that can be built by adding multiples of the other rows. (Counting independent columns instead gives the same number; that is a theorem of linear algebra, and it is why "rank" needs no "row" or "column" in front of it.) For the dimensional matrix, the rank r is the number of the balance equations that really constrain the exponents. If one row is a combination of the others, its equation is automatically satisfied whenever the others are, and it removes no freedom.

Two ways to compute it, both enough for exam-sized matrices.

1. *Elimination.* Add multiples of one row to another to create zeros, as when solving simultaneous equations. A row that becomes all zeros was redundant. The rank is the number of rows that do not vanish.
2. *Determinants.* The rank is the size of the largest square block, made of any chosen rows and columns, whose determinant is not zero. With three dimensions, find three columns whose 3×3 determinant is nonzero and the rank is 3. If every 3×3 determinant is zero, try 2×2 blocks.

Worked, for the sphere. Take the columns ρ, V, D:

$$\det \begin{pmatrix} 1 & 0 & 0 \\ -3 & 1 & 1 \\ 0 & -1 & 0 \end{pmatrix} = 1 \cdot \big(1 \cdot 0 - 1 \cdot (-1)\big) - 0 + 0 = 1 \neq 0.$$

A nonzero 3×3 determinant means rank 3, the most possible with three rows. So $n - r = 5 - 3 = 2$ groups (C_D and Re, as expected).

Worked, for a case where the rank is less than the number of dimensions. The force F needed to pull a thin wire ring of mean diameter D and wire diameter d off a liquid surface of surface tension Υ:

	F	Υ	D	d
M	1	1	0	0
L	1	0	1	1
T	-2	-2	0	0

Add twice the M row to the T row: $(-2+2, -2+2, 0, 0) = (0, 0, 0, 0)$. The T row was -2 times the M row: mass and time appear only in the fixed combination MT^{-2}, so balancing M automatically balances T. Two rows survive, so $r = 2$, even though three dimensions appear. The determinant test agrees: every 3×3 block contains the dependent rows and has determinant zero, while the block from rows M, L and columns Υ, D is $\det \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = 1$. Here $n - r = 4 - 2 = 2$ groups, not 1.

Why the theorem is true (a sketch). Let r be the rank and pick r of the variables, say $q_1, \ldots, q_r$, that are *dimensionally independent*: no product of powers of

them is dimensionless, which is the same as saying their columns have a nonzero $r \times r$ determinant. Because these columns already account for all r independent dimensions, every other variable q_k can be divided by a suitable product $q_1^{x_1} \cdots q_r^{x_r}$ to make it dimensionless; call the result Π_k. There are $n - r$ such Π_k. The physical relation $f(q_1, \ldots, q_n) = 0$ can therefore be rewritten as

$$g(q_1, \ldots, q_r, \Pi_{r+1}, \ldots, \Pi_n) = 0.$$

Now use the fact that the relation is a law of nature and holds in every system of units. Changing the size of the units of M, L and T changes the numerical values of $q_1, \ldots, q_r$ and, because they are independent, can be chosen to make each of them equal to 1, while every Π_k keeps its value (it has no dimensions to rescale). So g cannot depend on $q_1, \ldots, q_r$ at all, and the relation is $G(\Pi_{r+1}, \ldots, \Pi_n) = 0$, or, solving for the one containing the dependent variable, $\Pi_1 = \phi(\Pi_2, \ldots, \Pi_{n-r})$. The count $n - r$ is the number of independent solutions of the balance equations: n unknowns, r genuine constraints.

The repeating-variable method, step by step.

1. *List all n variables*, the dependent one included. The list is physics, not algebra: leave out a variable that matters and no algebra will put it back. Include g only if a free surface or buoyancy is involved; include Υ only if an interface curves; include the speed of sound (or the bulk modulus) only if the flow is compressible.

2. *Write each variable's dimensions* from its definition, and assemble the dimensional matrix.

3. *Find r, the rank*, by elimination or determinants. Do not just count the dimensions that appear: the ring shows they can differ. Then $n - r$ is the number of groups.

4. *Choose r repeating variables.* They must be dimensionally independent (check with a determinant), must between them contain every dimension that appears, and must not include the dependent variable (otherwise it would be smeared across every group, and you could not solve for it). Good habit: one geometric length, one velocity (or rate), one fluid property such as ρ.

5. *Form one group per remaining variable*: $\Pi = q_k\, q_1^a q_2^b q_3^c$. Write the M, L, T balances, which are r linear equations, and solve for a, b, c.

6. *Check each group is dimensionless* by substituting dimensions. It takes ten seconds and catches sign slips.

7. *Rename and rearrange* into standard groups, and write the result as $\Pi_1 = \phi(\Pi_2, \ldots)$ with the dependent variable in Π_1 only.

When the rank is less than the number of dimensions. Use r repeating variables, not the number of dimensions. With fewer than r you cannot make every group dimensionless; with more, you will find that no set of that size is independent (in the ring, any three of F, Υ, D, d contain either two lengths or a dimensionless product such as $F/(\Upsilon D)$). The balance equations for the redundant dimension are then satisfied automatically, which is a useful check. Equivalent shortcut: treat the fixed combination (MT^{-2} in the ring) as a single dimension.

Different repeating variables, same count. Any valid choice gives $n-r$ groups, but different groups. The two sets are always related by products and powers, so they contain the same information. Two rules make this precise and give you freedom.

1. Any power of a group is a group: Re^{-1} or $\sqrt{\mathrm{We}}$ is as good as Re or We.
2. A group may be replaced by its product with powers of the others, provided the set stays independent (each old group can still be recovered from the new set). For example $\Pi_1' = \Pi_1\Pi_2^2$ may replace Π_1.

Standard combinations made this way: $\mathrm{Ca} = \mathrm{We}/\mathrm{Re}$; the Ohnesorge number $\mathrm{Oh} = \mu/\sqrt{\rho\Upsilon L} = \sqrt{\mathrm{We}}/\mathrm{Re}$; the Bond number $\mathrm{Bo} = \rho g L^2/\Upsilon = \mathrm{We}/\mathrm{Fr}^2$; and the drag group $F\rho/\mu^2 = C_F\mathrm{Re}^2$, which is useful when the velocity is the unknown because V no longer appears in it.

Recognising standard groups. After solving, invert or take powers until the group matches the sheet. A group with μ and $\rho V L$ is a Reynolds number; one with g is a Froude number (often appearing as $gL/V^2 = \mathrm{Fr}^{-2}$); one with Υ and $\rho V^2 L$ is a Weber number; one with a pressure difference over ρV^2 is an Euler number or pressure coefficient; one with a frequency times L/V is a Strouhal number; one with the speed of sound or K/ρ is a Mach number ($V/\sqrt{K/\rho}$). Ratios of lengths, angles and roughness ratios ϵ/L are groups on their own.

The inspection method. For short lists you can write the groups down by knowing the standard ones: pick groups that each introduce one new variable not in the earlier ones (that guarantees independence), until there are $n-r$ of them, then check each is dimensionless. It is fast and acceptable on an exam if you state the count first, so the marker sees that you know how many to look for. White also describes a variant that removes one dimension at a time by dividing variables into each other; it lands on the same groups.

First moves. List every variable with its dimensions in a table. Build the dimensional matrix and compute the rank, showing the determinant or the elimination. State n, r and $n-r$. Choose repeating variables, show they are independent, and keep the dependent variable out. Solve the exponent equations one dimension at a time (start with the dimension that appears in only one repeating variable). Check every group. Rename to standard groups and write $\Pi_1 = \phi(\dots)$.

Worked examples

Example 18.1. In a drop-on-demand inkjet head, the diameter d of the main drop depends on the nozzle diameter D, the jet speed V, and the ink's density ρ, viscosity μ and surface tension Υ. (a) Find the dimensionless groups by the repeating-variable method and express them with standard names. (b) Evaluate them for $\rho = 1050\,\mathrm{kg/m^3}$, $\mu = 0.010\,\mathrm{Pa\,s}$, $\Upsilon = 0.035\,\mathrm{N/m}$, $D = 30\,\mu\mathrm{m}$, $V = 6\,\mathrm{m/s}$. (c) A new ink has twice the viscosity and the same ρ and Υ. What nozzle diameter and jet speed reproduce the same drop shape, and how large is the new drop if the old one was $d = 33\,\mu\mathrm{m}$?

Solution. (a) Variables: d (L), D (L), V (LT^{-1}), ρ (ML^{-3}), μ ($ML^{-1}T^{-1}$), Υ (MT^{-2}). $n = 6$. The columns ρ, V, D have the determinant 1 computed in the model above, so

$r = 3$ and there are 3 groups. Repeating variables ρ, V, D (independent, cover M, L, T, exclude d).

$\Pi_1 = d\,\rho^a V^b D^c$: M: $a = 0$; T: $-b = 0$; L: $1 + c = 0$. So $\Pi_1 = d/D$.

$\Pi_2 = \mu\,\rho^a V^b D^c$: M: $1 + a = 0$, $a = -1$; T: $-1 - b = 0$, $b = -1$; L: $-1 + 3 - 1 + c = 0$, $c = -1$. So $\Pi_2 = \mu/(\rho V D) = 1/\mathrm{Re}$.

$\Pi_3 = \Upsilon\,\rho^a V^b D^c$: M: $a = -1$; T: $-2 - b = 0$, $b = -2$; L: $0 + 3 - 2 + c = 0$, $c = -1$. So $\Pi_3 = \Upsilon/(\rho V^2 D) = 1/\mathrm{We}$.

Checks: $\{\mu/(\rho V D)\} = ML^{-1}T^{-1}/(ML^{-3} \cdot LT^{-1} \cdot L) = 1$; $\{\Upsilon/(\rho V^2 D)\} = MT^{-2}/(ML^{-3} \cdot L^2T^{-2} \cdot L) = 1$. Result: $d/D = \phi(\mathrm{Re}, \mathrm{We})$. Since Re contains V and so does We, the printing community prefers a speed-free fluid group, $\mathrm{Oh} = \sqrt{\mathrm{We}}/\mathrm{Re} = \mu/\sqrt{\rho \Upsilon D}$, giving the equivalent $d/D = \psi(\mathrm{We}, \mathrm{Oh})$.

(b) $\mathrm{Re} = 1050 \times 6 \times 30 \times 10^{-6}/0.010 = 18.9$; $\mathrm{We} = 1050 \times 36 \times 30 \times 10^{-6}/0.035 = 32.4$; $\mathrm{Oh} = 0.010/\sqrt{1050 \times 0.035 \times 30 \times 10^{-6}} = 0.301$, and $\sqrt{32.4}/18.9 = 0.301$ agrees.

(c) Same drop shape means the same Re and We. With ρ and Υ unchanged, We fixed means $V_2^2 D_2 = V_1^2 D_1$, and Re fixed with $\mu_2 = 2\mu_1$ means $V_2 D_2 = 2 V_1 D_1$. Divide the first by the second: $V_2 = V_1/2 = 3\,\mathrm{m/s}$. Then $D_2 = 2V_1 D_1/V_2 = 4D_1 = 120\,\mu\mathrm{m}$. Since d/D is unchanged, $d_2 = 4 \times 33 = 132\,\mu\mathrm{m}$. □

Sanity check: $\mathrm{Oh} = \mu/\sqrt{\rho \Upsilon D}$ with μ doubled and D quadrupled is unchanged, as it must be. A thicker ink needs a bigger, slower jet to behave the same, which costs resolution: the dimensional analysis has turned a material change into a design trade-off in one line.

Example 18.2. The ring example from the model: the force F to detach a horizontal wire ring of mean diameter D and wire diameter d from a liquid depends on Υ, D, d and the contact angle θ. (a) Find the rank and the groups. (b) A ring with $D = 20\,\mathrm{mm}$, $d = 0.4\,\mathrm{mm}$ needs $F = 9.1\,\mathrm{mN}$ in water ($\Upsilon = 0.0728\,\mathrm{N/m}$). Predict F for ethanol ($\Upsilon = 0.022\,\mathrm{N/m}$) with the same ring, both liquids wetting fully ($\theta = 0$). (c) In reality the ring also lifts a little meniscus of liquid, so ρ and g enter. Redo the count, find the new group and evaluate it for both liquids ($\rho = 998$ and $789\,\mathrm{kg/m^3}$). What does it say about the prediction in (b)?

Solution. (a) $n = 5$: $F, \Upsilon, D, d, \theta$. The angle is dimensionless, a column of zeros, which does not change the rank. The rank is 2, as computed in the model (T row $= -2 \times M$ row). So $5 - 2 = 3$ groups. Repeating variables Υ, D (independent: $\det \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ from the M, L rows).

$\Pi_1 = F\,\Upsilon^a D^b$: M: $1 + a = 0$; L: $1 + b = 0$; T: $-2 - 2a = 0$, which holds automatically with $a = -1$, as the rank said it would. $\Pi_1 = F/(\Upsilon D)$. $\Pi_2 = d/D$. $\Pi_3 = \theta$. Result: $F/(\Upsilon D) = \phi(d/D, \theta)$.

(b) Same ring and same θ, so d/D and θ are unchanged and so is Π_1: $F/(\Upsilon D) = 0.0091/(0.0728 \times 0.020) = 6.25$. For ethanol, $F = 6.25 \times 0.022 \times 0.020 = 2.75\,\mathrm{mN}$, the water force scaled by the ratio of surface tensions.

(c) Adding ρ (ML^{-3}) and g (LT^{-2}): the ρ column has M without T^{-2} and g has T without M, so the M and T rows are no longer proportional. The block of columns Υ, D, ρ has determinant $\det \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & -3 \\ -2 & 0 & 0 \end{pmatrix} = 1 \cdot (0) - 0 + 1 \cdot (0 + 2) = 2 \neq 0$, so $r = 3$ and $n - r = 7 - 3 = 4$ groups. The new one, with repeating Υ, D, ρ: $\Pi_4 = g\,\Upsilon^a D^b \rho^c$; T:

$-2-2a=0$, $a=-1$; M: $a+c=0$, $c=1$; L: $1+b-3c=0$, $b=2$. So $\Pi_4=\rho g D^2/\Upsilon$, the Bond number. Water: $998\times 9.81\times 0.020^2/0.0728=53.8$. Ethanol: $789\times 9.81\times 0.020^2/0.022=141$. The Bond numbers differ by a factor of 2.6, so the two tests are not fully similar; the prediction in (b) holds only to the extent that ϕ is insensitive to Bo. □

Sanity check: the ideal pull is surface tension along two contact circles, $F=2\pi D\Upsilon$, so $\Pi_1=2\pi=6.28$; the measured 6.25 is within 1%. Real ring tensiometers apply a correction table indexed by exactly the extra groups found in (c), which is why the count matters.

Example 18.3. The pressure drop Δp across a perforated plate in a duct depends on the approach velocity V, the fluid's ρ and μ, the hole diameter d, the plate thickness t and the open-area fraction ϕ. (a) Find the groups with repeating variables ρ, V, d. (b) Repeat with ρ, μ, d and show how the two sets are related. (c) A water test ($\rho=998\,\mathrm{kg/m^3}$, $\mu=1.0\times10^{-3}\,\mathrm{Pa\,s}$) with $d=3\,\mathrm{mm}$ gives $\Delta p=1.9\,\mathrm{kPa}$ at $V=0.8\,\mathrm{m/s}$. Assuming the loss coefficient is independent of Re in this range, find the velocity for $\Delta p=5.0\,\mathrm{kPa}$ and check the assumption is self-consistent.

Solution. (a) $n=7$, the dimensions M, L, T all appear, and ρ, V, d are independent (determinant 1), so $r=3$ and there are 4 groups. ϕ is already dimensionless; t/d is a length ratio. $\Pi_1=\Delta p\,\rho^a V^b d^c$: M: $a=-1$; T: $-2-b=0$, $b=-2$; L: $-1+3-2+c=0$, $c=0$. $\Pi_1=\Delta p/(\rho V^2)=\mathrm{Eu}$. $\Pi_2=\mu\,\rho^a V^b d^c$ gives $\mu/(\rho V d)=1/\mathrm{Re}$ exactly as in the first example. So $\mathrm{Eu}=\phi_1(\mathrm{Re}, t/d, \phi)$.

(b) With ρ, μ, d (determinant of columns $(1,-3,0),(1,-1,-1),(0,1,0)$ is $\det\begin{pmatrix}1&1&0\\-3&-1&1\\0&-1&0\end{pmatrix}=1\cdot(0+1)-1\cdot(0-0)+0=1$, independent): $\Pi_1'=\Delta p\,\rho^a\mu^b d^c$: T: $-2-b=0$, $b=-2$; M: $1+a+b=0$, $a=1$; L: $-1-3a-b+c=0$, $c=2$. $\Pi_1'=\Delta p\,\rho d^2/\mu^2$. $\Pi_2'=V\rho^a\mu^b d^c$: T: $-1-b=0$, $b=-1$; M: $a+b=0$, $a=1$; L: $1-3+1+c=0$, $c=1$. $\Pi_2'=\rho V d/\mu=\mathrm{Re}$. Still four groups. The relation: $\Pi_1'=\frac{\Delta p}{\rho V^2}\left(\frac{\rho V d}{\mu}\right)^2=\mathrm{Eu}\,\mathrm{Re}^2$. The second set has V only in Re, which is the form to use when V is the unknown and Δp is given.

(c) Test: $\mathrm{Re}=998\times0.8\times0.003/10^{-3}=2395$, $\mathrm{Eu}=1900/(998\times0.8^2)=2.97$. With Eu constant, $V=\sqrt{\Delta p/(\rho\,\mathrm{Eu})}=\sqrt{5000/(998\times2.97)}=1.30\,\mathrm{m/s}$, at $\mathrm{Re}=3890$. The new Re is 60% above the tested one and of the same order, so the assumption is a modest extrapolation rather than a jump across regimes; it is self-consistent, but only a test at $1.3\,\mathrm{m/s}$ can confirm it. Sharp-edged holes approach a constant loss coefficient as Re grows, which makes the error, if any, small. □

Sanity check: $\Delta p\propto V^2$ at constant Eu, so $V_2/V_1=\sqrt{5.0/1.9}=1.62$ and $0.8\times1.62=1.30\,\mathrm{m/s}$. In the second form, the new condition is $\Pi_1'=5000\times998\times(0.003)^2/(10^{-3})^2=4.49\times10^7$, against 1.71×10^7 for the test; if a full Π_1' versus Re curve were available, V could be read off directly without iteration.

Trap. Taking r as the number of dimensions that appear instead of the rank: the ring has three dimensions and rank 2. Including the dependent variable among the repeating variables. Choosing repeating variables that are not independent, such as two lengths, or ρ, V and ρV^2. Leaving out g in a free-surface problem or including it in a closed duct (there it only adds hydrostatic pressure). Forgetting that an angle or a ratio is already a group and counts toward n. Solving exponents with a sign slip and not checking. Writing $\phi(\mathrm{Re},\mathrm{We})$ and then claiming ϕ is known: the theorem gives the form of the relation, never the function.

Ladder.

1. White P5.18. Laminar flow in a triangular pore; the effect of doubling its size. One group is a scaling law.
2. White P5.32. Weir flow, given to be proportional to the width. A known proportionality removes a group.
3. White P5.38. Spray droplet size; Reynolds and Weber appear. Repeating variables with a surface tension.
4. White P5.20. Lift on a spinning cylinder in a crossflow. A rotation rate as a variable.
5. White P5.28. Beam deflection in a crossflow, simplified with given facts. Using physics to cut the group count.
6. White P5.24. Missile lift; the dimensional matrix and its rank. Rank and Mach.
7. White P5.26. A pendulum's period; what the analysis cannot say about amplitude. Where dimensional analysis stops.

Answers are in the back of White; settle each call the same sitting.

Where it hides. In every similarity question (Section 20), whose first step is a Pi analysis. In "find the form of the law" questions where the count gives a single group, so the group is a constant: a pendulum period $\propto \sqrt{\ell/g}$, a capillary wave speed. In chapter 6 pipe flow, where $f = \phi(\mathrm{Re}, \epsilon/D)$ is a Pi result, and in drag and lift coefficients. In heat-transfer and mixing problems with Θ in the matrix, where the rank may become 4.

19 Nondimensional equations

The Pi theorem finds groups from a list of variables you guessed. The second route starts from the equations of motion themselves: measure every length, speed, time and pressure in units natural to the problem, and the equations rewrite themselves with a few dimensionless numbers in front of their terms. Those numbers are the Reynolds, Froude, Euler and Strouhal numbers, and their sizes tell you which terms matter before you solve anything. This is how one decides that a flow is creeping, inviscid, quasi-steady or gravity-free, and it is the justification for every model test in Section 20.

Recognise it. "Nondimensionalize the equation (or boundary condition) using the scales L, U, ..." "Show that the flow depends only on Re and Fr." "Which terms can be neglected when ...?" "Estimate the thickness of the region where viscosity matters." A flow with a very small or very large Reynolds number and a question about what simplifies. A pulsating or oscillating flow with a given frequency. A free surface, droplet or wave and a question about the role of gravity or surface tension.

On the crib sheet. The groups: Re, Fr, Eu, St $= \omega L/V$, We, Ca, and the Stokes drag of a sphere in the form $C_D = 24/\mathrm{Re}$. Section 13 of the sheet prints only 2-D incompressible continuity and the vorticity and strain-rate definitions; the Navier–Stokes equations themselves are not on it, so start from the form in Section 13. Not on it either: the scaled equation, which group multiplies which term, the change of pressure scale at low Re, and the free-surface conditions.

The model

The starting equations. For an incompressible Newtonian fluid with constant ρ and μ (derived in Section 13):

$$\nabla \cdot \mathbf{V} = 0, \qquad \rho\left(\frac{\partial \mathbf{V}}{\partial t} + (\mathbf{V}\cdot\nabla)\mathbf{V}\right) = -\nabla p + \rho\mathbf{g} + \mu\nabla^2\mathbf{V}.$$

The left side is mass times acceleration per unit volume: the local (unsteady) part and the convective part. The right side is the pressure force, the weight and the net viscous force, each per unit volume.

Choosing scales. A *reference scale* is a constant with dimensions that sets the typical size of a variable in this problem: a length L (a diameter, a channel height), a speed U (the free-stream or mean velocity), a frequency ω if something oscillates, and a pressure difference Δp_0 if one is imposed. Define the starred, dimensionless variables

$$\mathbf{x}^* = \frac{\mathbf{x}}{L}, \quad \mathbf{V}^* = \frac{\mathbf{V}}{U}, \quad t^* = \omega t, \quad p^* = \frac{p - p_\infty}{\Delta p_0}, \quad \nabla^* = L\nabla.$$

The aim is that each starred quantity and each starred derivative is of order one where the action is, so that the size of each term is carried entirely by its coefficient. That is the only rule for choosing scales, and when it is broken the conclusions are wrong (see the Stokes limit below).

Substituting. By the chain rule, $\partial/\partial t = \omega\,\partial/\partial t^*$, $\nabla = \nabla^*/L$ and $\nabla^2 = \nabla^{*2}/L^2$. Write $\mathbf{g} = g\,\hat{\mathbf{g}}$ with $\hat{\mathbf{g}}$ a unit vector pointing down. Then

$$\rho U\omega\,\frac{\partial \mathbf{V}^*}{\partial t^*} + \frac{\rho U^2}{L}(\mathbf{V}^*\cdot\nabla^*)\mathbf{V}^* = -\frac{\Delta p_0}{L}\nabla^* p^* + \rho g\,\hat{\mathbf{g}} + \frac{\mu U}{L^2}\nabla^{*2}\mathbf{V}^*.$$

Divide through by $\rho U^2/L$, the size of the convective inertia, so that term has coefficient 1:

$$\boxed{\mathrm{St}\,\frac{\partial \mathbf{V}^*}{\partial t^*} + (\mathbf{V}^*\cdot\nabla^*)\mathbf{V}^* = -\mathrm{Eu}\,\nabla^* p^* + \frac{1}{\mathrm{Fr}^2}\,\hat{\mathbf{g}} + \frac{1}{\mathrm{Re}}\nabla^{*2}\mathbf{V}^*}$$

with

$$\mathrm{St} = \frac{\omega L}{U}, \quad \mathrm{Eu} = \frac{\Delta p_0}{\rho U^2}, \quad \mathrm{Fr} = \frac{U}{\sqrt{gL}}, \quad \mathrm{Re} = \frac{\rho U L}{\mu}.$$

Continuity becomes $\nabla^*\cdot\mathbf{V}^* = 0$, with no coefficient at all: it has only one kind of term. Each group is a ratio of two force sizes per unit volume. Re is convective inertia ($\rho U^2/L$) over viscous force ($\mu U/L^2$); Fr^2 is inertia over weight; Eu is the imposed pressure force over inertia; St is unsteady inertia over convective inertia.

The conclusion is the theorem behind model testing: two flows with geometrically similar boundaries and the same St, Eu, Fr and Re obey the same dimensionless equation with the same dimensionless boundary conditions, so they have the same dimensionless solution $\mathbf{V}^*(\mathbf{x}^*, t^*)$, $p^*(\mathbf{x}^*, t^*)$.

Three groups that often drop out.

1. *Strouhal.* If nothing imposes a frequency, the natural time scale is L/U, so $\omega = U/L$ and $\mathrm{St} = 1$. In steady flow the term vanishes.

2. *Euler.* If the pressure is not imposed but is whatever the flow makes it (flow past a body), the natural pressure scale is $\Delta p_0 = \rho U^2$, so $\mathrm{Eu} = 1$ in the equation and the measured Eu (or the pressure coefficient C_p) is an *output*, a function of the other groups.
3. *Froude, when there is no free surface.* Weight per unit volume is $\rho\mathbf{g} = -\nabla(\rho g z)$ with z upward. Combine it with pressure into the *modified pressure* $\mathcal{P} = p + \rho g z$: then $-\nabla p + \rho\mathbf{g} = -\nabla\mathcal{P}$ and gravity has vanished from the equation. In a pipe or around a submerged body, gravity only adds the hydrostatic pressure and does not change the velocity field, so Fr is irrelevant. It comes back only through a boundary condition where p itself, not $\mathcal{P}$, is prescribed: a free surface.

Small Reynolds number: Stokes (creeping) flow. If $\mathrm{Re} \ll 1$, the boxed equation seems to say the viscous term is huge and nothing balances it. The fault is the pressure scale: ρU^2 is the pressure inertia can create, and inertia is now negligible. Pressure is set by viscosity instead, so rescale with $\Delta p_0 = \mu U/L$. Writing $p^{**} = (p - p_\infty)L/(\mu U)$, using modified pressure, and multiplying the equation by Re:

$$\mathrm{Re}\left[\mathrm{St}\,\frac{\partial \mathbf{V}^*}{\partial t^*} + (\mathbf{V}^*\cdot\nabla^*)\mathbf{V}^*\right] = -\nabla^* p^{**} + \nabla^{*2}\mathbf{V}^*.$$

As $\mathrm{Re} \to 0$ the left side disappears: $\nabla\mathcal{P} = \mu\nabla^2\mathbf{V}$, the *Stokes equations*. Consequences that exam questions test: the equation is linear, so doubling the speed doubles every velocity, pressure and force; density has dropped out, so forces scale as $\mu U L$ times a shape factor (for a sphere, $F = 3\pi\mu U D$, which is the sheet's $C_D = 24/\mathrm{Re}$); and reversing the motion of the boundaries reverses the flow exactly. Examples: microfluidics, lubrication films, sedimenting fine particles, swimming bacteria, flow of very viscous resins and glass.

Large Reynolds number: inviscid flow and boundary layers. If $\mathrm{Re} \gg 1$, the viscous term looks negligible and the equation becomes Euler's equation, $\rho\, D\mathbf{V}/Dt = -\nabla p + \rho\mathbf{g}$, from which Bernoulli follows along streamlines. But this limit is *singular*: dropping the viscous term removes the highest derivatives, and with them the ability to satisfy no-slip at a wall. Near a wall the velocity changes over a thin distance δ, not L, so the true size of the viscous term there is $\mu U/\delta^2$. Balancing it against inertia $\rho U^2/L$ gives

$$\frac{\delta}{L} \sim \mathrm{Re}^{-1/2}.$$

So at large Re the flow is inviscid almost everywhere, with viscosity confined to thin boundary layers and wakes (chapter 7). The lesson generalises: whenever a term looks negligible, check whether some region has a smaller length scale that makes it large again.

Nondimensional boundary conditions. Boundary conditions must be scaled with the same scales, and they can bring their own groups.

1. *No slip* at a wall moving with velocity $\mathbf{V}_w$: $\mathbf{V}^* = \mathbf{V}_w/U$. No new group, only a velocity ratio.
2. *Kinematic free-surface condition*: a fluid particle on the surface $z = \eta(x, y, t)$ stays on it, $w = \partial\eta/\partial t + u\,\partial\eta/\partial x + v\,\partial\eta/\partial y$. Scaled with $\eta^* = \eta/L$ it gives $w^* = \mathrm{St}\,\partial\eta^*/\partial t^* + u^*\,\partial\eta^*/\partial x^* + v^*\,\partial\eta^*/\partial y^*$. No new group.

3. *Dynamic free-surface condition*: across a curved interface, surface tension makes the liquid pressure exceed the air pressure p_a by $\Upsilon(1/R_1 + 1/R_2)$, where R_1, R_2 are the principal radii of curvature (positive when the centre of curvature is on the liquid side) and viscous normal stresses are neglected. Write the curvature as $\kappa = 1/R_1 + 1/R_2$. In terms of the modified pressure with $p_\infty = p_a$, at $z = \eta$,

$$\mathcal{P} - p_a = \rho g \eta + \Upsilon \kappa.$$

   Scale with ρU^2, $\eta^* = \eta/L$ and $\kappa^* = L\kappa$:

$$\mathcal{P}^* = \frac{gL}{U^2}\,\eta^* + \frac{\Upsilon}{\rho U^2 L}\,\kappa^* = \frac{\eta^*}{\mathrm{Fr}^2} + \frac{\kappa^*}{\mathrm{We}}.$$

This is where gravity re-enters a flow whose interior equation lost it, and where surface tension enters at all. Their relative importance is the ratio of the two coefficients, the Bond number $\mathrm{Bo} = \mathrm{We}/\mathrm{Fr}^2 = \rho g L^2/\Upsilon$. For water, $\mathrm{Bo} = 1$ at $L = \sqrt{\Upsilon/(\rho g)} = 2.7\,\mathrm{mm}$, the *capillary length*: surface tension dominates below it, gravity above.

First moves. State the dimensional equation you start from. List the scales and say why each is the natural one. Define every starred variable, convert each derivative by the chain rule, divide by the coefficient of the term you choose as reference, and name each resulting group. Then evaluate the groups with numbers and order the terms by size. If a term you keep looks absurdly large, the pressure or length scale is wrong; rescale. For any free surface, scale the boundary condition as well and look for Fr and We there.

Worked examples

Example 19.1. A diaphragm pump drives water ($\nu = 1.0 \times 10^{-6}\,\mathrm{m^2/s}$) through a long cooling channel between parallel plates a distance $h = 2.0\,\mathrm{mm}$ apart, with mean speed $U = 0.30\,\mathrm{m/s}$ and a pulsation at $f = 25\,\mathrm{Hz}$. (a) Nondimensionalize the x-momentum equation using h, U and $\omega = 2\pi f$, and name the groups. (b) Evaluate Re, St, and the ratio of unsteady inertia to viscous force. (c) Which terms dominate in the core of the channel? Estimate the thickness of the wall layer in which viscosity matters. (d) Below what pulsation frequency would the flow be quasi-steady, that is, at each instant the steady laminar profile for the instantaneous pressure gradient?

Solution. (a) Far from the inlet the flow is fully developed: $u = u(y,t)$, $v = 0$. Continuity, $\partial u/\partial x = 0$, then makes the convective term $u\,\partial u/\partial x + v\,\partial u/\partial y$ vanish identically. With gravity absorbed into modified pressure, the x-momentum equation is $\rho\,\partial u/\partial t = -\partial \mathcal{P}/\partial x + \mu\,\partial^2 u/\partial y^2$. Scale $y^* = y/h$, $u^* = u/U$, $t^* = \omega t$, $x^* = x/h$, $\mathcal{P}^* = \mathcal{P}/(\rho U^2)$, and divide by $\rho U^2/h$:

$$\mathrm{St}\,\frac{\partial u^*}{\partial t^*} = -\frac{\partial \mathcal{P}^*}{\partial x^*} + \frac{1}{\mathrm{Re}}\,\frac{\partial^2 u^*}{\partial y^{*2}}, \qquad \mathrm{St} = \frac{\omega h}{U}, \quad \mathrm{Re} = \frac{Uh}{\nu}.$$

(b) $\omega = 2\pi \times 25 = 157\,\mathrm{rad/s}$. $\mathrm{Re} = 0.30 \times 0.002/10^{-6} = 600$. $\mathrm{St} = 157 \times 0.002/0.30 = 1.05$. The ratio unsteady/viscous is $\mathrm{St}/(1/\mathrm{Re}) = \mathrm{St}\,\mathrm{Re} = \omega h^2/\nu = 157 \times (0.002)^2/10^{-6} = 628$. (This product, the square of the Womersley number, is the natural single group for oscillating channel flow: it contains no U at all.)

(c) In the core, unsteady inertia is about 600 times the viscous force, so the core moves as a plug, accelerated back and forth by the pressure gradient, as an inviscid

fluid would. Near the walls no-slip forces $u = 0$, so viscosity must matter in a layer of thickness δ where the viscous term is as large as the unsteady one. Rescaling y by δ instead of h: $\rho U\omega \sim \mu U/\delta^2$, so

$$\delta \sim \sqrt{\nu/\omega} = \sqrt{10^{-6}/157} = 8.0 \times 10^{-5}\,\mathrm{m} = 0.080\,\mathrm{mm},$$

about 4% of the channel height. The velocity profile is flat across most of the gap with thin oscillating layers at the walls.

(d) Quasi-steady means the unsteady term is negligible: $\mathrm{St}\,\mathrm{Re} = \omega h^2/\nu \ll 1$. The boundary $\omega h^2/\nu = 1$ is at $f = \nu/(2\pi h^2) = 10^{-6}/(2\pi \times 4 \times 10^{-6}) = 0.040\,\mathrm{Hz}$, one cycle every $25\,\mathrm{s}$. Any pulsation faster than a few hundredths of a hertz is not quasi-steady in this channel.

Assumption to state: laminar flow. The Reynolds number based on the hydraulic diameter $2h$ is 1200, below the usual transition range for a channel. □

Sanity check: the viscous diffusion time across the half-gap is $(h/2)^2/\nu = 1\,\mathrm{s}$, while the pulsation period is $0.04\,\mathrm{s}$; viscosity has no time to spread the wall's influence into the core, which is what $\mathrm{St}\,\mathrm{Re} = 628$ says. The scaling estimate assigned the convective term a coefficient 1, yet it is exactly zero here: scaling bounds the size of a term, the geometry can still make it vanish.

Example 19.2. A photopolymer resin ($\rho = 1100\,\mathrm{kg/m^3}$, $\mu = 2.5\,\mathrm{Pa\,s}$) flows at $U = 5\,\mathrm{mm/s}$ past printed features of size $L = 1\,\mathrm{mm}$ during recoating. (a) Compute Re and Fr, and say which terms of the Navier–Stokes equation matter. (b) Show that the pressure scale ρU^2 is wrong here and give the right one, with numbers. (c) Estimate the force on a $1\,\mathrm{mm}$ spherical bead held fixed in this flow, two ways, and say how it changes if the speed doubles or the resin is replaced by one of the same viscosity and twice the density.

Solution. (a) $\mathrm{Re} = 1100 \times 0.005 \times 0.001/2.5 = 2.2 \times 10^{-3}$. $\mathrm{Fr} = 0.005/\sqrt{9.81 \times 0.001} = 0.050$. With no free surface near the feature, gravity goes into the modified pressure and Fr plays no role in the flow pattern. With $\mathrm{Re} \ll 1$, inertia (unsteady and convective) is negligible: the flow obeys the Stokes equations $\nabla\mathcal{P} = \mu\nabla^2\mathbf{V}$.

(b) $\rho U^2 = 1100 \times (0.005)^2 = 0.0275\,\mathrm{Pa}$, the pressure inertia could produce. The viscous stress scale is $\mu U/L = 2.5 \times 0.005/0.001 = 12.5\,\mathrm{Pa}$, about 450 times larger. Pressure differences must balance the viscous force, so they are of order $12.5\,\mathrm{Pa}$; scaling with ρU^2 would make p^* of order 450, not 1, violating the rule that starred variables be of order one. The ratio of the two scales is Re itself: $\rho U^2/(\mu U/L) = \mathrm{Re}$.

(c) Stokes drag: $F = 3\pi\mu U D = 3\pi \times 2.5 \times 0.005 \times 0.001 = 1.18 \times 10^{-4}\,\mathrm{N}$. From the sheet's $C_D = 24/\mathrm{Re}$ with $\mathrm{Re} = 2.2\times10^{-3}$ (based on D): $C_D = 1.09\times10^4$, and $F = C_D\cdot\frac{1}{2}\rho U^2\cdot\pi D^2/4 = 1.09 \times 10^4 \times 0.0138 \times 7.85 \times 10^{-7} = 1.18 \times 10^{-4}\,\mathrm{N}$. The two agree because they are the same formula. Doubling U doubles F (linear equations), to $2.36 \times 10^{-4}\,\mathrm{N}$. Doubling ρ at fixed μ changes nothing, since ρ does not appear in the Stokes equations. □

Sanity check: $1.2 \times 10^{-4}\,\mathrm{N}$ is about the weight of a $12\,\mathrm{mg}$ mass; a $1\,\mathrm{mm}$ steel bead weighs about $4 \times 10^{-5}\,\mathrm{N}$, so the flow could carry such a bead along, which matches the experience that thick resin drags particles with it. The force per unit area, $F/(\pi D^2) \approx 38\,\mathrm{Pa}$, is a few times the viscous scale $12.5\,\mathrm{Pa}$, as a shape factor of order π would predict.

Example 19.3. A harbour is to be studied in a wave tank. Prototype swell has wavelength $40\,\mathrm{m}$; local wind chop has wavelength $5\,\mathrm{m}$. Water: $\rho = 998\,\mathrm{kg/m^3}$, $\Upsilon =$

$0.073\,\mathrm{N/m}$. (a) Starting from the dynamic free-surface condition, show which group measures surface tension against gravity, and find the wavelength at which the two are equal for water. (b) The deep-water wave speed, including both effects, is $c^2 = g\lambda/(2\pi) + 2\pi\Upsilon/(\rho\lambda)$ (given). For models at $1{:}50$ and $1{:}400$, find the ratio of the surface-tension term to the gravity term for the scaled swell and chop, and the resulting error in the model wave speed relative to a gravity-only (Froude-scaled) wave.

Solution. (a) The scaled condition from the model above is $\mathcal{P}^* = \eta^*/\mathrm{Fr}^2 + \kappa^*/\mathrm{We}$. The ratio of the surface-tension coefficient to the gravity coefficient is $\mathrm{Fr}^2/\mathrm{We} = \Upsilon/(\rho g L^2) = 1/\mathrm{Bo}$. For a sinusoidal wave the curvature is $(2\pi/\lambda)^2$ times the amplitude, so the natural length is $\lambda/2\pi$ and the two effects are equal when $\rho g(\lambda/2\pi)^2 = \Upsilon$, that is

$$\lambda_c = 2\pi\sqrt{\frac{\Upsilon}{\rho g}} = 2\pi\sqrt{\frac{0.073}{998 \times 9.81}} = 2\pi \times 2.73\,\mathrm{mm} = 17.2\,\mathrm{mm}.$$

The given c^2 shows the same: its two terms are equal at $\lambda^2 = 4\pi^2\Upsilon/(\rho g)$.

(b) The ratio of the two terms in c^2 is $(2\pi\Upsilon/\rho\lambda)/(g\lambda/2\pi) = (\lambda_c/\lambda)^2$, and the speed error is $\sqrt{1 + (\lambda_c/\lambda)^2} - 1$.

Case	Model λ	Capillary/gravity	Speed error
Prototype swell	$40\,\mathrm{m}$	1.8×10^{-7}	0
Swell, $1{:}50$	$0.80\,\mathrm{m}$	4.6×10^{-4}	0.02%
Swell, $1{:}400$	$0.10\,\mathrm{m}$	0.029	1.5%
Chop, $1{:}50$	$0.10\,\mathrm{m}$	0.029	1.5%
Chop, $1{:}400$	$12.5\,\mathrm{mm}$	1.88	70%

At $1{:}50$ both wave systems are gravity waves in the model, as they are in the harbour, and Froude scaling is sound. At $1{:}400$ the swell is still acceptable, but the scaled chop is shorter than λ_c: it is a capillary ripple travelling 70% faster than Froude scaling assumes (0.237 against $0.140\,\mathrm{m/s}$), and anything that depends on it, such as wave agitation inside the harbour, is wrong in the model. □

Sanity check: wind ripples on a puddle, a few centimetres long, are visibly capillary (they move fast for their size and ignore gravity); ocean swell is not. The 1.5% entry appears twice because the $1{:}50$ chop and the $1{:}400$ swell have the same model wavelength, as they must.

Trap. Scaling pressure with ρU^2 in a creeping flow and concluding viscosity dominates "infinitely". Dropping the viscous term at high Re and then imposing no-slip. Keeping Fr in a closed-duct or fully submerged flow (it belongs only to free surfaces). Forgetting to scale the boundary conditions, which is where We and often Fr live. Writing St with a frequency in Hz when the sheet's form uses ω in rad/s (state which you use). Treating Eu as a parameter to match when it is the measured output.

Ladder.

1. White P5.44. A porous-medium energy equation; dimensions, then nondimensionalize. Choosing scales for an equation.
2. White P5.66. A torpedo's cavitation speed at a new depth. A pressure group in the boundary condition.
3. White P5.56. Square-cylinder data in water scaled to a chimney in wind. When matching the groups is enough.

Answers are in the back of White; settle each call the same sitting.

Where it hides. In the exact solutions of chapter 4, where the convective term vanishes for geometric reasons and the scaled equation shows why the solution depends on so few groups. In lubrication and thin-film problems, where the small group is $\mathrm{Re}\,(h/L)^2$ rather than Re. In any "can I use Bernoulli here" decision, which is a judgement that $1/\mathrm{Re}$ is small away from walls. In boundary layers (chapter 7), whose thickness estimate $\delta/L \sim \mathrm{Re}^{-1/2}$ is a scaling argument. In model testing, where the groups in the scaled equations are exactly the ones to match.

20 Modelling and similarity

A model test is a measurement on a small (or occasionally large) copy of a device, made so that the answer can be scaled to the full-size device, the *prototype*. The Pi theorem says why this can work: if every independent dimensionless group has the same value in the model and the prototype, the dependent group must too, and the scaling rule is just "hold the dependent group fixed". The difficulty is that the groups usually cannot all be matched at once, and most of the engineering, and most of the exam marks after the arithmetic, lies in deciding which group to give up and how to correct for it.

Recognise it. "A $1{:}20$ model of …is tested in …" "At what speed must the model be run?" "Predict the prototype force, power, pressure, discharge or time." "Can both Froude and Reynolds similarity be achieved?" "What fluid would be needed?" A ship, spillway, river or wave basin (Froude), a submerged body, pipe component or aircraft at low speed (Reynolds), a high-speed aircraft or compressor (Mach). Any question with two sets of fluid properties and two sizes.

On the crib sheet. "Similarity: $\Pi_{\text{model}} = \Pi_{\text{prototype}}$", and the groups Re, Fr, Ma, We, Eu, St, Ca, the cavitation number $(p - p_v)/(\frac{1}{2}\rho V^2)$ and C_p. The turbulent flat-plate drag $C_D = 0.031/\mathrm{Re}_L^{1/7}$ in section 8 is what the ship-resistance method below needs. Not on it: the scale laws for force, power, time and discharge, the Froude–Reynolds conflict and the ways round it.

The model

Notation. Subscript m for model and p for prototype. A "$1{:}\lambda$ model" has every length λ times smaller than the prototype, so the *length scale ratio* is $L_p/L_m = \lambda$.

Three levels of similarity.

1. *Geometric similarity*: every length in the model is the prototype length divided by the same λ, and every angle is the same. This includes the lengths people forget: surface roughness heights, gaps, fillet radii, the depth of water around a ship.
2. *Kinematic similarity*: at corresponding points (points at the same position relative to the geometry), the velocities in the model and prototype point in the same direction and have the same ratio everywhere. Streamline patterns are then geometrically similar, and all times scale by one ratio, $t_p/t_m = (L_p/L_m)(V_m/V_p)$.
3. *Dynamic similarity*: at corresponding points, every type of force (pressure, viscous, gravity, surface tension, compressibility) has the same ratio between model

and prototype. Since each standard group is a ratio of two force types (Section 19), this means every independent group is matched.

Geometric plus dynamic similarity gives kinematic similarity, because the nondimensional equations and boundary conditions are then identical and so is their solution.

From matching groups to scale laws. The Pi analysis gives $\Pi_1 = \phi(\Pi_2, \dots, \Pi_k)$ with the measured quantity in Π_1. If $\Pi_2, \dots, \Pi_k$ are all matched, then $\Pi_{1,m} = \Pi_{1,p}$ whatever ϕ is. For a force, Π_1 is a force coefficient $F/(\rho V^2 L^2)$, so

$$\frac{F_p}{F_m} = \frac{\rho_p}{\rho_m}\left(\frac{V_p}{V_m}\right)^2\left(\frac{L_p}{L_m}\right)^2, \qquad \frac{P_p}{P_m} = \frac{F_p V_p}{F_m V_m} = \frac{\rho_p}{\rho_m}\left(\frac{V_p}{V_m}\right)^3\left(\frac{L_p}{L_m}\right)^2,$$

$$\frac{\Delta p_p}{\Delta p_m} = \frac{\rho_p}{\rho_m}\left(\frac{V_p}{V_m}\right)^2, \qquad \frac{Q_p}{Q_m} = \frac{V_p}{V_m}\left(\frac{L_p}{L_m}\right)^2, \qquad \frac{f_p}{f_m} = \frac{V_p}{V_m}\,\frac{L_m}{L_p}.$$

The pressure law is Eu (or C_p) matched; the frequency law is St matched. These hold under every kind of similarity; what changes between Reynolds, Froude and Mach scaling is only the velocity ratio.

Reynolds scaling. When viscosity and inertia are the only forces that matter (no free surface, no compressibility: pipes, valves, submerged bodies, low-speed aircraft), match Re:

$$\frac{V_m L_m}{\nu_m} = \frac{V_p L_p}{\nu_p} \quad\Longrightarrow\quad \frac{V_m}{V_p} = \frac{\nu_m}{\nu_p}\,\lambda.$$

In the same fluid a $1{:}10$ model must run ten times faster. Substituting this into the force law gives $F_p/F_m = (\rho_p \nu_p^2)/(\rho_m \nu_m^2)$, independent of size: a Reynolds-scaled model in the same fluid feels the same force as the prototype. Ways to keep the speed reasonable: test in water instead of air (ν is about 15 times smaller), pressurise the tunnel ($\nu = \mu/\rho$ falls as ρ rises), or cool the gas.

Froude scaling. When a free surface is present and gravity drives the waves or the flow (ships, spillways, open channels, harbours), match Fr. With the same g,

$$\frac{V_m}{\sqrt{g L_m}} = \frac{V_p}{\sqrt{g L_p}} \quad\Longrightarrow\quad \frac{V_p}{V_m} = \lambda^{1/2}.$$

The other scales follow from the general laws above. For the same fluid:

Quantity	Length	Velocity	Time	Discharge	Pressure	Force	Power
Prototype/model	λ	$\lambda^{1/2}$	$\lambda^{1/2}$	$\lambda^{5/2}$	λ	λ^3	$\lambda^{7/2}$

Each entry is a product: time is $L/V = \lambda/\lambda^{1/2}$; discharge is $VL^2 = \lambda^{1/2}\lambda^2$; pressure is $\rho V^2 = \lambda$; force is pressure times area, $\lambda \cdot \lambda^2$; power is force times velocity, $\lambda^3 \cdot \lambda^{1/2}$. Accelerations are the same in both (g is), which is the physical content of Froude scaling. If the fluids differ (seawater prototype, fresh-water model), multiply force, pressure and power by ρ_p/ρ_m.

Mach scaling. When compressibility matters, roughly Ma > 0.3 where density changes exceed about 5%, match Ma $= V/a$: $V_m/V_p = a_m/a_p$, and the ratio of specific heats k should also match (air to air does). Below Ma ≈ 0.3 the Mach number can be ignored, which is what makes low-speed wind tunnels and water tunnels interchangeable.

The Froude–Reynolds conflict. A ship makes waves (gravity) and has skin friction (viscosity). Matching both requires $V_m/V_p = \lambda^{-1/2}$ and $V_m/V_p = (\nu_m/\nu_p)\lambda$, so

$$\frac{\nu_m}{\nu_p} = \lambda^{-3/2}.$$

For a $1{:}25$ model the model fluid would need a kinematic viscosity 125 times smaller than water's. No practical liquid comes close (mercury, one of the lowest, is about 9 times lower than water). So in water, with water, complete similarity is impossible. What is done:

1. *Use a different fluid* where the required ratio is achievable. Only possible for mild scale ratios; rare for free-surface work, more common in Reynolds–Mach conflicts in wind tunnels (pressurised or cryogenic gas).
2. *Accept the Reynolds mismatch when the flow is insensitive to* Re. Many flows become independent of Re once it is large enough: flows that separate at sharp edges (spillway crests, bluff bodies with sharp corners), and fully rough turbulent channel flow, where the friction factor depends on ϵ/D only. Then match Fr, keep the model large enough that its Re is in the same independent regime (turbulent, comfortably above transition), and state the assumption. This is standard for spillways, stilling basins and rivers.
3. *Split the effect into parts that each obey one group* (Froude's hull method). Assume the total resistance coefficient is a friction part that depends on Re only plus a wave (residual) part that depends on Fr only: $C_T = C_F(\mathrm{Re}) + C_R(\mathrm{Fr})$. Test at the prototype Fr, compute the model's friction from a flat-plate formula at the model Re, subtract to get C_R, carry C_R unchanged to the prototype, and add the prototype's friction at the prototype Re. The third worked example does this.

Distortion and scale effects. A *scale effect* is any difference between scaled model data and the prototype caused by a group that was not matched. The usual ones: a small model's boundary layer stays laminar where the prototype's is turbulent (cured with trip wires or sand strips near the leading edge); surface tension, negligible in the prototype, matters in water a few centimetres deep (We too small), and air entrainment in spillways is under-represented; surface roughness cannot be scaled down below the machining finish; cavitation depends on $(p - p_v)/(\frac{1}{2}\rho V^2)$, which a Froude model at atmospheric pressure does not match (cavitation tunnels lower the ambient pressure to fix it).

A *distorted model* deliberately uses different scale ratios in different directions. River and estuary models often have a vertical scale several times less severe than the horizontal, because a geometrically exact model of a wide shallow river would have water only millimetres deep: laminar, dominated by surface tension. The cost is that slopes are exaggerated and friction must be adjusted (often by adding roughness) until the model reproduces measured prototype water levels. A distorted model is calibrated, not simply scaled.

When a model test is the wrong tool. Model tests answer questions whose physics is captured by a few matched groups. They are a poor choice when:

1. the governing equations can be solved reliably by analysis or computation (laminar flows in simple geometries; much of chapter 4), so the test adds cost but not information;
2. a group that matters cannot be matched and its effect cannot be separated or corrected, as in a small model where surface tension, laminar flow and cavitation all depart from the prototype at once;
3. the phenomenon depends on material behaviour that does not scale with geometry: sediment and erosion (grain size and density), ice, combustion chemistry, biological growth;
4. the required accuracy is finer than the expected scale effect.

In practice models and computation are used together: the model validates the computation in a regime both can reach, and the computation extrapolates to full-scale Re.

First moves. Identify which forces matter (viscous, gravity, surface tension, compressibility) and hence which groups. Write the length ratio λ. Match the governing group to get the model velocity. Check the groups you did not match and say why they do or do not matter (evaluate the model Re, We or Ma with numbers). Scale the measured quantity by holding its coefficient fixed. Sanity-check the direction: a bigger, faster prototype in water has larger forces.

Worked examples

Example 20.1. A remotely operated underwater vehicle, $2.0\,\mathrm{m}$ long, will cruise deeply submerged at $1.5\,\mathrm{m/s}$ in seawater at $10^{\circ}\mathrm{C}$ ($\rho = 1026\,\mathrm{kg/m^3}$, $\nu = 1.35\times10^{-6}\,\mathrm{m^2/s}$). A $1{:}5$ model is to be tested. (a) Explain why Re is the only group to match. Find the model speed in a fresh-water tunnel ($\rho = 998\,\mathrm{kg/m^3}$, $\nu = 1.00\times10^{-6}\,\mathrm{m^2/s}$). (b) The model drag at that speed is $92\,\mathrm{N}$. Find the prototype drag and the power needed to overcome it. (c) Find the model speed in a wind tunnel ($\rho = 1.204\,\mathrm{kg/m^3}$, $\nu = 1.51\times10^{-5}\,\mathrm{m^2/s}$, $a = 343\,\mathrm{m/s}$) and check whether compressibility spoils the test. What drag will the balance read?

Solution. (a) Deeply submerged: no free surface, so gravity only adds hydrostatic pressure and Fr is irrelevant (Section 19); no interface, so no We; low speed, so no Ma. Geometric similarity plus Re match gives dynamic similarity. $L_m = 0.40\,\mathrm{m}$.

$$V_m = V_p\,\frac{L_p}{L_m}\,\frac{\nu_m}{\nu_p} = 1.5\times 5\times\frac{1.00}{1.35} = 5.56\,\mathrm{m/s}.$$

Check: $\mathrm{Re}_p = 1.5\times2.0/1.35\times10^{-6} = 2.22\times10^{6}$ and $\mathrm{Re}_m = 5.56\times0.40/10^{-6} = 2.22\times10^{6}$.

(b) Same drag coefficient $F/(\rho V^2 L^2)$:

$$F_p = 92\times\frac{1026}{998}\times\left(\frac{1.5}{5.56}\right)^2\times 5^2 = 172\,\mathrm{N}, \qquad P = F_p V_p = 172\times1.5 = 259\,\mathrm{W}.$$

(c) $V_m = 1.5\times5\times(1.51\times10^{-5}/1.35\times10^{-6}) = 83.9\,\mathrm{m/s}$, so $\mathrm{Ma} = 83.9/343 = 0.245 < 0.3$: density changes are a few percent at most and the test remains incompressible to good approximation. The balance would read $F_m = F_p\,(\rho_m\nu_m^2)/(\rho_p\nu_p^2) = 172\times(1.204\times(1.51\times10^{-5})^2)/(1026\times(1.35\times10^{-6})^2) = 25.3\,\mathrm{N}$. □

Sanity check: with Re matched, $F_p/F_m = (\rho_p \nu_p^2)/(\rho_m \nu_m^2) = 1026 \times 1.35^2/998 = 1.87$, and $92 \times 1.87 = 172\,\mathrm{N}$: the size has cancelled, as the Reynolds-scaling paragraph promised. The model runs faster than the prototype (5.6 against $1.5\,\mathrm{m/s}$), which is always so for a small Reynolds model in a similar fluid; the wind-tunnel speed is larger again because air's ν is 11 times seawater's.

Example 20.2. A $1{:}25$ model of an overflow spillway and stilling basin is built in a hydraulics laboratory with fresh water at $20^\circ\mathrm{C}$ ($\nu = 1.0 \times 10^{-6}\,\mathrm{m^2/s}$, $\Upsilon = 0.073\,\mathrm{N/m}$), the same water as the prototype. Measured in the model: discharge $0.085\,\mathrm{m^3/s}$, velocity $2.4\,\mathrm{m/s}$ and flow depth $40\,\mathrm{mm}$ at the toe of the spillway, a force of $38\,\mathrm{N}$ on a baffle block, and a lowest crest pressure of $-0.9\,\mathrm{kPa}$ gauge. (a) Which group governs, and why? (b) Find the prototype discharge, toe velocity, baffle force and crest pressure. (c) A design flood lasts $6\,\mathrm{h}$ in the prototype; how long must the model run? (d) Check the unmatched groups and comment on cavitation.

Solution. (a) The flow is driven by gravity over a free surface, so Fr governs. Matching Re as well would need $\nu_m = \nu_p/125$, impossible; we accept the Re mismatch because the flow separates at sharp edges and is fully turbulent, a regime where it is insensitive to Re (checked in (d)).

(b) With $\lambda = 25$, same fluid:

$$Q_p = 0.085 \times 25^{5/2} = 0.085 \times 3125 = 266\,\mathrm{m^3/s}, \qquad V_p = 2.4 \times 25^{1/2} = 12.0\,\mathrm{m/s},$$

$$F_p = 38 \times 25^3 = 5.94 \times 10^5\,\mathrm{N} = 594\,\mathrm{kN}, \qquad p_p = -0.9 \times 25 = -22.5\,\mathrm{kPa\ gauge}.$$

(c) Time scales as $\lambda^{1/2} = 5$: the model runs $6/5 = 1.2\,\mathrm{h}$.

(d) Model Re at the toe, on depth: $2.4 \times 0.040/10^{-6} = 9.6 \times 10^4$, turbulent. Prototype: $12.0 \times 1.0/10^{-6} = 1.2 \times 10^7$, a factor $\lambda^{3/2} = 125$ higher, but both are turbulent, which is what the Re-independence assumption needs. Model Weber number on depth: $998 \times 2.4^2 \times 0.040/0.073 = 3150$, so surface tension is negligible in the main flow (it would matter for thin sheets and air entrainment). Cavitation: atmospheric pressure is the same in both, not scaled by λ. The prototype crest is at $101.3 - 22.5 = 78.8\,\mathrm{kPa}$ absolute, far above the vapour pressure of about $2.3\,\mathrm{kPa}$, so no cavitation is predicted there; but the model's cavitation number, with $p - p_v \approx 99\,\mathrm{kPa}$ against $\tfrac{1}{2}\rho V^2$ twenty-five times smaller, is much larger than the prototype's, so the model could never have shown cavitation even if it occurred. □

Sanity check: $12\,\mathrm{m/s}$ at the toe is the free-fall speed $\sqrt{2gh}$ from $h = 7.3\,\mathrm{m}$, plausible for a spillway of this scale. The force grew by 25^3: forces scale with volume in Froude models.

Example 20.3. A cargo hull is $60\,\mathrm{m}$ long with a wetted area of $900\,\mathrm{m^2}$ and a design speed of $7.0\,\mathrm{m/s}$ in seawater ($\rho = 1025\,\mathrm{kg/m^3}$, $\nu = 1.19 \times 10^{-6}\,\mathrm{m^2/s}$). A $1{:}40$ model is towed in fresh water ($\rho = 998\,\mathrm{kg/m^3}$, $\nu = 1.00 \times 10^{-6}\,\mathrm{m^2/s}$) at the Froude-similar speed and its total resistance is $2.2\,\mathrm{N}$. (a) Find the towing speed and show that Reynolds similarity is impossible. (b) Predict the prototype resistance and effective power by Froude's method, using the turbulent flat-plate formula $C_F = 0.031/\mathrm{Re}_L^{1/7}$ for the friction. (c) Compare with the prediction from Froude scaling of the total resistance, and explain the difference.

Solution. (a) $V_m = 7.0/\sqrt{40} = 1.107\,\mathrm{m/s}$ ($\mathrm{Fr} = 7.0/\sqrt{9.81 \times 60} = 0.289$ for both). Matching Re in fresh water instead would need $V_m = 7.0 \times 40 \times 1.00/1.19 = 235\,\mathrm{m/s}$, or, at the Froude speed, a fluid with $\nu_m = 1.19 \times 10^{-6} \times 40^{-3/2} = 4.7 \times 10^{-9}\,\mathrm{m^2/s}$; neither exists.

(b) Model: $L_m = 1.5\,\text{m}$, $S_m = 900/40^2 = 0.5625\,\text{m}^2$, $\text{Re}_m = 1.107 \times 1.5/10^{-6} = 1.66 \times 10^6$.

$$C_{T,m} = \frac{R_m}{\frac{1}{2}\rho_m V_m^2 S_m} = \frac{2.2}{\frac{1}{2} \times 998 \times 1.107^2 \times 0.5625} = \frac{2.2}{343.8} = 6.40 \times 10^{-3}.$$

Friction: $C_{F,m} = 0.031/(1.66 \times 10^6)^{1/7} = 4.01 \times 10^{-3}$. Residual: $C_R = 6.40 \times 10^{-3} - 4.01 \times 10^{-3} = 2.39 \times 10^{-3}$, the same for the prototype because Fr is matched. Prototype: $\text{Re}_p = 7.0 \times 60/1.19 \times 10^{-6} = 3.53 \times 10^8$, $C_{F,p} = 0.031/(3.53 \times 10^8)^{1/7} = 1.86 \times 10^{-3}$, so $C_{T,p} = 2.39 \times 10^{-3} + 1.86 \times 10^{-3} = 4.25 \times 10^{-3}$, and

$$R_p = C_{T,p}\,\tfrac{1}{2}\rho_p V_p^2 S_p = 4.25 \times 10^{-3} \times \tfrac{1}{2} \times 1025 \times 7.0^2 \times 900 = 96.2\,\text{kN}, \qquad P_E = R_p V_p = 673\,\text{kW}.$$

(c) Scaling the total resistance as a Froude force: $R_p = 2.2 \times (1025/998) \times 40^3 = 145\,\text{kN}$, 50% too high. That treats the whole coefficient 6.40×10^{-3} as if it depended on Fr alone, but the friction part falls as Re rises, and the prototype's Re is 200 times the model's. The model's friction coefficient is more than twice the prototype's, and Froude scaling carries that excess to full size.

Assumptions to state: the resistance separates into independent friction and wave parts; the friction of the hull equals that of a flat plate of the same length and wetted area; the model boundary layer is turbulent from near the bow (towing tanks fit a trip wire to ensure it); and the sheet's formula is used at $\text{Re}_p = 3.5 \times 10^8$, beyond the range it was fitted to. Towing tanks use a dedicated correlation line for this step; the method is the same. □

Sanity check: $C_T \approx 4 \times 10^{-3}$ is in the usual range for a full-scale hull, and a residual of about 40% of the model total at $\text{Fr} = 0.29$ is a moderate wave-making share. The 0.67 MW is before propeller losses.

Trap. Matching Re for a ship or spillway, or Fr for a submerged body. Scaling force as λ^2 by habit: it is λ^3 under Froude scaling and size-independent under Reynolds scaling in the same fluid. Forgetting the density ratio between seawater and fresh water, or air and water. Using λ for model/prototype in one line and prototype/model in the next; write the ratio out. Assuming atmospheric pressure scales in a Froude model (it does not, so cavitation and gas effects are not modelled). Claiming similarity because the dominant group is matched without checking the others with numbers.

Ladder.

1. White P5.80. A ship and its towing-tank model; speed, drag and power ratios. Froude scaling laws.
2. White P5.4. A sphere in water scaled to a balloon in air; speed and force. Reynolds matching across fluids.
3. White P5.84. A wave-tank model of a platform; current, period and wave height. Froude scaling for time and length.
4. White P5.62. A small wind-turbine test scaled to a large machine. Speed and power together.
5. White P5.66. A torpedo's cavitation speed at a new depth. Cavitation number as the matched group.
6. White P5.60. Model propeller data reduced to coefficients; the prototype at altitude. A full reduce-then-predict.
7. White P5.76. Ship model drag split into a Reynolds part and a Froude part. The Froude and Reynolds conflict. Needs a flat-plate drag law from Chapter 7; read it, solve it only if time.

Answers are in the back of White; settle each call the same sitting.

Where it hides. In every pump and turbine scaling question, where the similarity laws for head, flow and power at fixed specific speed are this section with ωD as the velocity scale. In drag problems (chapter 7): a drag coefficient measured at one speed and used at another is a model-to-prototype scaling. In pipe flow (chapter 6), where a friction factor read at one Re serves another fluid.

21 Sources

The course. The date, place, format, examinable sections and the rule on aids are from the MIE312 course announcement for Midterm 2, which Adit pasted on 2026-11-10. The crib sheet described in Section 1 is the file *MIE312 Aid Sheet* that Adit supplied the same day. Its contents are summarized, not reproduced, and the version handed out on the day governs.

The textbook. F. M. White, *Fluid Mechanics*, 8th edition, McGraw-Hill. Its section numbers organize this document, and its end-of-chapter problems are named by number only, as in White P3.62. The ladders use only problems whose answers appear in the book's *Answers to Selected Problems*, so that every practice call can be settled. No text, figure, example or problem statement from the book is reproduced here; the one-line descriptions on the ladders are paraphrases written for this document.

The examples. Every worked example in this document was invented for it, and every number in them was computed when it was written. None is a textbook or lecture example with its numbers changed. The standard derivations are the common property of the subject.